Supplemental Material for "Chromatic Constraints on Quantum Observables: Contextuality and Separability"

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This Supplemental Material provides the complete vector and context enumerations, two-valued state tables, normalized strong coloring tables, odd-cycle nonextendability certificates, chromatic signature and separation profile tables, parent-embedding dictionaries, and supplementary proofs and finite verification certificates supporting the main text. Section S1 lists the explicit unnormalized coordinates and context tables for the completed Yu–Oh hypergraph H_{YO}^{+} and its faithful restrictions $H_{17,10}$ and $H_{21,13}$, the three- and four-dimensional logical gadgets, the 37-ray complex Hesse hypergraph H_{37}, the 26-ray suspension $H_{\mathrm{YO}}^{(4)}$ and 65-ray canonical linear completion H_{65}, the Cabello completions $C_{36,18}$ and $C_{45,21}$, and the specified finite KS parents (49-ray Schütte–Bub, 87-ray MUB-inventory union, 40- and 72-ray Peres unions, and 68-ray three-ray extensions). Section S2 gives the complete two-valued state tables, extendable-state classifications, normalized coloring tables, odd-cycle certificates, and separation profiles across both regimes. Section S3 presents the supplementary proofs and exact reductions, including Isaacs dot-product snarks and quantitative Δ_3 robustness bounds for G_{32}, the 540,672-candidate reduction for the minimum 18-state order-determining family on $L(H_{37})$, the canonical completion selection rule for H_{65}, the forced-zero and ray-criticality certificates for the four-dimensional Cabello completions, transverse criticality deletion colorings, and historical Kochen–Specker gadget certificates.

READING GUIDE AND VERIFICATION SCOPE

This supplement collects complete vector and context inventories (S1), state and coloring enumerations (S2), and extended proofs and certificates (S3). Labels are local to each configuration. "Exact palette" means $\chi(H) = d$, with no additional density or geometric assumption.

The five-state $H_{17,10}$ mixture and both-direction gadget witnesses use the complete state families. Every selected support satisfies every context equation, and rational arithmetic verifies every Born marginal and the attained 1/9 nonextendable weight. The ordinary implication $10 \Rightarrow 24$ on $H_{17,10}$ gives a false inclusion between distinct rays and hence a failure of order determination.

The companion electronic data include all coordinates, state enumerations (including the compressed 393,216-state H_{65} inventory), verification programs, and the exhaustive certificates for the eighteen-state Hesse minimum and the 65,535-subset Yu–Oh classification. Checking an eighteen-state family establishes an upper bound; the matching lower bound requires the exhaustive certificate in Sec. S3.3. The directories `part-a/data`, `part-a/verification`, `part-b/data`, and `part-b/verification` contain the underlying data and verifiers. Reports and source-integrity hashes are in `audit/`; Sec. S3.7 describes the joint verification entry point and its scope.

S1. EXPLICIT UNNORMALIZED VECTORS AND CONTEXTS

All numerical vertex labels in this section are local to the named configuration and match the figures and theorems of the main text.

S1.1. Completed Yu–Oh and the Petersen-derived contexts

The three-dimensional vectors below give all projective rays used in Sec. III C; signs are chosen to make the first nonzero integer coordinate positive. The 26-ray suspension appends zero and adds e_4. The abstract G_{32}, H_9, H_8 context lists in Eq. (10) and Theorem 11 specify their incidences completely. The geometric obstructions exclude a corresponding injective three-dimensional vector enumeration for these abstract examples.

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TABLE S1: All 25 unnormalized completed Yu–Oh vectors.

i	r_i	i	r_i
1	$(0,1,1)$	14	$(2,1,1)$
2	$(1,0,0)$	15	$(1,2,-1)$
3	$(0,1,-1)$	16	$(1,-2,1)$
4	$(1,1,0)$	17	$(1,1,-2)$
5	$(0,0,1)$	18	$(1,-1,-2)$
6	$(1,-1,0)$	19	$(2,1,-1)$
7	$(1,0,1)$	20	$(2,-1,1)$
8	$(0,1,0)$	21	$(1,-2,-1)$
9	$(1,0,-1)$	22	$(1,1,2)$
10	$(1,-1,1)$	23	$(1,2,1)$
11	$(1,-1,-1)$	24	$(1,-1,2)$
12	$(1,1,1)$	25	$(2,-1,-1)$
13	$(1,1,-1)$		

S1.2. Faithful Yu–Oh restrictions $H_{17,10}$ and $H_{21,13}$ and logical gadgets in dimensions three and four

Tables S2 and S3 record the restriction membership of the 25 parent rays and 16 parent contexts of H^{+}_{YO} in $H_{17,10}$ and $H_{21,13}$. Tables S4–S14 provide the explicit unnormalized integer coordinates and context lists for the faithful real 3-dimensional gadgets P_5, $B_{13,7}$, $D_{16,9}$, $I_{16,9}$, and $T_{19,11}$ and the faithful real 4-dimensional gadgets $W^{(4)}_{11,5}$, $W^{(4)}_{16,6}$, $B^{(4)}_{14,7}$, $L^{(4)}_{18,7}$, $D^{(4)}_{17,9}$, and $L^{(4)}_{22,9}$. Note that for the standalone gadgets P_5, $D_{16,9}$, $I_{16,9}$, and $T_{19,11}$, the vertex numbering is local to each gadget panel; in $L^{(4)}_{18,7}$ [Fig. 3(d)], $r_{13} = (1,0,1,1)$ and $r_{14} = (2,0,-1,-1)$ are chosen so that $r_{13} \not\perp r_{15}$, whereas in $L^{(4)}_{22,9}$ [Fig. 3(f)], $r_{13} = (1,0,4,4)$ and $r_{14} = (8,0,-1,-1)$ are used so that $r_{13} \perp r_{15}$ in the added context $\{13,20,15,19\}$.

TABLE S2: All 25 unnormalized parent vectors; membership of the two geometric examples.

i	r_i	$H_{17,10}$	$H_{21,13}$	i	r_i	$H_{17,10}$	$H_{21,13}$
1	$(0,1,1)$	yes	yes	14	$(2,1,1)$	yes	yes
2	$(1,0,0)$	yes	yes	15	$(1,2,-1)$	yes	yes
3	$(0,1,-1)$	yes	yes	16	$(1,-2,1)$	–	–
4	$(1,1,0)$	yes	yes	17	$(1,1,-2)$	–	–
5	$(0,0,1)$	yes	yes	18	$(1,-1,-2)$	yes	yes
6	$(1,-1,0)$	yes	yes	19	$(2,1,-1)$	yes	yes
7	$(1,0,1)$	yes	yes	20	$(2,-1,1)$	–	yes
8	$(0,1,0)$	yes	yes	21	$(1,-2,-1)$	–	yes
9	$(1,0,-1)$	yes	yes	22	$(1,1,2)$	–	yes
10	$(1,-1,1)$	yes	yes	23	$(1,2,1)$	yes	yes
11	$(1,-1,-1)$	yes	yes	24	$(1,-1,2)$	yes	yes
12	$(1,1,1)$	–	–	25	$(2,-1,-1)$	–	–
13	$(1,1,-1)$	–	yes				

TABLE S3: The 16 parent contexts and the two selected restrictions.

j	context	$H_{17,10}$	$H_{21,13}$
1	$\{1,2,3\}$	yes	yes
2	$\{7,8,9\}$	yes	yes
3	$\{4,5,6\}$	yes	yes
4	$\{2,8,5\}$	yes	yes
5	$\{3,12,25\}$	–	–
6	$\{9,12,16\}$	–	–
7	$\{6,12,17\}$	–	–
8	$\{3,11,14\}$	yes	yes
9	$\{7,11,15\}$	yes	yes
10	$\{4,11,24\}$	yes	yes
11	$\{9,10,23\}$	yes	yes
12	$\{1,10,19\}$	yes	yes
13	$\{4,10,18\}$	yes	yes
14	$\{1,13,20\}$	–	yes
15	$\{7,13,21\}$	–	yes
16	$\{6,13,22\}$	–	yes

S1.3. 3D panel (a): P_5

TABLE S4: Unnormalized vectors and all contexts of P_5.

i	r_i	j	Context
1	$(1,0,0)$	1	$\{1,6,2\}$
2	$(0,1,1)$	2	$\{2,7,3\}$
3	$(1,1,-1)$	3	$\{3,8,4\}$
4	$(1,-2,-1)$	4	$\{4,9,5\}$
5	$(0,1,-2)$	5	$\{5,10,1\}$
6	$(0,1,-1)$		
7	$(2,-1,1)$		
8	$(1,0,1)$		
9	$(5,2,1)$		
10	$(0,2,1)$		

S1.4. 3D panel (b): $B_{13,7}$

TABLE S5: Unnormalized vectors and all contexts of $B_{13,7}$.

i	r_i	j	Context
1	$(0,1,1)$	1	$\{1,2,3\}$
2	$(1,0,0)$	2	$\{7,8,9\}$
3	$(0,1,-1)$	3	$\{2,8,5\}$
5	$(0,0,1)$	4	$\{3,11,14\}$
7	$(1,0,1)$	5	$\{7,11,15\}$
8	$(0,1,0)$	6	$\{9,10,23\}$
9	$(1,0,-1)$	7	$\{1,10,19\}$
10	$(1,-1,1)$		
11	$(1,-1,-1)$		
14	$(2,1,1)$		
15	$(1,2,-1)$		
19	$(2,1,-1)$		
23	$(1,2,1)$		

S1.5. 3D panel (c): $D_{16,9}$

TABLE S6: Unnormalized vectors and all contexts of $D_{16,9}$.

i	r_i	j	Context
1	$(0,1,2)$	1	$\{1,2,3\}$
2	$(1,0,0)$	2	$\{7,8,9\}$
3	$(0,2,-1)$	3	$\{4,5,6\}$
4	$(1,2,0)$	4	$\{2,8,5\}$
5	$(0,0,1)$	5	$\{3,11,14\}$
6	$(2,-1,0)$	6	$\{7,11,15\}$
7	$(1,0,1)$	7	$\{4,11,24\}$
8	$(0,1,0)$	8	$\{9,10,23\}$
9	$(1,0,-1)$	9	$\{4,10,18\}$
10	$(2,-1,2)$		
11	$(2,-1,-2)$		
14	$(5,2,4)$		
15	$(1,4,-1)$		
18	$(4,-2,-5)$		
23	$(1,4,1)$		
24	$(4,-2,5)$		

S1.6. 3D panel (d): $I_{16,9}$

TABLE S7: Unnormalized vectors and all contexts of $I_{16,9}$.

i	r_i	j	Context
1	$(0,1,1)$	1	$\{1,2,3\}$
2	$(1,0,0)$	2	$\{7,8,9\}$
3	$(0,1,-1)$	3	$\{2,8,5\}$
4	$(4,5,3)$	4	$\{3,11,14\}$
5	$(0,0,1)$	5	$\{7,11,15\}$
7	$(1,0,2)$	6	$\{4,11,24\}$
8	$(0,1,0)$	7	$\{9,10,23\}$
9	$(2,0,-1)$	8	$\{1,10,19\}$
10	$(1,-2,2)$	9	$\{4,10,18\}$
11	$(2,-1,-1)$		
14	$(1,1,1)$		
15	$(2,5,-1)$		
18	$(16,-5,-13)$		
19	$(4,1,-1)$		
23	$(2,5,4)$		
24	$(1,-5,7)$		

S1.7. 3D panel (e): $T_{19,11}$

TABLE S8: Unnormalized vectors and all contexts of $T_{19,11}$.

i	r_i	j	Context
1	$(0, 1, 2)$	1	$\{1, 2, 3\}$
2	$(1, 0, 0)$	2	$\{7, 8, 9\}$
3	$(0, 2, -1)$	3	$\{4, 5, 6\}$
4	$(1, 2, 0)$	4	$\{2, 8, 5\}$
5	$(0, 0, 1)$	5	$\{3, 11, 14\}$
6	$(2, -1, 0)$	6	$\{7, 11, 15\}$
7	$(1, 0, 1)$	7	$\{4, 11, 24\}$
8	$(0, 1, 0)$	8	$\{9, 10, 23\}$
9	$(1, 0, -1)$	9	$\{4, 10, 18\}$
10	$(2, -1, 2)$	10	$\{14, 26, 27\}$
11	$(2, -1, -2)$	11	$\{18, 26, 28\}$
14	$(5, 2, 4)$		
15	$(1, 4, -1)$		
18	$(4, -2, -5)$		
23	$(1, 4, 1)$		
24	$(4, -2, 5)$		
26	$(2, -41, 18)$		
27	$(200, -82, -209)$		
28	$(241, 82, 160)$		

S1.8. 4D panel (a): $W^{(4)}_{11,5}$

TABLE S9: Unnormalized vectors and all contexts of $W^{(4)}_{11,5}$.

i	r_i	j	Context
0	$(0, 0, 0, 1)$	1	$\{0, 1, 6, 2\}$
1	$(1, 0, 0, 0)$	2	$\{0, 2, 7, 3\}$
2	$(0, 1, 1, 0)$	3	$\{0, 3, 8, 4\}$
3	$(1, 1, -1, 0)$	4	$\{0, 4, 9, 5\}$
4	$(1, -2, -1, 0)$	5	$\{0, 5, 10, 1\}$
5	$(0, 1, -2, 0)$		
6	$(0, 1, -1, 0)$		
7	$(2, -1, 1, 0)$		
8	$(1, 0, 1, 0)$		
9	$(5, 2, 1, 0)$		
10	$(0, 2, 1, 0)$		

S1.9. 4D panel (b): $W^{(4)}_{16,6}$

TABLE S10: Unnormalized vectors and all contexts of $W^{(4)}_{16,6}$.

i	r_i	j	Context
0	$(0,0,0,1)$	1	$\{2,0,6,1\}$
1	$(1,0,0,0)$	2	$\{2,7,13,3\}$
2	$(0,1,1,0)$	3	$\{3,0,8,4\}$
3	$(1,1,-1,0)$	4	$\{4,9,14,5\}$
4	$(1,-2,-1,0)$	5	$\{5,10,15,1\}$
5	$(0,1,-2,0)$	6	$\{0,11,12,5\}$
6	$(0,1,-1,0)$		
7	$(2,-1,1,1)$		
8	$(1,0,1,0)$		
9	$(5,2,1,1)$		
10	$(0,4,2,1)$		
11	$(1,2,1,0)$		
12	$(5,-2,-1,0)$		
13	$(2,-1,1,-6)$		
14	$(5,2,1,-30)$		
15	$(0,2,1,-10)$		

S1.10. 4D panel (c): $B^{(4)}_{14,7}$

TABLE S11: Unnormalized vectors and all contexts of $B^{(4)}_{14,7}$.

i	r_i	j	Context
0	$(0,0,0,1)$	1	$\{0,1,2,3\}$
1	$(0,1,1,0)$	2	$\{0,7,8,9\}$
2	$(1,0,0,0)$	3	$\{0,2,8,5\}$
3	$(0,1,-1,0)$	4	$\{0,3,11,14\}$
5	$(0,0,1,0)$	5	$\{0,7,11,15\}$
7	$(1,0,1,0)$	6	$\{0,1,10,19\}$
8	$(0,1,0,0)$	7	$\{0,9,10,23\}$
9	$(1,0,-1,0)$		
10	$(1,-1,1,0)$		
11	$(1,-1,-1,0)$		
14	$(2,1,1,0)$		
15	$(1,2,-1,0)$		
19	$(2,1,-1,0)$		
23	$(1,2,1,0)$		

S1.11. 4D panel (d): $L^{(4)}_{18,7}$

TABLE S12: Unnormalized vectors and all contexts of $L^{(4)}_{18,7}$.

i	r_i	j	Context
1	$(1,0,0,0)$	1	$\{1,2,4,3\}$
2	$(0,1,0,0)$	2	$\{5,8,7,6\}$
3	$(0,0,1,0)$	3	$\{4,9,10,8\}$
4	$(0,0,0,1)$	4	$\{11,1,12,5\}$
5	$(0,1,-1,-1)$	5	$\{11,14,13,2\}$
6	$(4,-5,-11,6)$	6	$\{15,3,16,7\}$
7	$(1,-4,0,-4)$	7	$\{15,17,18,6\}$
8	$(4,1,1,0)$		
9	$(1,-5,1,0)$		
10	$(2,-1,-7,0)$		
11	$(0,0,1,-1)$		
12	$(0,2,1,1)$		
13	$(1,0,1,1)$		
14	$(2,0,-1,-1)$		
15	$(4,2,0,-1)$		
16	$(4,-5,0,6)$		
17	$(5,-9,7,2)$		
18	$(2,113,77,234)$		

S1.12. 4D panel (e): $D^{(4)}_{17,9}$

TABLE S13: Unnormalized vectors and all contexts of $D^{(4)}_{17,9}$.

i	r_i	j	Context
0	$(0,0,0,1)$	1	$\{0,1,2,3\}$
1	$(0,1,2,0)$	2	$\{0,7,8,9\}$
2	$(1,0,0,0)$	3	$\{0,4,5,6\}$
3	$(0,2,-1,0)$	4	$\{0,2,8,5\}$
4	$(1,2,0,0)$	5	$\{0,3,11,14\}$
5	$(0,0,1,0)$	6	$\{0,7,11,15\}$
6	$(2,-1,0,0)$	7	$\{0,9,10,23\}$
7	$(1,0,1,0)$	8	$\{0,4,11,24\}$
8	$(0,1,0,0)$	9	$\{0,4,10,18\}$
9	$(1,0,-1,0)$		
10	$(2,-1,2,0)$		
11	$(2,-1,-2,0)$		
14	$(5,2,4,0)$		
15	$(1,4,-1,0)$		
18	$(4,-2,-5,0)$		
23	$(1,4,1,0)$		
24	$(4,-2,5,0)$		

S1.13. 4D panel (f): $L^{(4)}_{22,9}$

TABLE S14: Unnormalized vectors and all contexts of $L^{(4)}_{22,9}$.

i	r_i	j	Context
1	$(1,0,0,0)$	1	$\{1,2,4,3\}$
2	$(0,1,0,0)$	2	$\{5,8,7,6\}$
3	$(0,0,1,0)$	3	$\{4,9,10,8\}$
4	$(0,0,0,1)$	4	$\{11,1,12,5\}$
5	$(0,1,-1,-1)$	5	$\{11,14,13,2\}$
6	$(4,-5,-11,6)$	6	$\{15,3,16,7\}$
7	$(1,-4,0,-4)$	7	$\{15,17,18,6\}$
8	$(4,1,1,0)$	8	$\{11,21,22,19\}$
9	$(1,-5,1,0)$	9	$\{13,20,15,19\}$
10	$(2,-1,-7,0)$		
11	$(0,0,1,-1)$		
12	$(0,2,1,1)$		
13	$(1,0,4,4)$		
14	$(8,0,-1,-1)$		
15	$(4,2,0,-1)$		
16	$(4,-5,0,6)$		
17	$(5,-9,7,2)$		
18	$(2,113,77,234)$		
19	$(16,-33,-2,-2)$		
20	$(4,2,-21,20)$		
21	$(22,12,-11,-11)$		
22	$(129,44,153,153)$		

S1.14. Hesse H_{37}: vectors, contexts, and non-collinearity

This appendix uses exactly the vertex numbers and context numbers of Fig. 4. All indices a, k in the family notation are in $\mathbb{Z}_3$. The vectors are unnormalized and the inner product is Hermitian. Table S15 lists every vector, squared norm, color, and incident context. Table S16 lists all 24 contexts with their actual vectors. Together they specify the faithful orthogonal representation completely.

S1.14.1. All 37 vectors, their colors, and incident contexts

TABLE S15: Complete enumeration of the 37 vectors of H_{37}. The color $c(i) \in \{0,1,2,3\}$ is the explicit four-coloring in the main text. Context indices refer to Table S16.

i	Ray	v_i	$\|v_i\|^2$	$c(i)$	Incident contexts
1	z	$(1,0,0)$	1	0	$\{4,5,6\}$
2	M_{00}	$(1,1,1)$	3	0	$\{1,7,16\}$
3	M_{01}	$(1,\zeta,\zeta^2)$	3	1	$\{1,8,17\}$
4	M_{02}	$(1,\zeta^2,\zeta)$	3	2	$\{1,9,18\}$
5	M_{10}	$(1,\zeta,\zeta)$	3	0	$\{2,10,21\}$
6	M_{11}	$(1,\zeta^2,1)$	3	1	$\{2,11,19\}$
7	M_{12}	$(1,1,\zeta^2)$	3	2	$\{2,12,20\}$
8	M_{20}	$(1,\zeta^2,\zeta^2)$	3	0	$\{3,13,23\}$
9	M_{21}	$(1,1,\zeta)$	3	1	$\{3,14,24\}$
10	M_{22}	$(1,\zeta,1)$	3	2	$\{3,15,22\}$
11	P_0	$(0,1,-1)$	2	3	$\{4,7,10,13\}$
12	P_1	$(0,1,-\zeta)$	2	3	$\{5,8,11,14\}$
13	P_2	$(0,1,-\zeta^2)$	2	3	$\{6,9,12,15\}$
14	Q_0	$(-1,0,1)$	2	3	$\{16,19,22\}$

Continued on the next page.

TABLE S15: Complete enumeration of the 37 vectors of H_{37}. The color $c(i) \in \{0,1,2,3\}$ is the explicit four-coloring in the main text. Context indices refer to Table S16. (continued).

i	Ray	v_i	$\|v_i\|^2$	$c(i)$	Incident contexts
15	Q_1	$(-\zeta, 0, 1)$	2	3	$\{17, 20, 23\}$
16	Q_2	$(-\zeta^2, 0, 1)$	2	3	$\{18, 21, 24\}$
17	A_0	$(0, 1, 1)$	2	1	$\{4\}$
18	A_1	$(0, 1, \zeta)$	2	1	$\{5\}$
19	A_2	$(0, 1, \zeta^2)$	2	1	$\{6\}$
20	B_{00}	$(-2, 1, 1)$	6	1	$\{7\}$
21	B_{01}	$(-2, \zeta, \zeta^2)$	6	2	$\{8\}$
22	B_{02}	$(-2, \zeta^2, \zeta)$	6	0	$\{9\}$
23	B_{10}	$(-2, \zeta, \zeta)$	6	1	$\{10\}$
24	B_{11}	$(-2, \zeta^2, 1)$	6	2	$\{11\}$
25	B_{12}	$(-2, 1, \zeta^2)$	6	0	$\{12\}$
26	B_{20}	$(-2, \zeta^2, \zeta^2)$	6	1	$\{13\}$
27	B_{21}	$(-2, 1, \zeta)$	6	2	$\{14\}$
28	B_{22}	$(-2, \zeta, 1)$	6	0	$\{15\}$
29	C_{00}	$(1, -2, 1)$	6	1	$\{16\}$
30	C_{01}	$(1, -2\zeta, \zeta^2)$	6	2	$\{17\}$
31	C_{02}	$(1, -2\zeta^2, \zeta)$	6	0	$\{18\}$
32	C_{10}	$(1, -2\zeta^2, 1)$	6	2	$\{19\}$
33	C_{11}	$(1, -2, \zeta^2)$	6	0	$\{20\}$
34	C_{12}	$(1, -2\zeta, \zeta)$	6	1	$\{21\}$
35	C_{20}	$(1, -2\zeta, 1)$	6	0	$\{22\}$
36	C_{21}	$(1, -2\zeta^2, \zeta^2)$	6	1	$\{23\}$
37	C_{22}	$(1, -2, \zeta)$	6	2	$\{24\}$

S1.14.2. All 24 contexts and their orthogonal vectors

TABLE S16: All 24 orthogonal triads, numbered in the same order as the context colors in Fig. 4. Vectors are given in increasing order of their vertex numbers. Each triad contributes three orthogonality edges; the 72 resulting pairs are all the orthogonal pairs.

Context	Family	Vertices	Unnormalized vectors
e_1	$\mathcal{M}_0$	$\{2, 3, 4\}$	$\{(1,1,1),\ (1,\zeta,\zeta^2),\ (1,\zeta^2,\zeta)\}$
e_2	$\mathcal{M}_1$	$\{5, 6, 7\}$	$\{(1,\zeta,\zeta),\ (1,\zeta^2,1),\ (1,1,\zeta^2)\}$
e_3	$\mathcal{M}_2$	$\{8, 9, 10\}$	$\{(1,\zeta^2,\zeta^2),\ (1,1,\zeta),\ (1,\zeta,1)\}$
e_4	$\mathcal{A}_0$	$\{1, 11, 17\}$	$\{(1,0,0),\ (0,1,-1),\ (0,1,1)\}$
e_5	$\mathcal{A}_1$	$\{1, 12, 18\}$	$\{(1,0,0),\ (0,1,-\zeta),\ (0,1,\zeta)\}$
e_6	$\mathcal{A}_2$	$\{1, 13, 19\}$	$\{(1,0,0),\ (0,1,-\zeta^2),\ (0,1,\zeta^2)\}$
e_7	$\mathcal{B}_{00}$	$\{2, 11, 20\}$	$\{(1,1,1),\ (0,1,-1),\ (-2,1,1)\}$
e_8	$\mathcal{B}_{01}$	$\{3, 12, 21\}$	$\{(1,\zeta,\zeta^2),\ (0,1,-\zeta),\ (-2,\zeta,\zeta^2)\}$
e_9	$\mathcal{B}_{02}$	$\{4, 13, 22\}$	$\{(1,\zeta^2,\zeta),\ (0,1,-\zeta^2),\ (-2,\zeta^2,\zeta)\}$
e_{10}	$\mathcal{B}_{10}$	$\{5, 11, 23\}$	$\{(1,\zeta,\zeta),\ (0,1,-1),\ (-2,\zeta,\zeta)\}$
e_{11}	$\mathcal{B}_{11}$	$\{6, 12, 24\}$	$\{(1,\zeta^2,1),\ (0,1,-\zeta),\ (-2,\zeta^2,1)\}$
e_{12}	$\mathcal{B}_{12}$	$\{7, 13, 25\}$	$\{(1,1,\zeta^2),\ (0,1,-\zeta^2),\ (-2,1,\zeta^2)\}$
e_{13}	$\mathcal{B}_{20}$	$\{8, 11, 26\}$	$\{(1,\zeta^2,\zeta^2),\ (0,1,-1),\ (-2,\zeta^2,\zeta^2)\}$
e_{14}	$\mathcal{B}_{21}$	$\{9, 12, 27\}$	$\{(1,1,\zeta),\ (0,1,-\zeta),\ (-2,1,\zeta)\}$
e_{15}	$\mathcal{B}_{22}$	$\{10, 13, 28\}$	$\{(1,\zeta,1),\ (0,1,-\zeta^2),\ (-2,\zeta,1)\}$
e_{16}	$\mathcal{C}_{00}$	$\{2, 14, 29\}$	$\{(1,1,1),\ (-1,0,1),\ (1,-2,1)\}$

Continued on the next page.

TABLE S16: All 24 orthogonal triads, numbered in the same order as the context colors in Fig. 4. Vectors are given in increasing order of their vertex numbers. Each triad contributes three orthogonality edges; the 72 resulting pairs are all the orthogonal pairs. (continued).

Context	Family	Vertices	Unnormalized vectors
e_{17}	$\mathcal{C}_{01}$	$\{3,15,30\}$	$\{(1,\zeta,\zeta^2),\ (-\zeta,0,1),\ (1,-2\zeta,\zeta^2)\}$
e_{18}	$\mathcal{C}_{02}$	$\{4,16,31\}$	$\{(1,\zeta^2,\zeta),\ (-\zeta^2,0,1),\ (1,-2\zeta^2,\zeta)\}$
e_{19}	$\mathcal{C}_{10}$	$\{6,14,32\}$	$\{(1,\zeta^2,1),\ (-1,0,1),\ (1,-2\zeta^2,1)\}$
e_{20}	$\mathcal{C}_{11}$	$\{7,15,33\}$	$\{(1,1,\zeta^2),\ (-\zeta,0,1),\ (1,-2,\zeta^2)\}$
e_{21}	$\mathcal{C}_{12}$	$\{5,16,34\}$	$\{(1,\zeta,\zeta),\ (-\zeta^2,0,1),\ (1,-2\zeta,\zeta)\}$
e_{22}	$\mathcal{C}_{20}$	$\{10,14,35\}$	$\{(1,\zeta,1),\ (-1,0,1),\ (1,-2\zeta,1)\}$
e_{23}	$\mathcal{C}_{21}$	$\{8,15,36\}$	$\{(1,\zeta^2,\zeta^2),\ (-\zeta,0,1),\ (1,-2\zeta^2,\zeta^2)\}$
e_{24}	$\mathcal{C}_{22}$	$\{9,16,37\}$	$\{(1,1,\zeta),\ (-\zeta^2,0,1),\ (1,-2,\zeta)\}$

S1.14.3. Norm shells and a non-collinearity certificate

The four squared-norm shells have sizes

$$\#\{i:\|v_i\|^2=1\}=1,\quad \#\{i:\|v_i\|^2=2\}=9,\quad \#\{i:\|v_i\|^2=3\}=9,\quad \#\{i:\|v_i\|^2=6\}=18. \tag{S1}$$

Exact evaluation over the $\binom{37}{2}=666$ unordered pairs gives

$$\max_{i<j}\frac{|\langle v_i,v_j\rangle|^2}{\|v_i\|^2\|v_j\|^2}=\frac{3}{4}<1. \tag{S2}$$

For example, the maximum is attained by $v_{11}=P_0=(0,1,-1)$ and $v_{29}=C_{00}=(1,-2,1)$:

$$\frac{|\langle v_{11},v_{29}\rangle|^2}{\|v_{11}\|^2\|v_{29}\|^2}=\frac{(-3)^2}{2\cdot 6}=\frac{3}{4}. \tag{S3}$$

Thus no two printed vectors represent the same ray. The maximum and all 72 zero inner products are checked exactly by the accompanying verifier; the family-wise argument in the main text explains why there are no further orthogonalities.

S1.15. Four-dimensional $H_{\mathrm{YO}}^{(4)}$ and H_{65}: vectors and contexts

For direct human and computational verification, this appendix enumerates all unnormalized integer vectors in $\mathbb{Z}^4$ and all four-element contexts for the two four-dimensional examples H_{65} and $H_{\mathrm{YO}}^{(4)}$:

1. The *canonical linear hypergraph* $H_{65}=(V_{65},E_{29})$ (Theorem 22, Appendix S1.15.1), which has exactly 65 **pairwise non-collinear integer vectors** (17 core vectors $v_1,\dots,v_{17}$ and 48 degree-1 completion vectors $v_{18},\dots,v_{65}$), 29 **four-element contexts** (5 core contexts $C_1,\dots,C_5$ and 24 bridge-completion contexts $E_1,\dots,E_{24}$), and 174 **orthogonality edges** ($29\times\binom{4}{2}=174$).

2. The *non-linear warm-up apex suspension* $H_{\mathrm{YO}}^{(4)}=(V_{26},E_{16}^{(4)})$ (Proposition 20, Appendix S1.15.2), which has exactly 26 **pairwise non-collinear integer vectors** (13 Yu–Oh rays $v_1,\dots,v_{13}$, 12 completion rays $v_{14},\dots,v_{25}$ in $\mathbb{R}^3\times\{0\}$, and 1 apex ray v_{26}), 16 **four-element contexts**, and 73 **orthogonality edges** ($16\times\binom{3}{2}=48$ edges in $\mathbb{R}^3\times\{0\}$ plus 25 apex edges).

Canonical primitive normal form and human-verifiable non-collinearity. Every vector $v_i=(x_{i,1},x_{i,2},x_{i,3},x_{i,4})^T\in\mathbb{Z}^4$ listed below in both H_{65} and $H_{\mathrm{YO}}^{(4)}$ is written in *canonical primitive projective form*:

$$\begin{gathered}\gcd\big(|x_{i,1}|,|x_{i,2}|,|x_{i,3}|,|x_{i,4}|\big)=1,\\ \text{first nonzero entry of } v_i>0.\end{gathered} \tag{S4}$$

Because every projective ray $[v]\in\mathbb{P}(\mathbb{Q}^4)$ possesses a *unique* representative in $\mathbb{Z}^4$ satisfying Eq. (S4), two vectors v_i,v_j in canonical primitive form are collinear (proportional over $\mathbb{R}$ or $\mathbb{C}$) *if and only if they are identical* 4*-tuples* ($v_i=v_j$) and, in particular, have the same squared Euclidean norm $\|v_i\|^2=\|v_j\|^2$.

S1.15.1. The canonical linear 65-vertex, 29-context hypergraph H_{65}

Table S17 enumerates all 65 unnormalized integer vectors $v_1, \ldots, v_{65} \in \mathbb{Z}^4$ of H_{65}, together with their squared norms $\|v_i\|^2$, their colors $c(i) \in \{0, 1, 2, 3, 4\}$ in the strong 5-coloring extending Eq. (S28), and the contexts in E_{29} to which each vertex belongs. Table S18 enumerates all 29 contexts $C_1, \ldots, C_5, E_1, \ldots, E_{24} \in E_{29}$, displaying side by side both the vertex indices $\{i, j, k, \ell\}$ and the four mutually orthogonal integer vectors $\{v_i, v_j, v_k, v_\ell\} \subset \mathbb{Z}^4$.

To verify by hand that all 65 vectors in Table S17 are pairwise non-collinear across all $\binom{65}{2} = 2{,}080$ pairs, observe that all 65 tuples satisfy Eq. (S4) and partition by squared norm $\|v_i\|^2$ into seven small, disjoint norm shells ($4 + 6 + 7 + 12 + 12 + 12 + 12 = 65$):

- $\|v_i\|^2 = 1$ (4 vectors): indices $\{1, 2, 3, 4\}$, the 4 standard coordinate basis vectors;
- $\|v_i\|^2 = 2$ (6 vectors): indices $\{12, \ldots, 17\}$, one vector for each of the $\binom{4}{2} = 6$ coordinate-pair supports;
- $\|v_i\|^2 = 4$ (7 vectors): indices $\{5, \ldots, 11\}$, seven distinct ± 1 vectors with $x_{i,1} = +1$;
- $\|v_i\|^2 = 11$ (12 vectors): indices $\{18, 20, 22, 25, 26, 28, 30, 33, 34, 36, 39, 41\}$, each having a single zero coordinate (three each with $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$) and distinct sign/entry patterns of $\{0, \pm 1, \pm 1, \pm 3\}$;
- $\|v_i\|^2 = 18$ (12 vectors): indices $\{43, 45, 46, 49, 51, 53, 54, 56, 59, 61, 62, 65\}$, twelve distinct permutations and sign patterns of $\{\pm 1, \pm 2, \pm 2, \pm 3\}$;
- $\|v_i\|^2 = 22$ (12 vectors): indices $\{19, 21, 23, 24, 27, 29, 31, 32, 35, 37, 38, 40\}$, each having a single zero coordinate (three each with $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$) and distinct patterns of $\{0, \pm 2, \pm 3, \pm 3\}$;
- $\|v_i\|^2 = 36$ (12 vectors): indices $\{42, 44, 47, 48, 50, 52, 55, 57, 58, 60, 63, 64\}$, twelve distinct permutations and sign patterns of $\{\pm 1, \pm 1, \pm 3, \pm 5\}$.

Because no two 4-tuples within any of the seven norm shells coincide, no two of the 65 vectors are collinear; quantitatively, the maximum squared cosine over all $\binom{65}{2} = 2{,}080$ pairs is $\max_{1 \le i < j \le 65} (v_i \cdot v_j)^2 / (\|v_i\|^2 \|v_j\|^2) = 19^2/(11 \times 36) = 361/396 \approx 0.9116 < 1$, attained by (v_{41}, v_{47}), whose squared norms are 11 and 36 and whose inner product is 19.

TABLE S17: Complete enumeration of the 65 unnormalized integer vectors $v_i \in \mathbb{Z}^4$ of H_{65}, their squared norms $\|v_i\|^2$, their colors $c(i) \in \{0, 1, 2, 3, 4\}$, and the contexts in E_{29} containing vertex i.

i	$v_i \in \mathbb{Z}^4$	$\|v_i\|^2$	$c(i)$	Contexts containing i	i	$v_i \in \mathbb{Z}^4$	$\|v_i\|^2$	$c(i)$	Contexts containing i
1	$(1,0,0,0)$	1	0	C_1, E_3, E_7, E_{11}	34	$(1,-1,0,3)$	11	1	$E_9 = \{3, 16, 34, 35\}$
2	$(0,1,0,0)$	1	1	C_1, E_1, E_5, E_{12}	35	$(3,-3,0,-2)$	22	3	$E_9 = \{3, 16, 34, 35\}$
3	$(0,0,1,0)$	1	2	C_1, E_2, E_8, E_9	36	$(1,-1,3,0)$	11	1	$E_{10} = \{4, 16, 36, 37\}$
4	$(0,0,0,1)$	1	3	C_1, E_4, E_6, E_{10}	37	$(3,-3,-2,0)$	22	2	$E_{10} = \{4, 16, 36, 37\}$
5	$(1,1,1,1)$	4	0	$C_2, E_{13}, E_{19}, E_{23}$	38	$(0,2,3,3)$	22	1	$E_{11} = \{1, 17, 38, 39\}$
6	$(1,1,-1,-1)$	4	1	$C_2, E_{15}, E_{17}, E_{24}$	39	$(0,3,-1,-1)$	11	3	$E_{11} = \{1, 17, 38, 39\}$
7	$(1,-1,1,-1)$	4	2	$C_2, E_{16}, E_{20}, E_{21}$	40	$(2,0,-3,-3)$	22	0	$E_{12} = \{2, 17, 40, 41\}$
8	$(1,-1,-1,1)$	4	4	$C_2, E_{14}, E_{18}, E_{22}$	41	$(3,0,1,1)$	11	3	$E_{12} = \{2, 17, 40, 41\}$
9	$(1,1,-1,1)$	4	0	C_3, C_4	42	$(1,-5,3,1)$	36	1	$E_{13} = \{5, 12, 42, 43\}$
10	$(1,-1,1,1)$	4	1	C_3, C_5	43	$(2,-1,-3,2)$	18	2	$E_{13} = \{5, 12, 42, 43\}$
11	$(1,-1,-1,-1)$	4	3	C_4, C_5	44	$(1,-3,5,1)$	36	0	$E_{14} = \{8, 12, 44, 45\}$
12	$(1,0,0,-1)$	2	3	$C_3, E_1, E_2, E_{13}, E_{14}$	45	$(2,3,1,2)$	18	1	$E_{14} = \{8, 12, 44, 45\}$
13	$(0,1,1,0)$	2	4	$C_3, E_3, E_4, E_{15}, E_{16}$	46	$(1,-2,2,-3)$	18	0	$E_{15} = \{6, 13, 46, 47\}$
14	$(1,0,1,0)$	2	2	$C_4, E_5, E_6, E_{17}, E_{18}$	47	$(5,-1,1,3)$	36	2	$E_{15} = \{6, 13, 46, 47\}$
15	$(0,1,0,-1)$	2	1	$C_4, E_7, E_8, E_{19}, E_{20}$	48	$(3,-1,1,5)$	36	0	$E_{16} = \{7, 13, 48, 49\}$
16	$(1,1,0,0)$	2	0	$C_5, E_9, E_{10}, E_{21}, E_{22}$	49	$(3,2,-2,-1)$	18	1	$E_{16} = \{7, 13, 48, 49\}$
17	$(0,0,1,-1)$	2	2	$C_5, E_{11}, E_{12}, E_{23}, E_{24}$	50	$(1,3,-1,5)$	36	0	$E_{17} = \{6, 14, 50, 51\}$
18	$(1,0,-3,1)$	11	0	$E_1 = \{2, 12, 18, 19\}$	51	$(2,-3,-2,1)$	18	3	$E_{17} = \{6, 14, 50, 51\}$
19	$(3,0,2,3)$	22	2	$E_1 = \{2, 12, 18, 19\}$	52	$(1,5,-1,3)$	36	0	$E_{18} = \{8, 14, 52, 53\}$
20	$(1,3,0,1)$	11	0	$E_2 = \{3, 12, 20, 21\}$	53	$(2,1,-2,-3)$	18	1	$E_{18} = \{8, 14, 52, 53\}$
21	$(3,-2,0,3)$	22	1	$E_2 = \{3, 12, 20, 21\}$	54	$(3,-2,1,-2)$	18	2	$E_{19} = \{5, 15, 54, 55\}$
22	$(0,1,-1,3)$	11	1	$E_3 = \{1, 13, 22, 23\}$	55	$(3,1,-5,1)$	36	3	$E_{19} = \{5, 15, 54, 55\}$
23	$(0,3,-3,-2)$	22	2	$E_3 = \{1, 13, 22, 23\}$	56	$(1,2,3,2)$	18	0	$E_{20} = \{7, 15, 56, 57\}$
24	$(2,-3,3,0)$	22	0	$E_4 = \{4, 13, 24, 25\}$	57	$(5,1,-3,1)$	36	3	$E_{20} = \{7, 15, 56, 57\}$

Continued on the next page.

TABLE S17: All 65 vectors of H_{65} (continued).

i	$v_i \in \mathbb{Z}^4$	$\|v_i\|^2$	$c(i)$	Contexts containing i	i	$v_i \in \mathbb{Z}^4$	$\|v_i\|^2$	$c(i)$	Contexts containing i
25	$(3,1,-1,0)$	11	1	$E_4 = \{4,13,24,25\}$	58	$(1,-1,-5,-3)$	36	1	$E_{21} = \{7,16,58,59\}$
26	$(1,0,-1,-3)$	11	0	$E_5 = \{2,14,26,27\}$	59	$(2,-2,-1,3)$	18	3	$E_{21} = \{7,16,58,59\}$
27	$(3,0,-3,2)$	22	3	$E_5 = \{2,14,26,27\}$	60	$(1,-1,-3,-5)$	36	1	$E_{22} = \{8,16,60,61\}$
28	$(1,-3,-1,0)$	11	0	$E_6 = \{4,14,28,29\}$	61	$(2,-2,3,-1)$	18	2	$E_{22} = \{8,16,60,61\}$
29	$(3,2,-3,0)$	22	1	$E_6 = \{4,14,28,29\}$	62	$(1,3,-2,-2)$	18	1	$E_{23} = \{5,17,62,63\}$
30	$(0,1,3,1)$	11	2	$E_7 = \{1,15,30,31\}$	63	$(5,-3,-1,-1)$	36	3	$E_{23} = \{5,17,62,63\}$
31	$(0,3,-2,3)$	22	3	$E_7 = \{1,15,30,31\}$	64	$(3,-5,-1,-1)$	36	0	$E_{24} = \{6,17,64,65\}$
32	$(2,3,0,3)$	22	0	$E_8 = \{3,15,32,33\}$	65	$(3,1,2,2)$	18	3	$E_{24} = \{6,17,64,65\}$
33	$(3,-1,0,-1)$	11	3	$E_8 = \{3,15,32,33\}$					

TABLE S18: Complete enumeration of the 29 four-element contexts of $H_{65} = (V_{65}, E_{29})$, listing for each context both its vertex indices $\{i,j,k,\ell\}$ and the four mutually orthogonal unnormalized integer vectors $\{v_i, v_j, v_k, v_\ell\} \subset \mathbb{Z}^4$.

Context	Vertex set	v_i	v_j	v_k	v_ℓ
Five core 4*-contexts* $C_1, \ldots, C_5$ *on* $V_{\text{core}} = \{1, \ldots, 17\}$*:*					
C_1	$\{1,2,3,4\}$	$v_1 = (1,0,0,0)$	$v_2 = (0,1,0,0)$	$v_3 = (0,0,1,0)$	$v_4 = (0,0,0,1)$
C_2	$\{5,6,7,8\}$	$v_5 = (1,1,1,1)$	$v_6 = (1,1,-1,-1)$	$v_7 = (1,-1,1,-1)$	$v_8 = (1,-1,-1,1)$
C_3	$\{9,10,12,13\}$	$v_9 = (1,1,-1,1)$	$v_{10} = (1,-1,1,1)$	$v_{12} = (1,0,0,-1)$	$v_{13} = (0,1,1,0)$
C_4	$\{9,11,14,15\}$	$v_9 = (1,1,-1,1)$	$v_{11} = (1,-1,-1,-1)$	$v_{14} = (1,0,1,0)$	$v_{15} = (0,1,0,-1)$
C_5	$\{10,11,16,17\}$	$v_{10} = (1,-1,1,1)$	$v_{11} = (1,-1,-1,-1)$	$v_{16} = (1,1,0,0)$	$v_{17} = (0,0,1,-1)$
Twelve coordinate-to-bridge completion contexts $E_1, \ldots, E_{12}$ *(between* C_1 *and* $B_1 \cup B_2 \cup B_3$*):*					
E_1	$\{2,12,18,19\}$	$v_2 = (0,1,0,0)$	$v_{12} = (1,0,0,-1)$	$v_{18} = (1,0,-3,1)$	$v_{19} = (3,0,2,3)$
E_2	$\{3,12,20,21\}$	$v_3 = (0,0,1,0)$	$v_{12} = (1,0,0,-1)$	$v_{20} = (1,3,0,1)$	$v_{21} = (3,-2,0,3)$
E_3	$\{1,13,22,23\}$	$v_1 = (1,0,0,0)$	$v_{13} = (0,1,1,0)$	$v_{22} = (0,1,-1,3)$	$v_{23} = (0,3,-3,-2)$
E_4	$\{4,13,24,25\}$	$v_4 = (0,0,0,1)$	$v_{13} = (0,1,1,0)$	$v_{24} = (2,-3,3,0)$	$v_{25} = (3,1,-1,0)$
E_5	$\{2,14,26,27\}$	$v_2 = (0,1,0,0)$	$v_{14} = (1,0,1,0)$	$v_{26} = (1,0,-1,-3)$	$v_{27} = (3,0,-3,2)$
E_6	$\{4,14,28,29\}$	$v_4 = (0,0,0,1)$	$v_{14} = (1,0,1,0)$	$v_{28} = (1,-3,-1,0)$	$v_{29} = (3,2,-3,0)$
E_7	$\{1,15,30,31\}$	$v_1 = (1,0,0,0)$	$v_{15} = (0,1,0,-1)$	$v_{30} = (0,1,3,1)$	$v_{31} = (0,3,-2,3)$
E_8	$\{3,15,32,33\}$	$v_3 = (0,0,1,0)$	$v_{15} = (0,1,0,-1)$	$v_{32} = (2,3,0,3)$	$v_{33} = (3,-1,0,-1)$
E_9	$\{3,16,34,35\}$	$v_3 = (0,0,1,0)$	$v_{16} = (1,1,0,0)$	$v_{34} = (1,-1,0,3)$	$v_{35} = (3,-3,0,-2)$
E_{10}	$\{4,16,36,37\}$	$v_4 = (0,0,0,1)$	$v_{16} = (1,1,0,0)$	$v_{36} = (1,-1,3,0)$	$v_{37} = (3,-3,-2,0)$
E_{11}	$\{1,17,38,39\}$	$v_1 = (1,0,0,0)$	$v_{17} = (0,0,1,-1)$	$v_{38} = (0,2,3,3)$	$v_{39} = (0,3,-1,-1)$
E_{12}	$\{2,17,40,41\}$	$v_2 = (0,1,0,0)$	$v_{17} = (0,0,1,-1)$	$v_{40} = (2,0,-3,-3)$	$v_{41} = (3,0,1,1)$
Twelve Hadamard-to-bridge completion contexts $E_{13}, \ldots, E_{24}$ *(between* C_2 *and* $B_1 \cup B_2 \cup B_3$*):*					
E_{13}	$\{5,12,42,43\}$	$v_5 = (1,1,1,1)$	$v_{12} = (1,0,0,-1)$	$v_{42} = (1,-5,3,1)$	$v_{43} = (2,-1,-3,2)$
E_{14}	$\{8,12,44,45\}$	$v_8 = (1,-1,-1,1)$	$v_{12} = (1,0,0,-1)$	$v_{44} = (1,-3,5,1)$	$v_{45} = (2,3,1,2)$
E_{15}	$\{6,13,46,47\}$	$v_6 = (1,1,-1,-1)$	$v_{13} = (0,1,1,0)$	$v_{46} = (1,-2,2,-3)$	$v_{47} = (5,-1,1,3)$
E_{16}	$\{7,13,48,49\}$	$v_7 = (1,-1,1,-1)$	$v_{13} = (0,1,1,0)$	$v_{48} = (3,-1,1,5)$	$v_{49} = (3,2,-2,-1)$
E_{17}	$\{6,14,50,51\}$	$v_6 = (1,1,-1,-1)$	$v_{14} = (1,0,1,0)$	$v_{50} = (1,3,-1,5)$	$v_{51} = (2,-3,-2,1)$
E_{18}	$\{8,14,52,53\}$	$v_8 = (1,-1,-1,1)$	$v_{14} = (1,0,1,0)$	$v_{52} = (1,5,-1,3)$	$v_{53} = (2,1,-2,-3)$
E_{19}	$\{5,15,54,55\}$	$v_5 = (1,1,1,1)$	$v_{15} = (0,1,0,-1)$	$v_{54} = (3,-2,1,-2)$	$v_{55} = (3,1,-5,1)$
E_{20}	$\{7,15,56,57\}$	$v_7 = (1,-1,1,-1)$	$v_{15} = (0,1,0,-1)$	$v_{56} = (1,2,3,2)$	$v_{57} = (5,1,-3,1)$
E_{21}	$\{7,16,58,59\}$	$v_7 = (1,-1,1,-1)$	$v_{16} = (1,1,0,0)$	$v_{58} = (1,-1,-5,-3)$	$v_{59} = (2,-2,-1,3)$
E_{22}	$\{8,16,60,61\}$	$v_8 = (1,-1,-1,1)$	$v_{16} = (1,1,0,0)$	$v_{60} = (1,-1,-3,-5)$	$v_{61} = (2,-2,3,-1)$
E_{23}	$\{5,17,62,63\}$	$v_5 = (1,1,1,1)$	$v_{17} = (0,0,1,-1)$	$v_{62} = (1,3,-2,-2)$	$v_{63} = (5,-3,-1,-1)$
E_{24}	$\{6,17,64,65\}$	$v_6 = (1,1,-1,-1)$	$v_{17} = (0,0,1,-1)$	$v_{64} = (3,-5,-1,-1)$	$v_{65} = (3,1,2,2)$

S1.15.2. The non-linear warm-up 26-vertex, 16-context apex suspension $H^{(4)}_{\mathrm{YO}}$

For completeness, the 26 unnormalized integer vectors $v_1, \dots, v_{26} \in \mathbb{Z}^4$ of the non-linear warm-up hypergraph $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$ in Proposition 20 (written in canonical primitive form satisfying Eq. (S4)) are:

$$\begin{aligned}
v_1 &= (0,1,1,0), & v_2 &= (1,0,0,0),\\
v_3 &= (0,1,-1,0), & v_4 &= (1,1,0,0),\\
v_5 &= (0,0,1,0), & v_6 &= (1,-1,0,0),\\
v_7 &= (1,0,1,0), & v_8 &= (0,1,0,0),\\
v_9 &= (1,0,-1,0), & v_{10} &= (1,-1,1,0),\\
v_{11} &= (1,-1,-1,0), & v_{12} &= (1,1,1,0),\\
v_{13} &= (1,1,-1,0), & v_{14} &= (2,1,1,0),\\
v_{15} &= (1,2,-1,0), & v_{16} &= (1,-2,1,0),\\
v_{17} &= (1,1,-2,0), & v_{18} &= (1,-1,-2,0),\\
v_{19} &= (2,1,-1,0), & v_{20} &= (2,-1,1,0),\\
v_{21} &= (1,-2,-1,0), & v_{22} &= (1,1,2,0),\\
v_{23} &= (1,2,1,0), & v_{24} &= (1,-1,2,0),\\
v_{25} &= (2,-1,-1,0), & v_{26} &= (0,0,0,1).
\end{aligned} \tag{S5}$$

These 26 canonical primitive vectors partition by squared norm $\|v_i\|^2$ into four disjoint norm shells ($4+6+4+12 = 26$):

- $\|v_i\|^2 = 1$ (4 vectors): $\{v_2, v_5, v_8, v_{26}\}$, the four coordinate basis vectors of $\mathbb{R}^4$;
- $\|v_i\|^2 = 2$ (6 vectors): $\{v_1, v_3, v_4, v_6, v_7, v_9\}$, the six binary rays in $\mathbb{R}^3 \times \{0\}$;
- $\|v_i\|^2 = 3$ (4 vectors): $\{v_{10}, v_{11}, v_{12}, v_{13}\}$, the four triad rays $(1, \pm 1, \pm 1, 0)^T$;
- $\|v_i\|^2 = 6$ (12 vectors): $\{v_{14}, \dots, v_{25}\}$, all $3 \times 4 = 12$ canonical primitive permutations and sign patterns of $(\pm 2, \pm 1, \pm 1, 0)^T$.

Since all 26 canonical 4-tuples are distinct, all $\binom{26}{2} = 325$ pairs of vectors are strictly non-collinear, with maximum squared cosine $\max_{1 \le i < j \le 26} (v_i \cdot v_j)^2 / (\|v_i\|^2 \|v_j\|^2) = 4^2/(3 \times 6) = 8/9 \approx 0.8889 < 1$. Table S19 lists all 16 four-element contexts of $E^{(4)}_{16}$ side by side with their four mutually orthogonal integer vectors.

TABLE S19: Complete enumeration of the 16 four-element contexts of the non-linear warm-up apex suspension $H^{(4)}_{\mathrm{YO}} = (V_{26}, E^{(4)}_{16})$, listing for each context both its vertex indices $\{i, j, k, 26\}$ and the four mutually orthogonal unnormalized integer vectors $\{v_i, v_j, v_k, v_{26}\} \subset \mathbb{Z}^4$.

Context	Vertex set	v_i	v_j	v_k	v_{26}
$e^{(4)}_1$	$\{1,2,3,26\}$	$v_1 = (0,1,1,0)$	$v_2 = (1,0,0,0)$	$v_3 = (0,1,-1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_2$	$\{7,8,9,26\}$	$v_7 = (1,0,1,0)$	$v_8 = (0,1,0,0)$	$v_9 = (1,0,-1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_3$	$\{4,5,6,26\}$	$v_4 = (1,1,0,0)$	$v_5 = (0,0,1,0)$	$v_6 = (1,-1,0,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_4$	$\{2,5,8,26\}$	$v_2 = (1,0,0,0)$	$v_5 = (0,0,1,0)$	$v_8 = (0,1,0,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_5$	$\{3,12,25,26\}$	$v_3 = (0,1,-1,0)$	$v_{12} = (1,1,1,0)$	$v_{25} = (2,-1,-1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_6$	$\{9,12,16,26\}$	$v_9 = (1,0,-1,0)$	$v_{12} = (1,1,1,0)$	$v_{16} = (1,-2,1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_7$	$\{6,12,17,26\}$	$v_6 = (1,-1,0,0)$	$v_{12} = (1,1,1,0)$	$v_{17} = (1,1,-2,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_8$	$\{3,11,14,26\}$	$v_3 = (0,1,-1,0)$	$v_{11} = (1,-1,-1,0)$	$v_{14} = (2,1,1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_9$	$\{7,11,15,26\}$	$v_7 = (1,0,1,0)$	$v_{11} = (1,-1,-1,0)$	$v_{15} = (1,2,-1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{10}$	$\{4,11,24,26\}$	$v_4 = (1,1,0,0)$	$v_{11} = (1,-1,-1,0)$	$v_{24} = (1,-1,2,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{11}$	$\{9,10,23,26\}$	$v_9 = (1,0,-1,0)$	$v_{10} = (1,-1,1,0)$	$v_{23} = (1,2,1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{12}$	$\{1,10,19,26\}$	$v_1 = (0,1,1,0)$	$v_{10} = (1,-1,1,0)$	$v_{19} = (2,1,-1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{13}$	$\{4,10,18,26\}$	$v_4 = (1,1,0,0)$	$v_{10} = (1,-1,1,0)$	$v_{18} = (1,-1,-2,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{14}$	$\{1,13,20,26\}$	$v_1 = (0,1,1,0)$	$v_{13} = (1,1,-1,0)$	$v_{20} = (2,-1,1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{15}$	$\{7,13,21,26\}$	$v_7 = (1,0,1,0)$	$v_{13} = (1,1,-1,0)$	$v_{21} = (1,-2,-1,0)$	$v_{26} = (0,0,0,1)$
$e^{(4)}_{16}$	$\{6,13,22,26\}$	$v_6 = (1,-1,0,0)$	$v_{13} = (1,1,-1,0)$	$v_{22} = (1,1,2,0)$	$v_{26} = (0,0,0,1)$

S1.16. Explicit parent configurations and inclusion maps

S1.16.1. The 49-ray real Yu–Oh parent and published Bub labels

Table S20 lists all rays, including the original 25. The final column gives the exact labels in the published full 49-ray Bub encoding [1, p. 14, Sec. 5(xi)]. Under this map the 36 bases in Table S21 equal all 36 published contexts; their 108 pairs equal all geometric orthogonalities. In this labeling 26–33 are Eq. (S31), and 34–49 complete the sixteen remaining orthogonal pairs. The first 25 vectors are those of Table S1.

TABLE S20: All unnormalized vectors of the real 49-ray parent, with the verified published Bub labels.

i	v_i	Published label
1	$(0,1,1)$	G
2	$(1,0,0)$	1
3	$(0,1,-1)$	P
4	$(1,1,0)$	H
5	$(0,0,1)$	3
6	$(1,-1,0)$	U
7	$(1,0,1)$	Q
8	$(0,1,0)$	2
9	$(1,0,-1)$	e
10	$(1,-1,1)$	D
11	$(1,-1,-1)$	h
12	$(1,1,1)$	S
13	$(1,1,-1)$	a
14	$(2,1,1)$	g
15	$(1,2,-1)$	n
16	$(1,-2,1)$	d
17	$(1,1,-2)$	T
18	$(1,-1,-2)$	F
19	$(2,1,-1)$	E
20	$(2,-1,1)$	I
21	$(1,-2,-1)$	k
22	$(1,1,2)$	b
23	$(1,2,1)$	A
24	$(1,-1,2)$	N
25	$(2,-1,-1)$	R
26	$(0,1,2)$	i
27	$(0,2,-1)$	O
28	$(0,1,-2)$	6
29	$(0,2,1)$	7
30	$(1,2,0)$	5
31	$(2,-1,0)$	4
32	$(1,-2,0)$	V
33	$(2,1,0)$	j
34	$(2,1,-5)$	Z
35	$(5,-2,1)$	m
36	$(1,2,5)$	W
37	$(5,2,-1)$	c
38	$(1,-2,-5)$	f
39	$(5,-1,2)$	L
40	$(5,1,2)$	9
41	$(2,1,5)$	8
42	$(2,-1,-5)$	J
43	$(5,2,1)$	X
44	$(1,-2,5)$	l
45	$(5,-1,-2)$	Y
46	$(5,-2,-1)$	B
47	$(1,2,-5)$	C
48	$(5,1,-2)$	M
49	$(2,-1,5)$	K

TABLE S21: All 36 complete bases of the real parent.

e	Context	e	Context	e	Context
1	$\{1,2,3\}$	2	$\{1,10,19\}$	3	$\{1,13,20\}$
4	$\{2,5,8\}$	5	$\{2,26,27\}$	6	$\{2,28,29\}$
7	$\{3,11,14\}$	8	$\{3,12,25\}$	9	$\{4,5,6\}$
10	$\{4,10,18\}$	11	$\{4,11,24\}$	12	$\{5,30,31\}$
13	$\{5,32,33\}$	14	$\{6,12,17\}$	15	$\{6,13,22\}$
16	$\{7,8,9\}$	17	$\{7,11,15\}$	18	$\{7,13,21\}$
19	$\{9,10,23\}$	20	$\{9,12,16\}$	21	$\{14,32,34\}$
22	$\{15,26,35\}$	23	$\{15,31,36\}$	24	$\{16,26,37\}$
25	$\{16,33,38\}$	26	$\{17,29,39\}$	27	$\{18,27,40\}$
28	$\{19,32,41\}$	29	$\{20,30,42\}$	30	$\{21,28,43\}$
31	$\{21,33,44\}$	32	$\{22,27,45\}$	33	$\{23,28,46\}$
34	$\{23,31,47\}$	35	$\{24,29,48\}$	36	$\{25,30,49\}$

S1.16.2. The 87-ray Hesse parent

Use $\zeta = \exp(2\pi i/3)$ and $\zeta^2+\zeta+1=0$. The first 37 rays retain exactly the Hesse numbering; projective representatives have their first nonzero coordinate one. The inventory labels use the generation convention of Sec. VII B. The six additional orthogonal pairs outside the 64 complete bases are listed separately: this larger parent's full orthogonality graph need not equal its context two-section. The retained Hesse restriction has all its original orthogonalities covered by contexts.

TABLE S22: All 87 unnormalized rays of the first Hesse/lineage union.

i	v_i	Inventory label
1	$(1,0,0)$	a_{11}
2	$(1,1,1)$	u_1
3	$(1,\zeta,-1-\zeta)$	u_2
4	$(1,-1-\zeta,\zeta)$	u_3
5	$(1,\zeta,\zeta)$	u_4
6	$(1,-1-\zeta,1)$	u_5
7	$(1,1,-1-\zeta)$	u_6
8	$(1,-1-\zeta,-1-\zeta)$	u_7
9	$(1,1,\zeta)$	u_9
10	$(1,\zeta,1)$	u_8
11	$(0,1,-1)$	c_{11}
12	$(0,1,-\zeta)$	c_{12}
13	$(0,1,1+\zeta)$	c_{13}
14	$(1,0,-1)$	c_{21}
15	$(1,0,1+\zeta)$	c_{22}
16	$(1,0,-\zeta)$	c_{23}
17	$(0,1,1)$	b_{11}
18	$(0,1,\zeta)$	b_{12}
19	$(0,1,-1-\zeta)$	b_{13}
20	$(1,\frac{-1}{2},\frac{-1}{2})$	d_{11}
21	$(1,-\frac{1}{2}\zeta,\frac{1}{2}+\frac{1}{2}\zeta)$	d_{12}
22	$(1,\frac{1}{2}+\frac{1}{2}\zeta,-\frac{1}{2}\zeta)$	d_{13}
23	$(1,-\frac{1}{2}\zeta,-\frac{1}{2}\zeta)$	d_{14}
24	$(1,\frac{1}{2}+\frac{1}{2}\zeta,\frac{-1}{2})$	d_{15}
25	$(1,\frac{-1}{2},\frac{1}{2}+\frac{1}{2}\zeta)$	d_{16}
26	$(1,\frac{1}{2}+\frac{1}{2}\zeta,\frac{1}{2}+\frac{1}{2}\zeta)$	d_{17}
27	$(1,\frac{-1}{2},-\frac{1}{2}\zeta)$	d_{19}
28	$(1,-\frac{1}{2}\zeta,\frac{-1}{2})$	d_{18}
29	$(1,-2,1)$	d_{21}
30	$(1,-2\zeta,-1-\zeta)$	d_{22}
31	$(1,2+2\zeta,\zeta)$	d_{23}
32	$(1,2+2\zeta,1)$	d_{25}
33	$(1,-2,-1-\zeta)$	d_{26}

i	v_i	Inventory label
34	$(1, -2\zeta, \zeta)$	d_{24}
35	$(1, -2\zeta, 1)$	d_{28}
36	$(1, 2+2\zeta, -1-\zeta)$	d_{27}
37	$(1, -2, \zeta)$	d_{29}
38	$(0, 1, 0)$	a_{21}
39	$(0, 0, 1)$	a_{31}
40	$(1, \frac{-1}{2}, \frac{1}{2})$	b_{1121}
41	$(1, -\frac{1}{2}\zeta, \frac{-1}{2} - \frac{1}{2}\zeta)$	b_{1122}
42	$(1, \frac{1}{2} + \frac{1}{2}\zeta, \frac{1}{2}\zeta)$	b_{1123}
43	$(1, \frac{1}{2}, \frac{-1}{2})$	b_{1131}
44	$(1, \frac{1}{2}\zeta, \frac{1}{2} + \frac{1}{2}\zeta)$	b_{1132}
45	$(1, \frac{-1}{2} - \frac{1}{2}\zeta, -\frac{1}{2}\zeta)$	b_{1133}
46	$(1, 1, -1)$	b_{121}
47	$(1, \zeta, 1+\zeta)$	b_{122}
48	$(1, -1-\zeta, -\zeta)$	b_{123}
49	$(1, -1, 1)$	b_{131}
50	$(1, -\zeta, -1-\zeta)$	b_{132}
51	$(1, 1+\zeta, \zeta)$	b_{133}
52	$(1, 0, 1)$	b_{21}
53	$(1, -2, -1)$	b_{2121}
54	$(1, -2\zeta, 1+\zeta)$	b_{2122}
55	$(1, 2+2\zeta, -\zeta)$	b_{2123}
56	$(1, 0, -1-\zeta)$	b_{22}
57	$(1, 2, -1)$	b_{2231}
58	$(1, 2\zeta, 1+\zeta)$	b_{2232}
59	$(1, -2-2\zeta, -\zeta)$	b_{2233}
60	$(1, 0, \zeta)$	b_{23}
61	$(1, -1, -1)$	b_{231}
62	$(1, -\zeta, 1+\zeta)$	b_{232}
63	$(1, 1+\zeta, -\zeta)$	b_{233}
64	$(1, 1, 0)$	b_{31}
65	$(1, -1, -2)$	b_{3131}
66	$(1, -\zeta, 2+2\zeta)$	b_{3132}
67	$(1, 1+\zeta, -2\zeta)$	b_{3133}
68	$(1, \zeta, 0)$	b_{32}
69	$(1, -1, 2)$	b_{3231}
70	$(1, -\zeta, -2-2\zeta)$	b_{3232}
71	$(1, 1+\zeta, 2\zeta)$	b_{3233}
72	$(1, -1-\zeta, 0)$	b_{33}
73	$(1, -1, 0)$	c_{31}
74	$(1, -\zeta, 0)$	c_{32}
75	$(1, 1+\zeta, 0)$	c_{33}
76	$(1, 1, -2)$	d_{31}
77	$(1, \zeta, 2+2\zeta)$	d_{32}
78	$(1, -1-\zeta, -2\zeta)$	d_{33}
79	$(1, 2, 1)$	e_{11}
80	$(1, 2\zeta, -1-\zeta)$	e_{12}
81	$(1, -2-2\zeta, \zeta)$	e_{13}
82	$(1, 1, 2)$	e_{21}
83	$(1, \zeta, -2-2\zeta)$	e_{22}
84	$(1, -1-\zeta, 2\zeta)$	e_{23}
85	$(1, \frac{1}{2}, \frac{1}{2})$	e_{31}
86	$(1, \frac{1}{2}\zeta, \frac{-1}{2} - \frac{1}{2}\zeta)$	e_{32}
87	$(1, \frac{-1}{2} - \frac{1}{2}\zeta, \frac{1}{2}\zeta)$	e_{33}

TABLE S23: All 64 complete bases of the first 87-ray Hesse parent.

e	Context	e	Context	e	Context
1	$\{1,11,17\}$	2	$\{1,12,18\}$	3	$\{1,13,19\}$
4	$\{1,38,39\}$	5	$\{2,3,4\}$	6	$\{2,11,20\}$
7	$\{2,14,29\}$	8	$\{2,73,76\}$	9	$\{3,12,21\}$
10	$\{3,15,30\}$	11	$\{3,74,77\}$	12	$\{4,13,22\}$
13	$\{4,16,31\}$	14	$\{4,75,78\}$	15	$\{5,6,7\}$
16	$\{5,11,23\}$	17	$\{5,16,34\}$	18	$\{6,12,24\}$
19	$\{6,14,32\}$	20	$\{7,13,25\}$	21	$\{7,15,33\}$
22	$\{8,9,10\}$	23	$\{8,11,26\}$	24	$\{8,15,36\}$
25	$\{9,12,27\}$	26	$\{9,16,37\}$	27	$\{10,13,28\}$
28	$\{10,14,35\}$	29	$\{11,61,85\}$	30	$\{12,62,86\}$
31	$\{13,63,87\}$	32	$\{14,38,52\}$	33	$\{14,49,79\}$
34	$\{15,38,56\}$	35	$\{15,50,80\}$	36	$\{16,38,60\}$
37	$\{16,51,81\}$	38	$\{17,40,46\}$	39	$\{17,43,49\}$
40	$\{18,41,47\}$	41	$\{18,44,50\}$	42	$\{19,42,48\}$
43	$\{19,45,51\}$	44	$\{39,64,73\}$	45	$\{39,68,74\}$
46	$\{39,72,75\}$	47	$\{46,47,48\}$	48	$\{46,52,53\}$
49	$\{46,73,82\}$	50	$\{47,54,56\}$	51	$\{47,74,83\}$
52	$\{48,55,60\}$	53	$\{48,75,84\}$	54	$\{49,50,51\}$
55	$\{49,64,65\}$	56	$\{50,66,68\}$	57	$\{51,67,72\}$
58	$\{52,57,61\}$	59	$\{56,58,62\}$	60	$\{59,60,63\}$
61	$\{61,62,63\}$	62	$\{61,64,69\}$	63	$\{62,68,70\}$
64	$\{63,71,72\}$				

Its six further orthogonal pairs are $\{5,74\},\ \{6,75\},\ \{7,73\},\ \{8,75\},\ \{9,73\},\ \{10,74\}$.

S1.16.3. Peres24 and the complete four-dimensional extensions

The coordinates of H_{65} and $H_{\mathrm{YO}}^{(4)}$ are those already listed above. Table S24 explicitly enumerates the known Peres rays. In each union, retain the original local labels and append the missing rays in the order displayed in Table S25. Tables S26 and S27 list every available complete basis. Full orthogonality-pair lists accompany the coordinates in the data; the 40-ray union has additional pairs outside its complete contexts.

TABLE S24: All 24 primitive unnormalized Peres rays, in the certificate order.

i	p_i
1	$(1,0,0,0)$
2	$(0,1,0,0)$
3	$(0,0,1,0)$
4	$(0,0,0,1)$
5	$(1,-1,0,0)$
6	$(1,1,0,0)$
7	$(1,0,-1,0)$
8	$(1,0,1,0)$
9	$(1,0,0,-1)$
10	$(1,0,0,1)$
11	$(0,1,-1,0)$
12	$(0,1,1,0)$
13	$(0,1,0,-1)$
14	$(0,1,0,1)$
15	$(0,0,1,-1)$
16	$(0,0,1,1)$

i	p_i
17	$(1,-1,-1,-1)$
18	$(1,-1,-1,1)$
19	$(1,-1,1,-1)$
20	$(1,-1,1,1)$
21	$(1,1,-1,-1)$
22	$(1,1,-1,1)$
23	$(1,1,1,-1)$
24	$(1,1,1,1)$

TABLE S25: Every ray added to the original four-dimensional examples to obtain the stated parents.

Original	New label	Added vector
$H_{\mathrm{YO}}^{(4)}$	27	$(1,0,0,-1)$
$H_{\mathrm{YO}}^{(4)}$	28	$(1,0,0,1)$
$H_{\mathrm{YO}}^{(4)}$	29	$(0,1,0,-1)$
$H_{\mathrm{YO}}^{(4)}$	30	$(0,1,0,1)$
$H_{\mathrm{YO}}^{(4)}$	31	$(0,0,1,-1)$
$H_{\mathrm{YO}}^{(4)}$	32	$(0,0,1,1)$
$H_{\mathrm{YO}}^{(4)}$	33	$(1,-1,-1,-1)$
$H_{\mathrm{YO}}^{(4)}$	34	$(1,-1,-1,1)$
$H_{\mathrm{YO}}^{(4)}$	35	$(1,-1,1,-1)$
$H_{\mathrm{YO}}^{(4)}$	36	$(1,-1,1,1)$
$H_{\mathrm{YO}}^{(4)}$	37	$(1,1,-1,-1)$
$H_{\mathrm{YO}}^{(4)}$	38	$(1,1,-1,1)$
$H_{\mathrm{YO}}^{(4)}$	39	$(1,1,1,-1)$
$H_{\mathrm{YO}}^{(4)}$	40	$(1,1,1,1)$
H_{65}	66	$(1,-1,0,0)$
H_{65}	67	$(1,0,-1,0)$
H_{65}	68	$(1,0,0,1)$
H_{65}	69	$(0,1,-1,0)$
H_{65}	70	$(0,1,0,1)$
H_{65}	71	$(0,0,1,1)$
H_{65}	72	$(1,1,1,-1)$

TABLE S26: All 36 complete bases of the 40-ray parent.

e	Context	e	Context
1	$\{1,2,3,26\}$	2	$\{1,3,27,28\}$
3	$\{1,10,19,26\}$	4	$\{1,13,20,26\}$
5	$\{1,27,36,38\}$	6	$\{1,28,35,37\}$
7	$\{2,5,8,26\}$	8	$\{2,5,29,30\}$
9	$\{2,8,31,32\}$	10	$\{3,11,14,26\}$
11	$\{3,12,25,26\}$	12	$\{3,27,34,40\}$
13	$\{3,28,33,39\}$	14	$\{4,5,6,26\}$
15	$\{4,6,31,32\}$	16	$\{4,10,18,26\}$
17	$\{4,11,24,26\}$	18	$\{4,31,33,36\}$
19	$\{4,32,34,35\}$	20	$\{5,8,27,28\}$
21	$\{6,12,17,26\}$	22	$\{6,13,22,26\}$
23	$\{6,31,37,40\}$	24	$\{6,32,38,39\}$
25	$\{7,8,9,26\}$	26	$\{7,9,29,30\}$
27	$\{7,11,15,26\}$	28	$\{7,13,21,26\}$
29	$\{7,29,33,38\}$	30	$\{7,30,34,37\}$
31	$\{9,10,23,26\}$	32	$\{9,12,16,26\}$
33	$\{9,29,35,40\}$	34	$\{9,30,36,39\}$
35	$\{33,36,38,39\}$	36	$\{34,35,37,40\}$

TABLE S27: All 48 complete bases of the 72-ray parent.

e	Context	e	Context
1	$\{1,2,3,4\}$	2	$\{1,2,17,71\}$
3	$\{1,3,15,70\}$	4	$\{1,4,13,69\}$
5	$\{1,13,22,23\}$	6	$\{1,15,30,31\}$
7	$\{1,17,38,39\}$	8	$\{2,3,12,68\}$
9	$\{2,4,14,67\}$	10	$\{2,12,18,19\}$
11	$\{2,14,26,27\}$	12	$\{2,17,40,41\}$
13	$\{3,4,16,66\}$	14	$\{3,12,20,21\}$
15	$\{3,15,32,33\}$	16	$\{3,16,34,35\}$
17	$\{4,13,24,25\}$	18	$\{4,14,28,29\}$
19	$\{4,16,36,37\}$	20	$\{5,6,7,8\}$
21	$\{5,6,17,66\}$	22	$\{5,7,15,67\}$
23	$\{5,8,12,69\}$	24	$\{5,12,42,43\}$
25	$\{5,15,54,55\}$	26	$\{5,17,62,63\}$
27	$\{6,7,13,68\}$	28	$\{6,8,14,70\}$
29	$\{6,13,46,47\}$	30	$\{6,14,50,51\}$
31	$\{6,17,64,65\}$	32	$\{7,8,16,71\}$
33	$\{7,13,48,49\}$	34	$\{7,15,56,57\}$
35	$\{7,16,58,59\}$	36	$\{8,12,44,45\}$
37	$\{8,14,52,53\}$	38	$\{8,16,60,61\}$
39	$\{9,10,11,72\}$	40	$\{9,10,12,13\}$
41	$\{9,11,14,15\}$	42	$\{9,66,71,72\}$
43	$\{10,11,16,17\}$	44	$\{10,67,70,72\}$
45	$\{11,68,69,72\}$	46	$\{12,13,68,69\}$
47	$\{14,15,67,70\}$	48	$\{16,17,66,71\}$

S1.17. Cabello completions and the two 68-ray parents

S1.17.1. The eighteen-ray input and faithful 36-ray completion

All labels in this subsection are local. In $C_{18,9}$ and $C_{36,18}$, $18 = a$, $19 = b$, and $20 = c$; labels 4 and 5 are absent. They do not denote the completion rays of H_{65}. The core is the first eighteen rows of Table S28; labels 21–38 are fresh. The contexts of $C_{18,9}$ are the first nine rows of Table S29.

TABLE S28: All 36 unnormalized vectors of the faithful Cabello completion.

i	v_i
1	$(1,0,0,0)$
2	$(0,1,0,0)$
3	$(0,0,1,0)$
6	$(1,1,-1,-1)$
7	$(1,-1,1,-1)$
8	$(1,-1,-1,1)$
9	$(1,1,-1,1)$
10	$(1,-1,1,1)$
11	$(1,-1,-1,-1)$
12	$(1,0,0,-1)$
13	$(0,1,1,0)$
14	$(1,0,1,0)$
15	$(0,1,0,-1)$
16	$(1,1,0,0)$
17	$(0,0,1,-1)$
18	$(1,0,0,1)$
19	$(0,1,0,1)$
20	$(0,0,1,1)$
21	$(0,1,-1,3)$
22	$(0,3,-3,-2)$
23	$(1,0,-1,3)$
24	$(3,0,-3,-2)$
25	$(1,-1,0,3)$
26	$(3,-3,0,-2)$
27	$(3,1,2,2)$
28	$(3,-5,-1,-1)$
29	$(3,2,1,2)$
30	$(3,-1,-5,-1)$
31	$(2,3,1,2)$
32	$(1,-3,5,1)$
33	$(4,2,3,-3)$
34	$(5,-7,-1,1)$
35	$(4,3,2,-3)$
36	$(5,-1,-7,1)$
37	$(3,4,2,-3)$
38	$(1,-5,7,-1)$

TABLE S29: All eighteen full bases of the faithful Cabello completion.

e	Context
1	$\{9, 10, 12, 13\}$
2	$\{9, 11, 14, 15\}$
3	$\{10, 11, 16, 17\}$
4	$\{2, 3, 12, 18\}$
5	$\{1, 3, 15, 19\}$
6	$\{1, 2, 17, 20\}$
7	$\{6, 7, 13, 18\}$
8	$\{6, 8, 14, 19\}$
9	$\{7, 8, 16, 20\}$
10	$\{1, 13, 21, 22\}$
11	$\{2, 14, 23, 24\}$
12	$\{3, 16, 25, 26\}$
13	$\{6, 17, 27, 28\}$
14	$\{7, 15, 29, 30\}$
15	$\{8, 12, 31, 32\}$
16	$\{9, 20, 33, 34\}$
17	$\{10, 19, 35, 36\}$
18	$\{11, 18, 37, 38\}$

The orthogonal matrix $Q = M/\sqrt{2}$, with

$$M = \begin{pmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad M^T M = 2I_4, \tag{S6}$$

maps the published eighteen-ray set and all nine bases of Ref. [2, Eqs. (1)–(9)] onto this realization. Table S30 gives the complete ray map; the input representatives are unnormalized and signs are immaterial.

TABLE S30: Exact ray map from the originally published Cabello coordinates.

Published vector u	Local label of $[Qu]$
$(0, 0, 0, 1)$	12
$(0, 0, 1, 0)$	13
$(0, 0, 1, 1)$	11
$(0, 1, -1, 0)$	8
$(0, 1, 0, -1)$	1
$(0, 1, 0, 0)$	18
$(1, -1, -1, -1)$	19
$(1, -1, -1, 1)$	16
$(1, -1, 0, 0)$	9
$(1, -1, 1, -1)$	17
$(1, 0, -1, 0)$	2
$(1, 0, 0, -1)$	7
$(1, 0, 0, 1)$	6
$(1, 0, 1, 0)$	3
$(1, 1, -1, 1)$	15
$(1, 1, 0, 0)$	10
$(1, 1, 1, -1)$	14
$(1, 1, 1, 1)$	20

S1.17.2. An injective 45-ray completion after deleting c

Delete local ray 20 and contexts 6 and 9 from the first nine contexts above. The seven remaining bases are followed by fourteen fresh-pair contexts in Table S32. The next table prints all 45 vectors; 4, 5, and 20 are absent and 21–48 are fresh. Its only uncovered orthogonal pairs are $\{16, 21\}, \{16, 22\}, \{17, 35\}, \{17, 36\}$.

TABLE S31: All 45 unnormalized vectors of the vertex-deleted completion.

i	v_i
1	$(1, 0, 0, 0)$
2	$(0, 1, 0, 0)$
3	$(0, 0, 1, 0)$
6	$(1, 1, -1, -1)$
7	$(1, -1, 1, -1)$
8	$(1, -1, -1, 1)$
9	$(1, 1, -1, 1)$
10	$(1, -1, 1, 1)$
11	$(1, -1, -1, -1)$
12	$(1, 0, 0, -1)$
13	$(0, 1, 1, 0)$
14	$(1, 0, 1, 0)$
15	$(0, 1, 0, -1)$
16	$(1, 1, 0, 0)$
17	$(0, 0, 1, -1)$
18	$(1, 0, 0, 1)$
19	$(0, 1, 0, 1)$
21	$(0, 0, 1, 2)$
22	$(0, 0, 2, -1)$
23	$(0, 1, -1, 3)$
24	$(0, 3, -3, -2)$
25	$(0, 1, 2, 2)$
26	$(0, 4, -1, -1)$
27	$(1, 0, -1, 3)$
28	$(3, 0, -3, -2)$
29	$(1, 0, 2, 2)$
30	$(4, 0, -1, -1)$
31	$(1, -1, 0, 3)$
32	$(3, -3, 0, -2)$
33	$(5, 3, 4, 4)$
34	$(7, -9, -1, -1)$
35	$(1, 1, 3, 3)$
36	$(3, 3, -1, -1)$
37	$(5, 4, 3, 4)$
38	$(7, -1, -9, -1)$
39	$(2, -2, -1, 3)$
40	$(1, -1, -5, -3)$
41	$(4, 5, 3, 4)$
42	$(1, -7, 9, 1)$
43	$(4, -4, -1, -9)$
44	$(5, -5, 13, 3)$
45	$(6, 5, 4, -5)$
46	$(9, -1, -11, 1)$
47	$(3, 4, 2, -3)$
48	$(1, -5, 7, -1)$

TABLE S32: All 21 full bases of the vertex-deleted completion.

e	Context
1	$\{9, 10, 12, 13\}$
2	$\{9, 11, 14, 15\}$
3	$\{10, 11, 16, 17\}$
4	$\{2, 3, 12, 18\}$
5	$\{1, 3, 15, 19\}$
6	$\{6, 7, 13, 18\}$
7	$\{6, 8, 14, 19\}$
8	$\{1, 2, 21, 22\}$
9	$\{1, 13, 23, 24\}$
10	$\{1, 17, 25, 26\}$
11	$\{2, 14, 27, 28\}$
12	$\{2, 17, 29, 30\}$
13	$\{3, 16, 31, 32\}$
14	$\{6, 17, 33, 34\}$
15	$\{7, 8, 35, 36\}$
16	$\{7, 15, 37, 38\}$
17	$\{7, 16, 39, 40\}$
18	$\{8, 12, 41, 42\}$
19	$\{8, 16, 43, 44\}$
20	$\{10, 19, 45, 46\}$
21	$\{11, 18, 47, 48\}$

S1.17.3. All vectors and bases of the two 68-ray parents

In each parent retain all 65 vectors of Appendix S1.15.1, with their original labels, and add the three vectors in the corresponding rows below. Thus the six rows specify every added ray; the two parent inventories use distinct local meanings for labels 66–68. These labels also differ from the 72-ray parent numbering in Appendix S1.16.

TABLE S33: The three added rays in each local 68-ray parent inventory.

Parent variant	i	v_i
18–9 subsystem	66	$(1, 0, 0, 1)$
18–9 subsystem	67	$(0, 1, 0, 1)$
18–9 subsystem	68	$(0, 0, 1, 1)$
20–11 subsystem	66	$(0, 1, -1, 0)$
20–11 subsystem	67	$(1, 0, -1, 0)$
20–11 subsystem	68	$(1, -1, 0, 0)$

TABLE S34: All 35 full bases of 68-ray parent variant 1.

e	Context	e	Context
1	$\{1,2,3,4\}$	2	$\{1,2,17,68\}$
3	$\{1,3,15,67\}$	4	$\{1,13,22,23\}$
5	$\{1,15,30,31\}$	6	$\{1,17,38,39\}$
7	$\{2,3,12,66\}$	8	$\{2,12,18,19\}$
9	$\{2,14,26,27\}$	10	$\{2,17,40,41\}$
11	$\{3,12,20,21\}$	12	$\{3,15,32,33\}$
13	$\{3,16,34,35\}$	14	$\{4,13,24,25\}$
15	$\{4,14,28,29\}$	16	$\{4,16,36,37\}$
17	$\{5,6,7,8\}$	18	$\{5,12,42,43\}$
19	$\{5,15,54,55\}$	20	$\{5,17,62,63\}$
21	$\{6,7,13,66\}$	22	$\{6,8,14,67\}$
23	$\{6,13,46,47\}$	24	$\{6,14,50,51\}$
25	$\{6,17,64,65\}$	26	$\{7,8,16,68\}$
27	$\{7,13,48,49\}$	28	$\{7,15,56,57\}$
29	$\{7,16,58,59\}$	30	$\{8,12,44,45\}$
31	$\{8,14,52,53\}$	32	$\{8,16,60,61\}$
33	$\{9,10,12,13\}$	34	$\{9,11,14,15\}$
35	$\{10,11,16,17\}$		

TABLE S35: All 35 full bases of 68-ray parent variant 2.

e	Context	e	Context
1	$\{1,2,3,4\}$	2	$\{1,4,13,66\}$
3	$\{1,13,22,23\}$	4	$\{1,15,30,31\}$
5	$\{1,17,38,39\}$	6	$\{2,4,14,67\}$
7	$\{2,12,18,19\}$	8	$\{2,14,26,27\}$
9	$\{2,17,40,41\}$	10	$\{3,4,16,68\}$
11	$\{3,12,20,21\}$	12	$\{3,15,32,33\}$
13	$\{3,16,34,35\}$	14	$\{4,13,24,25\}$
15	$\{4,14,28,29\}$	16	$\{4,16,36,37\}$
17	$\{5,6,7,8\}$	18	$\{5,6,17,68\}$
19	$\{5,7,15,67\}$	20	$\{5,8,12,66\}$
21	$\{5,12,42,43\}$	22	$\{5,15,54,55\}$
23	$\{5,17,62,63\}$	24	$\{6,13,46,47\}$
25	$\{6,14,50,51\}$	26	$\{6,17,64,65\}$
27	$\{7,13,48,49\}$	28	$\{7,15,56,57\}$
29	$\{7,16,58,59\}$	30	$\{8,12,44,45\}$
31	$\{8,14,52,53\}$	32	$\{8,16,60,61\}$
33	$\{9,10,12,13\}$	34	$\{9,11,14,15\}$
35	$\{10,11,16,17\}$		

For variant 1 the nine-context parity subsystem consists of exactly $C_3, C_4, C_5, D_1, \ldots, D_6$ with $a = 66, b = 67, c = 68$. For variant 2 the twenty-ray subsystem consists of all rows whose labels lie in $\{1, \ldots, 17, 66, 67, 68\}$. The machine-readable inventories repeat every coordinate, basis, and inclusion map.

S2. TWO-VALUED STATES, EXTENDABILITY CERTIFICATES, COLORINGS, AND SEPARATION PROFILES

S2.1. State, coloring, odd-cycle certificate, and profile tables for chromatically complete configurations ($\chi(H) = d$)

This subsection provides the complete state and coloring enumerations and odd-cycle nonextendability certificates for $H_{6,3}$ (Table S36), P_5 (Table S37), H_8 (Tables S38 and S39), $H_{17,10}$ (Tables S40 and S41), $H_{21,13}$ (Tables S42, S43, and S44), the vertex separation profiles and Born probability decomposition on $H_{17,10}$ and $H_{21,13}$ (Tables S45–S47), the complete state and normalized coloring tables for all faithful 3- and 4-dimensional logical gadgets (Tables S48–S69 and S70), and the ten chromatically inseparable faithful restrictions of H_{YO}^{+} (Table S71).

TABLE S36: All two-valued states of $H_{6,3}$. The last column gives an odd cycle in the complement of each nonextendable support.

j	support of s_j	extendable	odd cycle
1	$\{1,4\}$	yes	–
2	$\{2,4,6\}$	no	$(3,1,5)$
3	$\{2,5\}$	yes	–
4	$\{3,6\}$	yes	–

TABLE S37: All two-valued states of P_5. The last column gives an odd cycle in the complement of each nonextendable support.

j	support of s_j	extendable	odd cycle
1	$\{1,3,9\}$	yes	–
2	$\{1,4,7\}$	yes	–
3	$\{1,7,8,9\}$	yes	–
4	$\{2,4,10\}$	yes	–
5	$\{2,5,8\}$	yes	–
6	$\{2,8,9,10\}$	yes	–
7	$\{3,5,6\}$	yes	–
8	$\{3,6,9,10\}$	yes	–
9	$\{4,6,7,10\}$	yes	–
10	$\{5,6,7,8\}$	yes	–
11	$\{6,7,8,9,10\}$	no	$(3,2,1,5,4)$

TABLE S38: All two-valued states of H_8. The last column gives an odd cycle in the complement of each nonextendable support.

j	support of s_j	extendable	odd cycle
1	$\{1,4,6,9\}$	yes	–
2	$\{1,5,8,10,14\}$	no	$(9,7,6,12,11)$
3	$\{1,5,9,13,14\}$	yes	–
4	$\{2,4,6,8,11\}$	yes	–
5	$\{2,4,7,11,14\}$	no	$(5,3,1,12,6)$
6	$\{2,5,8,10,12\}$	yes	–
7	$\{2,5,8,11,13,14\}$	yes	–
8	$\{2,5,9,12,13\}$	no	$(4,3,1,11,10)$
9	$\{3,6,8,11,13\}$	no	$(7,5,4,10,9)$
10	$\{3,7,10,12\}$	yes	–
11	$\{3,7,11,13,14\}$	yes	–

TABLE S39. All strong three-colorings of H_8 up to color permutation. The anchor is $(1,2,3) = (0,1,2)$.

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14
c_1	0	1	2	0	1	0	2	1	0	1	2	1	2	2
c_2	0	1	2	0	1	0	2	1	0	2	1	2	1	1
c_3	0	1	2	1	0	1	2	1	0	2	1	2	0	0

TABLE S40: All two-valued states of $H_{17,10}$. The last column gives an odd cycle in the complement of each nonextendable support.

j	support of s_j	extendable	odd cycle
1	$\{1,4,8,14,15,23\}$	yes	–
2	$\{1,5,7,14,18,23,24\}$	no	$(4,11,3,2,8,9,10)$
3	$\{1,5,9,11,18\}$	yes	–
4	$\{1,5,9,14,15,18,24\}$	no	$(11,3,2,8,7)$
5	$\{1,6,8,11,18,23\}$	yes	–
6	$\{1,6,8,14,15,18,23,24\}$	no	$(11,3,2,5,4)$
7	$\{2,4,7,14,19,23\}$	yes	–
8	$\{2,4,9,14,15,19\}$	yes	–
9	$\{2,6,7,10,14,24\}$	yes	–
10	$\{2,6,7,14,18,19,23,24\}$	no	$(11,3,1,10,4)$
11	$\{2,6,9,11,18,19\}$	yes	–
12	$\{2,6,9,14,15,18,19,24\}$	no	$(11,3,1,10,4)$
13	$\{3,4,8,15,19,23\}$	yes	–
14	$\{3,5,7,10,24\}$	yes	–
15	$\{3,5,7,18,19,23,24\}$	no	$(8,2,1,10,9)$
16	$\{3,5,9,15,18,19,24\}$	no	$(7,8,2,1,10,4,11)$
17	$\{3,6,8,10,15,24\}$	yes	–
18	$\{3,6,8,15,18,19,23,24\}$	no	$(5,2,1,10,4)$

TABLE S41. All strong three-colorings of $H_{17,10}$ up to color permutation. The anchor is $(2,8,5)=(0,1,2)$.

i	1	2	3	4	5	6	7	8	9	10	11	14	15	18	19	23	24
c_1	1	0	2	0	2	1	2	1	0	2	1	0	0	1	0	1	2
c_2	1	0	2	1	2	0	2	1	0	2	0	1	1	0	0	1	2
c_3	2	0	1	0	2	1	0	1	2	1	2	0	1	2	0	0	1
c_4	2	0	1	1	2	0	0	1	2	0	2	0	1	2	1	1	0

TABLE S42: All two-valued states of $H_{21,13}$. The last column gives an odd cycle in the complement of each nonextendable support.

j	support of s_j	extendable	odd cycle
1	$\{1,4,8,14,15,21,22,23\}$	no	$(7,11,3,2,5,6,13)$
2	$\{1,5,7,14,18,22,23,24\}$	no	$(4,11,3,2,8,9,10)$
3	$\{1,5,9,11,18,21,22\}$	yes	–
4	$\{1,5,9,14,15,18,21,22,24\}$	no	$(11,3,2,8,7)$
5	$\{1,6,8,11,18,21,23\}$	yes	–
6	$\{1,6,8,14,15,18,21,23,24\}$	no	$(11,3,2,5,4)$
7	$\{2,4,7,14,19,20,22,23\}$	no	$(8,9,10,1,13,6,5)$
8	$\{2,4,9,13,14,15,19\}$	yes	–
9	$\{2,4,9,14,15,19,20,21,22\}$	no	$(11,3,1,13,7)$
10	$\{2,6,7,10,14,20,24\}$	yes	–
11	$\{2,6,7,14,18,19,20,23,24\}$	no	$(11,3,1,10,4)$
12	$\{2,6,9,11,18,19,20,21\}$	no	$(5,4,10,1,13,7,8)$
13	$\{2,6,9,14,15,18,19,20,21,24\}$	no	$(11,3,1,10,4)$
14	$\{3,4,8,13,15,19,23\}$	yes	–
15	$\{3,4,8,15,19,20,21,22,23\}$	no	$(5,2,1,13,6)$
16	$\{3,5,7,10,20,22,24\}$	yes	–
17	$\{3,5,7,18,19,20,22,23,24\}$	no	$(8,2,1,10,9)$
18	$\{3,5,9,13,15,18,19,24\}$	no	$(7,8,2,1,10,4,11)$
19	$\{3,5,9,15,18,19,20,21,22,24\}$	no	$(8,2,1,13,7)$
20	$\{3,6,8,10,15,20,21,24\}$	no	$(4,5,2,1,13,7,11)$
21	$\{3,6,8,15,18,19,20,21,23,24\}$	no	$(5,2,1,10,4)$

TABLE S43. All strong three-colorings of $H_{21,13}$ up to color permutation. The anchor is $(2, 8, 5) = (0, 1, 2)$.

i	1	2	3	4	5	6	7	8	9	10	11	13	14	15	18	19	20	21	22	23	24
c_1	1	0	2	0	2	1	2	1	0	2	1	0	0	0	1	0	2	1	2	1	2
c_2	2	0	1	1	2	0	0	1	2	0	2	1	0	1	2	1	0	2	2	1	0

TABLE S44. The nine chromatic signatures of $H_{21,13}$. Columns give c_1 and rows give c_2 for the two normalized colorings in Table S43. Each cell is one inseparability class: its vertices have the same color in every admissible coloring. Ordinary two-valued states distinguish every pair, including vertices in the same cell.

$c_2 \backslash c_1$	0	1	2
2	$\{9\}$	$\{1, 11, 18, 21\}$	$\{5, 22\}$
1	$\{4, 13, 15, 19\}$	$\{8, 23\}$	$\{3\}$
0	$\{2, 14\}$	$\{6\}$	$\{7, 10, 20, 24\}$

TABLE S45: Complete state profiles in hexadecimal, and all Born probabilities for the 17-ray preparation in Eq. (26). The probabilities are also reproduced by Eq. (27).

i	$H_{17,10}$ profile	$H_{21,13}$ profile	p_i on $H_{17,10}$
1	3F	3F	1/12
2	FC0	1FC0	1/6
3	3F000	1FE000	3/4
4	10C1	61C1	0
5	E00E	7800E	2/3
6	30F30	181E30	1/3
7	6342	18642	3/4
8	31031	186031	1/6
9	8C8C	6198C	1/12
10	12100	88200	8/9
11	414	814	0
13	-	22080	–
14	BEB	17EB	1/4
15	398A9	1E71A9	1/4
18	2CE3E	171C3E	1/9
19	2DEC0	177DC0	1/36
20	-	1DDF40	–
21	-	1C593D	–
22	-	5C14F	–
23	25273	116473	1/36
24	3EB2A	1F962A	1

TABLE S46. The five valid states and exact weights reproducing the $H_{17,10}$ preparation P_{24}. State numbering is that of Table S40. Every context equation and every Born marginal are checked with rational arithmetic.

State	Weight	Support	Extendable?
s_4	$\frac{1}{12}$	$\{1,5,9,14,15,18,24\}$	no
s_9	$\frac{5}{36}$	$\{2,6,7,10,14,24\}$	yes
s_{10}	$\frac{1}{36}$	$\{2,6,7,14,18,19,23,24\}$	no
s_{14}	$\frac{7}{12}$	$\{3,5,7,10,24\}$	yes
s_{17}	$\frac{1}{6}$	$\{3,6,8,10,15,24\}$	yes

TABLE S47. Exact Born probabilities $p_i = (r_{24} \cdot r_i)^2/(6\|r_i\|^2)$, also reproduced by the five-state mixture.

i	p_i	i	p_i
1	$\frac{1}{12}$	10	$\frac{8}{9}$
2	$\frac{1}{6}$	11	0
3	$\frac{3}{4}$	14	$\frac{1}{4}$
4	0	15	$\frac{1}{4}$
5	$\frac{2}{3}$	18	$\frac{1}{9}$
6	$\frac{1}{3}$	19	$\frac{1}{36}$
7	$\frac{3}{4}$	23	$\frac{1}{36}$
8	$\frac{1}{6}$	24	1
9	$\frac{1}{12}$		

S2.2. State conventions for the additional gadgets

The state tables below are local to each labeled configuration. Supports are in lexicographic order; E is yes precisely for color-class supports. The three-dimensional pentagon and $H_{21,13}$ use the complete earlier tables and are not repeated. The four-dimensional apex-only states are included explicitly. All ordinary state families listed here are unital and separating.

S2.3. All states of $B_{13,7}$

TABLE S48: All two-valued states of $B_{13,7}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{1,5,7,14,23\}$	yes	8	$\{2,9,11,19\}$	yes
2	$\{1,5,9,11\}$	yes	9	$\{2,9,14,15,19\}$	yes
3	$\{1,5,9,14,15\}$	no	10	$\{3,5,7,10\}$	yes
4	$\{1,8,11,23\}$	yes	11	$\{3,5,7,19,23\}$	no
5	$\{1,8,14,15,23\}$	yes	12	$\{3,5,9,15,19\}$	yes
6	$\{2,7,10,14\}$	yes	13	$\{3,8,10,15\}$	yes
7	$\{2,7,14,19,23\}$	yes	14	$\{3,8,15,19,23\}$	yes

S2.4. All states of $D_{16,9}$

TABLE S49: All two-valued states of $D_{16,9}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{1,4,8,14,15,23\}$	yes	12	$\{2,6,7,10,14,24\}$	yes
2	$\{1,5,7,10,14,24\}$	yes	13	$\{2,6,7,14,18,23,24\}$	no
3	$\{1,5,7,14,18,23,24\}$	no	14	$\{2,6,9,11,18\}$	yes
4	$\{1,5,9,11,18\}$	yes	15	$\{2,6,9,14,15,18,24\}$	no
5	$\{1,5,9,14,15,18,24\}$	no	16	$\{3,4,8,15,23\}$	yes
6	$\{1,6,8,10,11\}$	yes	17	$\{3,5,7,10,24\}$	yes
7	$\{1,6,8,10,14,15,24\}$	no	18	$\{3,5,7,18,23,24\}$	yes
8	$\{1,6,8,11,18,23\}$	yes	19	$\{3,5,9,15,18,24\}$	yes
9	$\{1,6,8,14,15,18,23,24\}$	no	20	$\{3,6,8,10,15,24\}$	yes
10	$\{2,4,7,14,23\}$	yes	21	$\{3,6,8,15,18,23,24\}$	no
11	$\{2,4,9,14,15\}$	yes			

S2.5. All states of $I_{16,9}$

TABLE S50: All two-valued states of $I_{16,9}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{1,4,5,7,14,23\}$	yes	12	$\{2,7,14,18,19,23,24\}$	no
2	$\{1,4,5,9,14,15\}$	no	13	$\{2,9,11,18,19\}$	yes
3	$\{1,4,8,14,15,23\}$	yes	14	$\{2,9,14,15,18,19,24\}$	no
4	$\{1,5,7,14,18,23,24\}$	no	15	$\{3,4,5,7,19,23\}$	no
5	$\{1,5,9,11,18\}$	yes	16	$\{3,4,5,9,15,19\}$	yes
6	$\{1,5,9,14,15,18,24\}$	no	17	$\{3,4,8,15,19,23\}$	yes
7	$\{1,8,11,18,23\}$	yes	18	$\{3,5,7,10,24\}$	yes
8	$\{1,8,14,15,18,23,24\}$	no	19	$\{3,5,7,18,19,23,24\}$	no
9	$\{2,4,7,14,19,23\}$	yes	20	$\{3,5,9,15,18,19,24\}$	no
10	$\{2,4,9,14,15,19\}$	yes	21	$\{3,8,10,15,24\}$	yes
11	$\{2,7,10,14,24\}$	yes	22	$\{3,8,15,18,19,23,24\}$	no

S2.6. All states of $T_{19,11}$

TABLE S51: All two-valued states of $T_{19,11}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{1,4,8,14,15,23,28\}$	yes	14	$\{2,6,7,14,18,23,24\}$	no
2	$\{1,5,7,10,14,24,28\}$	yes	15	$\{2,6,9,11,18,27\}$	yes
3	$\{1,5,7,14,18,23,24\}$	no	16	$\{2,6,9,14,15,18,24\}$	no
4	$\{1,5,9,11,18,27\}$	yes	17	$\{3,4,8,15,23,26\}$	yes
5	$\{1,5,9,14,15,18,24\}$	no	18	$\{3,4,8,15,23,27,28\}$	no
6	$\{1,6,8,10,11,26\}$	yes	19	$\{3,5,7,10,24,26\}$	yes
7	$\{1,6,8,10,11,27,28\}$	no	20	$\{3,5,7,10,24,27,28\}$	no
8	$\{1,6,8,10,14,15,24,28\}$	no	21	$\{3,5,7,18,23,24,27\}$	yes
9	$\{1,6,8,11,18,23,27\}$	yes	22	$\{3,5,9,15,18,24,27\}$	yes
10	$\{1,6,8,14,15,18,23,24\}$	no	23	$\{3,6,8,10,15,24,26\}$	yes
11	$\{2,4,7,14,23,28\}$	yes	24	$\{3,6,8,10,15,24,27,28\}$	no
12	$\{2,4,9,14,15,28\}$	yes	25	$\{3,6,8,15,18,23,24,27\}$	no
13	$\{2,6,7,10,14,24,28\}$	yes			

S2.7. All states of $W^{(4)}_{11,5}$

TABLE S52: All two-valued states of $W^{(4)}_{11,5}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{0\}$	yes	7	$\{2,8,9,10\}$	yes
2	$\{1,3,9\}$	yes	8	$\{3,5,6\}$	yes
3	$\{1,4,7\}$	yes	9	$\{3,6,9,10\}$	yes
4	$\{1,7,8,9\}$	yes	10	$\{4,6,7,10\}$	yes
5	$\{2,4,10\}$	yes	11	$\{5,6,7,8\}$	yes
6	$\{2,5,8\}$	yes	12	$\{6,7,8,9,10\}$	no

S2.8. All states of $W^{(4)}_{16,6}$

TABLE S53: All two-valued states of $W^{(4)}_{16,6}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{0,7,9,10\}$	yes	37	$\{2,8,12,14,15\}$	yes
2	$\{0,7,9,15\}$	yes	38	$\{3,5,6\}$	yes
3	$\{0,7,10,14\}$	yes	39	$\{3,6,9,10,11\}$	yes
4	$\{0,7,14,15\}$	yes	40	$\{3,6,9,10,12\}$	yes
5	$\{0,9,10,13\}$	yes	41	$\{3,6,9,11,15\}$	yes
6	$\{0,9,13,15\}$	yes	42	$\{3,6,9,12,15\}$	yes
7	$\{0,10,13,14\}$	yes	43	$\{3,6,10,11,14\}$	yes
8	$\{0,13,14,15\}$	yes	44	$\{3,6,10,12,14\}$	yes
9	$\{1,3,9,11\}$	yes	45	$\{3,6,11,14,15\}$	yes
10	$\{1,3,9,12\}$	yes	46	$\{3,6,12,14,15\}$	yes
11	$\{1,3,11,14\}$	yes	47	$\{4,6,7,10,11\}$	yes
12	$\{1,3,12,14\}$	yes	48	$\{4,6,7,10,12\}$	yes
13	$\{1,4,7,11\}$	yes	49	$\{4,6,7,11,15\}$	yes
14	$\{1,4,7,12\}$	yes	50	$\{4,6,7,12,15\}$	yes
15	$\{1,4,11,13\}$	yes	51	$\{4,6,10,11,13\}$	yes
16	$\{1,4,12,13\}$	yes	52	$\{4,6,10,12,13\}$	yes
17	$\{1,7,8,9,11\}$	yes	53	$\{4,6,11,13,15\}$	yes
18	$\{1,7,8,9,12\}$	yes	54	$\{4,6,12,13,15\}$	yes
19	$\{1,7,8,11,14\}$	yes	55	$\{5,6,7,8\}$	yes
20	$\{1,7,8,12,14\}$	yes	56	$\{5,6,8,13\}$	yes
21	$\{1,8,9,11,13\}$	yes	57	$\{6,7,8,9,10,11\}$	no
22	$\{1,8,9,12,13\}$	yes	58	$\{6,7,8,9,10,12\}$	no
23	$\{1,8,11,13,14\}$	yes	59	$\{6,7,8,9,11,15\}$	no
24	$\{1,8,12,13,14\}$	yes	60	$\{6,7,8,9,12,15\}$	no
25	$\{2,4,10,11\}$	yes	61	$\{6,7,8,10,11,14\}$	no
26	$\{2,4,10,12\}$	yes	62	$\{6,7,8,10,12,14\}$	no
27	$\{2,4,11,15\}$	yes	63	$\{6,7,8,11,14,15\}$	no
28	$\{2,4,12,15\}$	yes	64	$\{6,7,8,12,14,15\}$	no
29	$\{2,5,8\}$	yes	65	$\{6,8,9,10,11,13\}$	no
30	$\{2,8,9,10,11\}$	yes	66	$\{6,8,9,10,12,13\}$	no
31	$\{2,8,9,10,12\}$	yes	67	$\{6,8,9,11,13,15\}$	no
32	$\{2,8,9,11,15\}$	yes	68	$\{6,8,9,12,13,15\}$	no
33	$\{2,8,9,12,15\}$	yes	69	$\{6,8,10,11,13,14\}$	no
34	$\{2,8,10,11,14\}$	yes	70	$\{6,8,10,12,13,14\}$	no
35	$\{2,8,10,12,14\}$	yes	71	$\{6,8,11,13,14,15\}$	no
36	$\{2,8,11,14,15\}$	yes	72	$\{6,8,12,13,14,15\}$	no

S2.9. All states of $B^{(4)}_{14,7}$

TABLE S54: All two-valued states of $B^{(4)}_{14,7}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{0\}$	yes	9	$\{2,9,11,19\}$	yes
2	$\{1,5,7,14,23\}$	yes	10	$\{2,9,14,15,19\}$	yes
3	$\{1,5,9,11\}$	yes	11	$\{3,5,7,10\}$	yes
4	$\{1,5,9,14,15\}$	no	12	$\{3,5,7,19,23\}$	no
5	$\{1,8,11,23\}$	yes	13	$\{3,5,9,15,19\}$	yes
6	$\{1,8,14,15,23\}$	yes	14	$\{3,8,10,15\}$	yes
7	$\{2,7,10,14\}$	yes	15	$\{3,8,15,19,23\}$	yes
8	$\{2,7,14,19,23\}$	yes			

S2.10. All states of $L^{(4)}_{18,7}$

TABLE S55: All two-valued states of $L^{(4)}_{18,7}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{1,6,9,13,16\}$	yes	35	$\{3,5,9,13,18\}$	yes
2	$\{1,6,9,14,16\}$	yes	36	$\{3,5,9,14,17\}$	yes
3	$\{1,6,10,13,16\}$	yes	37	$\{3,5,9,14,18\}$	yes
4	$\{1,6,10,14,16\}$	yes	38	$\{3,5,10,13,17\}$	yes
5	$\{1,7,9,13,17\}$	yes	39	$\{3,5,10,13,18\}$	yes
6	$\{1,7,9,13,18\}$	yes	40	$\{3,5,10,14,17\}$	yes
7	$\{1,7,9,14,17\}$	yes	41	$\{3,5,10,14,18\}$	yes
8	$\{1,7,9,14,18\}$	yes	42	$\{3,6,9,11\}$	yes
9	$\{1,7,10,13,17\}$	yes	43	$\{3,6,9,12,13\}$	yes
10	$\{1,7,10,13,18\}$	yes	44	$\{3,6,9,12,14\}$	yes
11	$\{1,7,10,14,17\}$	yes	45	$\{3,6,10,11\}$	yes
12	$\{1,7,10,14,18\}$	yes	46	$\{3,6,10,12,13\}$	yes
13	$\{1,8,13,15\}$	yes	47	$\{3,6,10,12,14\}$	yes
14	$\{1,8,13,16,17\}$	yes	48	$\{3,8,11,17\}$	yes
15	$\{1,8,13,16,18\}$	yes	49	$\{3,8,11,18\}$	yes
16	$\{1,8,14,15\}$	yes	50	$\{3,8,12,13,17\}$	yes
17	$\{1,8,14,16,17\}$	yes	51	$\{3,8,12,13,18\}$	yes
18	$\{1,8,14,16,18\}$	yes	52	$\{3,8,12,14,17\}$	yes
19	$\{2,5,9,15\}$	yes	53	$\{3,8,12,14,18\}$	yes
20	$\{2,5,9,16,17\}$	yes	54	$\{4,5,13,15\}$	yes
21	$\{2,5,9,16,18\}$	yes	55	$\{4,5,13,16,17\}$	yes
22	$\{2,5,10,15\}$	yes	56	$\{4,5,13,16,18\}$	yes
23	$\{2,5,10,16,17\}$	yes	57	$\{4,5,14,15\}$	yes
24	$\{2,5,10,16,18\}$	yes	58	$\{4,5,14,16,17\}$	yes
25	$\{2,6,9,12,16\}$	yes	59	$\{4,5,14,16,18\}$	yes
26	$\{2,6,10,12,16\}$	yes	60	$\{4,6,11,16\}$	yes
27	$\{2,7,9,12,17\}$	yes	61	$\{4,6,12,13,16\}$	yes
28	$\{2,7,9,12,18\}$	yes	62	$\{4,6,12,14,16\}$	yes
29	$\{2,7,10,12,17\}$	yes	63	$\{4,7,11,17\}$	yes
30	$\{2,7,10,12,18\}$	yes	64	$\{4,7,11,18\}$	yes
31	$\{2,8,12,15\}$	yes	65	$\{4,7,12,13,17\}$	yes
32	$\{2,8,12,16,17\}$	yes	66	$\{4,7,12,13,18\}$	yes
33	$\{2,8,12,16,18\}$	yes	67	$\{4,7,12,14,17\}$	yes
34	$\{3,5,9,13,17\}$	yes	68	$\{4,7,12,14,18\}$	yes

S2.11. All states of $D^{(4)}_{17,9}$

TABLE S56: All two-valued states of $D^{(4)}_{17,9}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{0\}$	yes	12	$\{2,4,9,14,15\}$	yes
2	$\{1,4,8,14,15,23\}$	yes	13	$\{2,6,7,10,14,24\}$	yes
3	$\{1,5,7,10,14,24\}$	yes	14	$\{2,6,7,14,18,23,24\}$	no
4	$\{1,5,7,14,18,23,24\}$	no	15	$\{2,6,9,11,18\}$	yes
5	$\{1,5,9,11,18\}$	yes	16	$\{2,6,9,14,15,18,24\}$	no
6	$\{1,5,9,14,15,18,24\}$	no	17	$\{3,4,8,15,23\}$	yes
7	$\{1,6,8,10,11\}$	yes	18	$\{3,5,7,10,24\}$	yes
8	$\{1,6,8,10,14,15,24\}$	no	19	$\{3,5,7,18,23,24\}$	yes
9	$\{1,6,8,11,18,23\}$	yes	20	$\{3,5,9,15,18,24\}$	yes
10	$\{1,6,8,14,15,18,23,24\}$	no	21	$\{3,6,8,10,15,24\}$	yes
11	$\{2,4,7,14,23\}$	yes	22	$\{3,6,8,15,18,23,24\}$	no

S2.12. All states of $L^{(4)}_{22,9}$

TABLE S57: All two-valued states of $L^{(4)}_{22,9}$. E indicates extendability.

j	Support	E	j	Support	E
1	$\{1,6,9,13,16,21\}$	yes	80	$\{2,8,12,16,17,20,21\}$	no
2	$\{1,6,9,13,16,22\}$	yes	81	$\{2,8,12,16,17,20,22\}$	no
3	$\{1,6,9,14,16,19\}$	yes	82	$\{2,8,12,16,18,19\}$	yes
4	$\{1,6,9,14,16,20,21\}$	no	83	$\{2,8,12,16,18,20,21\}$	no
5	$\{1,6,9,14,16,20,22\}$	no	84	$\{2,8,12,16,18,20,22\}$	no
6	$\{1,6,10,13,16,21\}$	yes	85	$\{3,5,9,13,17,21\}$	yes
7	$\{1,6,10,13,16,22\}$	yes	86	$\{3,5,9,13,17,22\}$	yes
8	$\{1,6,10,14,16,19\}$	yes	87	$\{3,5,9,13,18,21\}$	yes
9	$\{1,6,10,14,16,20,21\}$	no	88	$\{3,5,9,13,18,22\}$	yes
10	$\{1,6,10,14,16,20,22\}$	no	89	$\{3,5,9,14,17,19\}$	yes
11	$\{1,7,9,13,17,21\}$	yes	90	$\{3,5,9,14,17,20,21\}$	no
12	$\{1,7,9,13,17,22\}$	yes	91	$\{3,5,9,14,17,20,22\}$	no
13	$\{1,7,9,13,18,21\}$	yes	92	$\{3,5,9,14,18,19\}$	yes
14	$\{1,7,9,13,18,22\}$	yes	93	$\{3,5,9,14,18,20,21\}$	no
15	$\{1,7,9,14,17,19\}$	yes	94	$\{3,5,9,14,18,20,22\}$	no
16	$\{1,7,9,14,17,20,21\}$	no	95	$\{3,5,10,13,17,21\}$	yes
17	$\{1,7,9,14,17,20,22\}$	no	96	$\{3,5,10,13,17,22\}$	yes
18	$\{1,7,9,14,18,19\}$	yes	97	$\{3,5,10,13,18,21\}$	yes
19	$\{1,7,9,14,18,20,21\}$	no	98	$\{3,5,10,13,18,22\}$	yes
20	$\{1,7,9,14,18,20,22\}$	no	99	$\{3,5,10,14,17,19\}$	yes
21	$\{1,7,10,13,17,21\}$	yes	100	$\{3,5,10,14,17,20,21\}$	no
22	$\{1,7,10,13,17,22\}$	yes	101	$\{3,5,10,14,17,20,22\}$	no
23	$\{1,7,10,13,18,21\}$	yes	102	$\{3,5,10,14,18,19\}$	yes
24	$\{1,7,10,13,18,22\}$	yes	103	$\{3,5,10,14,18,20,21\}$	no
25	$\{1,7,10,14,17,19\}$	yes	104	$\{3,5,10,14,18,20,22\}$	no
26	$\{1,7,10,14,17,20,21\}$	no	105	$\{3,6,9,11,20\}$	yes
27	$\{1,7,10,14,17,20,22\}$	no	106	$\{3,6,9,12,13,21\}$	yes
28	$\{1,7,10,14,18,19\}$	yes	107	$\{3,6,9,12,13,22\}$	yes
29	$\{1,7,10,14,18,20,21\}$	no	108	$\{3,6,9,12,14,19\}$	yes
30	$\{1,7,10,14,18,20,22\}$	no	109	$\{3,6,9,12,14,20,21\}$	no
31	$\{1,8,13,16,17,21\}$	yes	110	$\{3,6,9,12,14,20,22\}$	no
32	$\{1,8,13,16,17,22\}$	yes	111	$\{3,6,10,11,20\}$	yes
33	$\{1,8,13,16,18,21\}$	yes	112	$\{3,6,10,12,13,21\}$	yes
34	$\{1,8,13,16,18,22\}$	yes	113	$\{3,6,10,12,13,22\}$	yes
35	$\{1,8,14,15,21\}$	yes	114	$\{3,6,10,12,14,19\}$	yes
36	$\{1,8,14,15,22\}$	yes	115	$\{3,6,10,12,14,20,21\}$	no
37	$\{1,8,14,16,17,19\}$	yes	116	$\{3,6,10,12,14,20,22\}$	no
38	$\{1,8,14,16,17,20,21\}$	no	117	$\{3,8,11,17,20\}$	yes
39	$\{1,8,14,16,17,20,22\}$	no	118	$\{3,8,11,18,20\}$	yes

j	Support	E	j	Support	E
40	$\{1,8,14,16,18,19\}$	yes	119	$\{3,8,12,13,17,21\}$	yes
41	$\{1,8,14,16,18,20,21\}$	no	120	$\{3,8,12,13,17,22\}$	yes
42	$\{1,8,14,16,18,20,22\}$	no	121	$\{3,8,12,13,18,21\}$	yes
43	$\{2,5,9,15,21\}$	yes	122	$\{3,8,12,13,18,22\}$	yes
44	$\{2,5,9,15,22\}$	yes	123	$\{3,8,12,14,17,19\}$	yes
45	$\{2,5,9,16,17,19\}$	yes	124	$\{3,8,12,14,17,20,21\}$	no
46	$\{2,5,9,16,17,20,21\}$	no	125	$\{3,8,12,14,17,20,22\}$	no
47	$\{2,5,9,16,17,20,22\}$	no	126	$\{3,8,12,14,18,19\}$	yes
48	$\{2,5,9,16,18,19\}$	yes	127	$\{3,8,12,14,18,20,21\}$	no
49	$\{2,5,9,16,18,20,21\}$	no	128	$\{3,8,12,14,18,20,22\}$	no
50	$\{2,5,9,16,18,20,22\}$	no	129	$\{4,5,13,16,17,21\}$	yes
51	$\{2,5,10,15,21\}$	yes	130	$\{4,5,13,16,17,22\}$	yes
52	$\{2,5,10,15,22\}$	yes	131	$\{4,5,13,16,18,21\}$	yes
53	$\{2,5,10,16,17,19\}$	yes	132	$\{4,5,13,16,18,22\}$	yes
54	$\{2,5,10,16,17,20,21\}$	no	133	$\{4,5,14,15,21\}$	yes
55	$\{2,5,10,16,17,20,22\}$	no	134	$\{4,5,14,15,22\}$	yes
56	$\{2,5,10,16,18,19\}$	yes	135	$\{4,5,14,16,17,19\}$	yes
57	$\{2,5,10,16,18,20,21\}$	no	136	$\{4,5,14,16,17,20,21\}$	no
58	$\{2,5,10,16,18,20,22\}$	no	137	$\{4,5,14,16,17,20,22\}$	no
59	$\{2,6,9,12,16,19\}$	yes	138	$\{4,5,14,16,18,19\}$	yes
60	$\{2,6,9,12,16,20,21\}$	no	139	$\{4,5,14,16,18,20,21\}$	no
61	$\{2,6,9,12,16,20,22\}$	no	140	$\{4,5,14,16,18,20,22\}$	no
62	$\{2,6,10,12,16,19\}$	yes	141	$\{4,6,11,16,20\}$	yes
63	$\{2,6,10,12,16,20,21\}$	no	142	$\{4,6,12,13,16,21\}$	yes
64	$\{2,6,10,12,16,20,22\}$	no	143	$\{4,6,12,13,16,22\}$	yes
65	$\{2,7,9,12,17,19\}$	yes	144	$\{4,6,12,14,16,19\}$	yes
66	$\{2,7,9,12,17,20,21\}$	no	145	$\{4,6,12,14,16,20,21\}$	no
67	$\{2,7,9,12,17,20,22\}$	no	146	$\{4,6,12,14,16,20,22\}$	no
68	$\{2,7,9,12,18,19\}$	yes	147	$\{4,7,11,17,20\}$	yes
69	$\{2,7,9,12,18,20,21\}$	no	148	$\{4,7,11,18,20\}$	yes
70	$\{2,7,9,12,18,20,22\}$	no	149	$\{4,7,12,13,17,21\}$	yes
71	$\{2,7,10,12,17,19\}$	yes	150	$\{4,7,12,13,17,22\}$	yes
72	$\{2,7,10,12,17,20,21\}$	no	151	$\{4,7,12,13,18,21\}$	yes
73	$\{2,7,10,12,17,20,22\}$	no	152	$\{4,7,12,13,18,22\}$	yes
74	$\{2,7,10,12,18,19\}$	yes	153	$\{4,7,12,14,17,19\}$	yes
75	$\{2,7,10,12,18,20,21\}$	no	154	$\{4,7,12,14,17,20,21\}$	no
76	$\{2,7,10,12,18,20,22\}$	no	155	$\{4,7,12,14,17,20,22\}$	no
77	$\{2,8,12,15,21\}$	yes	156	$\{4,7,12,14,18,19\}$	yes
78	$\{2,8,12,15,22\}$	yes	157	$\{4,7,12,14,18,20,21\}$	no
79	$\{2,8,12,16,17,19\}$	yes	158	$\{4,7,12,14,18,20,22\}$	no

S2.13. Complete normalized colorings of the gadget panels

Each color word gives one representative under global color permutation; the vertices and the anchor convention are specified separately for each table. The full labeled family is recovered by all palette permutations. Equal color labels across different colorings carry no extra constraint.

S2.14. Colorings of P_5

The columns of each word correspond, in order, to vertices $1,2,3,4,5,6,7,8,9,10$. The anchor context $(1,6,2)$ has colors $(0,\dots,2)$.

TABLE S58: Every normalized coloring of P_5.

j	Color word	j	Color word
1	0201211201	4	0210210211
2	0202111102	5	0212110002
3	0210110222		

S2.15. Colorings of $B_{13,7}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 5, 7, 8, 9, 10, 11, 14, 15, 19, 23$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S59: Every normalized coloring of $B_{13,7}$.

j	Color word	j	Color word
1	1022012010221	4	2012012020111
2	1022210201101	5	2012012120100
3	1022210210001	6	2012210102102

S2.16. Colorings of $D_{16,9}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 18, 23, 24$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S60: Every normalized coloring of $D_{16,9}$.

j	Color word	j	Color word
1	1020210121102202	5	2010210121201201
2	1020212101100222	6	2011200120201210
3	1020212102100112	7	2011202102021012
4	1021202102011012		

S2.17. Colorings of $I_{16,9}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 14, 15, 18, 19, 23, 24$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S61: Every normalized coloring of $I_{16,9}$.

j	Color word	j	Color word
1	1020221021001012	4	2010201212012001
2	1021221020110012	5	2011201202012110
3	1022201201021210	6	2012221010210021

S2.18. Colorings of $T_{19,11}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 18, 23, 24, 26, 27, 28$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S62: Every normalized coloring of $T_{19,11}$.

j	Color word	j	Color word
1	1020210121102202120	5	2010210121201201120
2	1020212101100222120	6	2011200120201210120
3	1020212102100112210	7	2011202102021012102
4	1021202102011012201		

S2.19. Colorings of $H_{21,13}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 18, 19, 20, 21, 22, 23, 24$. The anchor context $(2, 8, 5)$ has colors $(0, \ldots, 2)$.

TABLE S63: Every normalized coloring of $H_{21,13}$.

j	Color word	j	Color word
1	102021210210001021212	2	201120012021012102210

S2.20. Colorings of $W^{(4)}_{11,5}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$. The anchor context $(0, 1, 6, 2)$ has colors $(0, \ldots, 3)$.

TABLE S64: Every normalized coloring of $W^{(4)}_{11,5}$.

j	Color word	j	Color word
1	01312322312	4	01321321322
2	01313222213	5	01323221113
3	01321221333		

S2.21. Colorings of $W^{(4)}_{16,6}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$. The anchor context $(2, 0, 6, 1)$ has colors $(0, \ldots, 3)$.

TABLE S65: Every normalized coloring of $W^{(4)}_{16,6}$.

j	Color word	j	Color word
1	1302302101123322	41	1302022331003131
2	1302302101132322	42	1302022331030131
3	1302302101223321	43	1302022331103130
4	1302302101232321	44	1302022331130130
5	1302302102123312	45	1302022333003111
6	1302302102132312	46	1302022333030111
7	1302302102223311	47	1302022333103110
8	1302302102232311	48	1302022333130110
9	1302302301123122	49	1302322100003311
10	1302302301132122	50	1302322100030311
11	1302302301223121	51	1302322100103310
12	1302302301232121	52	1302322100130310
13	1302302302123112	53	1302322101003301
14	1302302302132112	54	1302322101030301
15	1302302302223111	55	1302322101103300
16	1302302302232111	56	1302322101130300
17	1303202101123232	57	1302322300003111
18	1303202101132232	58	1302322300030111
19	1303202101223231	59	1302322300103110
20	1303202101232231	60	1302322300130110
21	1303202103123212	61	1302322301003101
22	1303202103132212	62	1302322301030101
23	1303202103223211	63	1302322301103100
24	1303202103232211	64	1302322301130100
25	1303202201123132	65	1303022121003231
26	1303202201132132	66	1303022121030231
27	1303202201223131	67	1303022121103230
28	1303202201232131	68	1303022121130230
29	1303202203123112	69	1303022123003211

j	Color word	j	Color word
30	1303202203132112	70	1303022123030211
31	1303202203223111	71	1303022123103210
32	1303202203232111	72	1303022123130210
33	1302022131003331	73	1303022221003131
34	1302022131030331	74	1303022221030131
35	1302022131103330	75	1303022221103130
36	1302022131130330	76	1303022221130130
37	1302022133003311	77	1303022223003111
38	1302022133030311	78	1303022223030111
39	1302022133103310	79	1303022223103110
40	1302022133130310	80	1303022223130110

S2.22. Colorings of $B^{(4)}_{14,7}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 5, 7, 8, 9, 10, 11, 14, 15, 19, 23$. The anchor context $(0, 1, 2, 3)$ has colors $(0, \ldots, 3)$.

TABLE S66: Every normalized coloring of $B^{(4)}_{14,7}$.

j	Color word	j	Color word
1	01233213212331	4	01231132321321
2	01233312312221	5	01231231212333
3	01233312321121	6	01231231312322

S2.23. Colorings of $L^{(4)}_{18,7}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18$. The anchor context $(1, 2, 4, 3)$ has colors $(0, \ldots, 3)$.

TABLE S67: Every normalized coloring of $L^{(4)}_{18,7}$.

j	Color word	j	Color word
1	013212030123031203	73	013210230132021023
2	013212030123031230	74	013210230132021032
3	013212030123301203	75	013210230132201023
4	013212030123301230	76	013210230132201032
5	013212031023031203	77	013210231032021023
6	013212031023031230	78	013210231032021032
7	013212031023301203	79	013210231032201023
8	013212031023301230	80	013210231032201032
9	013210230123031023	81	013213201332020112
10	013210230123031032	82	013213201332020121
11	013210230123301023	83	013213201332200112
12	013210230123301032	84	013213201332200121
13	013210231023031023	85	013213203132020112
14	013210231023031032	86	013213203132020121
15	013210231023301023	87	013213203132200112
16	013210231023301032	88	013213203132200121
17	013213201323030112	89	013213201332021002
18	013213201323030121	90	013213201332021020
19	013213201323300112	91	013213201332201002
20	013213201323300121	92	013213201332201020
21	013213203123030112	93	013213203132021002
22	013213203123030121	94	013213203132021020
23	013213203123300112	95	013213203132201002
24	013213203123300121	96	013213203132201020
25	013213201323031002	97	013221030131022103

j	Color word	j	Color word
26	013213201323031020	98	013221030131022130
27	013213201323301002	99	013221030131202103
28	013213201323301020	100	013221030131202130
29	013213203123031002	101	013221031031022103
30	013213203123031020	102	013221031031022130
31	013213203123301002	103	013221031031202103
32	013213203123301020	104	013221031031202130
33	013231201321030123	105	013223010331021202
34	013231201321030132	106	013223010331021220
35	013231201321300123	107	013223010331201202
36	013231201321300132	108	013223010331201220
37	013231203121030123	109	013223013031021202
38	013231203121030132	110	013223013031021220
39	013231203121300123	111	013223013031201202
40	013231203121300132	112	013223013031201220
41	013232101321030213	113	013223010331022101
42	013232101321030231	114	013223010331022110
43	013232101321300213	115	013223010331202101
44	013232101321300231	116	013223010331202110
45	013232103121030213	117	013223013031022101
46	013232103121030231	118	013223013031022110
47	013232103121300213	119	013223013031202101
48	013232103121300231	120	013223013031202110
49	013230210321031023	121	013220130131022013
50	013230210321031032	122	013220130131022031
51	013230210321301023	123	013220130131202013
52	013230210321301032	124	013220130131202031
53	013230213021031023	125	013220131031022013
54	013230213021031032	126	013220131031022031
55	013230213021301023	127	013220131031202013
56	013230213021301032	128	013220131031202031
57	013232010321031203	129	013223101331020212
58	013232010321031230	130	013223101331020221
59	013232010321301203	131	013223101331200212
60	013232010321301230	132	013223101331200221
61	013232013021031203	133	013223103131020212
62	013232013021031230	134	013223103131020221
63	013232013021301203	135	013223103131200212
64	013232013021301230	136	013223103131200221
65	013212030132021203	137	013223101331022001
66	013212030132021230	138	013223101331022010
67	013212030132201203	139	013223101331202001
68	013212030132201230	140	013223101331202010
69	013212031032021203	141	013223103131022001
70	013212031032021230	142	013223103131022010
71	013212031032201203	143	013223103131202001
72	013212031032201230	144	013223103131202010

S2.24. Colorings of $D^{(4)}_{17,9}$

The columns of each word correspond, in order, to vertices $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 18, 23, 24$. The anchor context $(0, 1, 2, 3)$ has colors $(0, \ldots, 3)$.

TABLE S68: Every normalized coloring of $D^{(4)}_{17,9}$.

j	Color word	j	Color word
1	01232132313123123	5	01233122312123132
2	01232312131123323	6	01231323123211213
3	01232313121122333	7	01233121321213231
4	01232313123122113		

S2.25. Colorings of $L^{(4)}_{22,9}$

The columns of each word correspond, in order, to vertices $1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22$. The anchor context $(1, 2, 4, 3)$ has colors $(0, \ldots, 3)$.

TABLE S69: Every normalized coloring of $L^{(4)}_{22,9}$.

j	Color word	j	Color word
1	0132120301230312033201	109	0132120310320212032301
2	0132120301230312033210	110	0132120310320212032310
3	0132120301230312303201	111	0132120310320212302301
4	0132120301230312303210	112	0132120310320212302310
5	0132120310230312033201	113	0132120301322012030312
6	0132120310230312033210	114	0132120301322012030321
7	0132120310230312303201	115	0132120301322012300312
8	0132120310230312303210	116	0132120301322012300321
9	0132120301233012030213	117	0132120310322012030312
10	0132120301233012030231	118	0132120310322012030321
11	0132120301233012300213	119	0132120310322012300312
12	0132120301233012300231	120	0132120310322012300321
13	0132120310233012030213	121	0132132013322001121302
14	0132120310233012030231	122	0132132013322001121320
15	0132120310233012300213	123	0132132013322001211302
16	0132120310233012300231	124	0132132013322001211320
17	0132132013233001121203	125	0132132031322001121302
18	0132132013233001121230	126	0132132031322001121320
19	0132132013233001211203	127	0132132031322001211302
20	0132132013233001211230	128	0132132031322001211320
21	0132132031233001121203	129	0132102301320210232301
22	0132132031233001121230	130	0132102301320210232310
23	0132132031233001211203	131	0132102301320210322301
24	0132132031233001211230	132	0132102301320210322310
25	0132102301230310233201	133	0132102310320210232301
26	0132102301230310233210	134	0132102310320210232310
27	0132102301230310323201	135	0132102310320210322301
28	0132102301230310323210	136	0132102310320210322310
29	0132102310230310233201	137	0132102301322010230312
30	0132102310230310233210	138	0132102301322010230321
31	0132102310230310323201	139	0132102301322010320312
32	0132102310230310323210	140	0132102301322010320321
33	0132102301233010230213	141	0132102310322010230312
34	0132102301233010230231	142	0132102310322010230321
35	0132102301233010320213	143	0132102310322010320312
36	0132102301233010320231	144	0132102310322010320321
37	0132102310233010230213	145	0132132013320210022301
38	0132102310233010230231	146	0132132013320210022310
39	0132102310233010320213	147	0132132013320210202301
40	0132102310233010320231	148	0132132013320210202310

j	Color word	j	Color word
41	013213201323031002 3201	149	013213203132021002 2301
42	013213201323031002 3210	150	013213203132021002 2310
43	013213201323031020 3201	151	013213203132021020 2301
44	013213201323031020 3210	152	013213203132021020 2310
45	013213203123031002 3201	153	013213201332201002 0312
46	013213203123031002 3210	154	013213201332201002 0321
47	013213203123031020 3201	155	013213201332201020 0312
48	013213203123031020 3210	156	013213201332201020 0321
49	013213201323301002 0213	157	013213203132201002 0312
50	013213201323301002 0231	158	013213203132201002 0321
51	013213201323301020 0213	159	013213203132201020 0312
52	013213201323301020 0231	160	013213203132201020 0321
53	013213203123301002 0213	161	013223010331021202 2301
54	013213203123301002 0231	162	013223010331021202 2310
55	013213203123301020 0213	163	013223010331021220 2301
56	013213203123301020 0231	164	013223010331021220 2310
57	013231201321300123 1203	165	013223013031021202 2301
58	013231201321300123 1230	166	013223013031021202 2310
59	013231201321300132 1203	167	013223013031021220 2301
60	013231201321300132 1230	168	013223013031021220 2310
61	013231203121300123 1203	169	013223010331201202 0312
62	013231203121300123 1230	170	013223010331201202 0321
63	013231203121300132 1203	171	013223010331201220 0312
64	013231203121300132 1230	172	013223010331201220 0321
65	013232101321300213 1203	173	013223013031201202 0312
66	013232101321300213 1230	174	013223013031201202 0321
67	013232101321300231 1203	175	013223013031201220 0312
68	013232101321300231 1230	176	013223013031201220 0321
69	013232103121300213 1203	177	013221030131022103 1302
70	013232103121300213 1230	178	013221030131022103 1320
71	013232103121300231 1203	179	013221030131022130 1302
72	013232103121300231 1230	180	013221030131022130 1320
73	013230210321031023 3201	181	013221031031022103 1302
74	013230210321031023 3210	182	013221031031022103 1320
75	013230210321031032 3201	183	013221031031022130 1302
76	013230210321031032 3210	184	013221031031022130 1320
77	013230213021031023 3201	185	013223010331022101 1302
78	013230213021031023 3210	186	013223010331022101 1320
79	013230213021031032 3201	187	013223010331022110 1302
80	013230213021031032 3210	188	013223010331022110 1320
81	013230210321301023 0213	189	013223013031022101 1302
82	013230210321301023 0231	190	013223013031022101 1320
83	013230210321301032 0213	191	013223013031022110 1302
84	013230210321301032 0231	192	013223013031022110 1320
85	013230213021301023 0213	193	013223101331200212 1302
86	013230213021301023 0231	194	013223101331200212 1320
87	013230213021301032 0213	195	013223101331200221 1302
88	013230213021301032 0231	196	013223101331200221 1320
89	013232010321031203 3201	197	013223103131200212 1302
90	013232010321031203 3210	198	013223103131200212 1320
91	013232010321031230 3201	199	013223103131200221 1302
92	013232010321031230 3210	200	013223103131200221 1320
93	013232013021031203 3201	201	013220130131022013 1302
94	013232013021031203 3210	202	013220130131022013 1320
95	013232013021031230 3201	203	013220130131022031 1302
96	013232013021031230 3210	204	013220130131022031 1320
97	013232010321301203 0213	205	013220131031022013 1302
98	013232010321301203 0231	206	013220131031022013 1320
99	013232010321301230 0213	207	013220131031022031 1302
100	013232010321301230 0231	208	013220131031022031 1320
101	013232013021301203 0213	209	013223101331022001 1302
102	013232013021301203 0231	210	013223101331022001 1320

TABLE S70. Ordinary states violating each directed truth implication for color-identical terminals. Both witness patterns and all context normalizations are verified.

Configuration	Pair	10 witness	01 witness
$T_{19,11}$	$(14, 28)$	$\{1, 5, 7, 14, 18, 23, 24\}$	$\{1, 6, 8, 10, 11, 27, 28\}$
$T_{19,11}$	$(18, 27)$	$\{1, 5, 7, 14, 18, 23, 24\}$	$\{1, 6, 8, 10, 11, 27, 28\}$
$H_{21,13}$	$(2, 14)$	$\{2, 6, 9, 11, 18, 19, 20, 21\}$	$\{1, 4, 8, 14, 15, 21, 22, 23\}$
$H_{21,13}$	$(5, 22)$	$\{3, 5, 9, 13, 15, 18, 19, 24\}$	$\{1, 4, 8, 14, 15, 21, 22, 23\}$
$H_{21,13}$	$(8, 23)$	$\{3, 6, 8, 10, 15, 20, 21, 24\}$	$\{1, 5, 7, 14, 18, 22, 23, 24\}$

j	Color word	j	Color word
103	0132320130213012300213	211	013223101331022010 1302
104	0132320130213012300231	212	013223101331022010 1320
105	0132120301320212032301	213	013223103131022001 1302
106	0132120301320212032310	214	013223103131022001 1320
107	0132120301320212302301	215	013223103131022010 1302
108	0132120301320212302310	216	013223103131022010 1320

TABLE S71: All ten connected, edge-faithful, colorable restrictions with a forced color identity in the specified inventory.

retained diagonals	rays	contexts	colorings	inseparable pairs
$\{10, 11\}$	17	10	4	$(10, 24)$, $(11, 18)$
$\{10, 12\}$	17	10	4	$(10, 16)$, $(12, 23)$
$\{10, 13\}$	17	10	4	$(10, 20)$, $(13, 19)$
$\{11, 12\}$	17	10	4	$(11, 25)$, $(12, 14)$
$\{11, 13\}$	17	10	4	$(11, 21)$, $(13, 15)$
$\{12, 13\}$	17	10	4	$(12, 22)$, $(13, 17)$
$\{10, 11, 12\}$	21	13	2	21 pairs
$\{10, 11, 13\}$	21	13	2	21 pairs
$\{10, 12, 13\}$	21	13	2	21 pairs
$\{11, 12, 13\}$	21	13	2	21 pairs

TABLE S72. The 24 states are indexed by (k, b), where $1 \le k \le 12$ and $b \in \{0, 1\}$. Their support on vertices $1, \ldots, 13$ is T_k when $b = 0$ and $T_k \cup \{\eta_k\}$ when $b = 1$. Equation (S7) determines all remaining values.

k	T_k	η_k
1	$\{1, 4, 8\}$	12
2	$\{1, 5, 7\}$	12
3	$\{1, 5, 9\}$	11
4	$\{1, 6, 8\}$	11
5	$\{2, 4, 7\}$	12
6	$\{2, 4, 9\}$	13
7	$\{2, 6, 7\}$	10
8	$\{2, 6, 9\}$	11
9	$\{3, 4, 8\}$	13
10	$\{3, 5, 7\}$	10
11	$\{3, 5, 9\}$	13
12	$\{3, 6, 8\}$	10

S2.26. An explicit state certificate for the completed Yu–Oh hypergraph

The first four contexts of Eq. (15) force exactly one of vertices 2, 5, and 8 to have value one. In the triad containing that coordinate vertex, both other vertices have value zero. In each of the other two triads, one of the two remaining vertices has value one. There are therefore $3 \cdot 2^2 = 12$ possible supports T_k on vertices $1, \ldots, 9$, listed in Table S72.

For each such support, exactly one of vertices $10, 11, 12, 13$ has no orthogonal neighbor in T_k; call it η_k. The other three are forced to zero, while $s(\eta_k)$ may be either zero or one. This gives precisely 24 assignments on the original 13 vertices. Every completion vertex w belongs to one context $\{i, j, w\}$ and has the uniquely determined value

$$s(w) = 1 - s(i) - s(j). \tag{S7}$$

The listed assignments have $s(i) + s(j) \leq 1$ on every original orthogonal pair, so these extensions are valid two-valued states. Conversely, every state of the completed hypergraph restricts to one of these assignments.

For a compact separation certificate, encode the profile of vertex i as the nonnegative integer

$$m_i = \sum_{k=1}^{12} \sum_{b=0}^{1} s_{k,b}(i)\, 2^{2(k-1)+b}.$$

In hexadecimal notation these 25 integers are

i	1	2	3	4	5
m_i	0000FF	00FF00	FF0000	030F03	3C003C

i	6	7	8	9	10
m_i	C0F0C0	0C330C	C300C3	30CC30	882000

i	11	12	13	14	15
m_i	0080A0	00020A	220800	007F5F	F34C53

i	16	17	18	19	20
m_i	CF31C5	3F0D35	74D0FC	77DF00	DDF700

i	21	22	23	24	25
m_i	D1C4F3	1D073F	4713CF	FC705C	00FDF5

(S8)

All 25 integers are nonzero and pairwise distinct, certifying unitality and separation. Notably, the Hamming weights (number of states in which $s(i) = 1$) partition the vertices into three classes: 8 for $i \in \{1, \ldots, 9\}$, 3 for $i \in \{10, \ldots, 13\}$, and 13 for the completion vertices $i \in \{14, \ldots, 25\}$ (with $8 + 3 + 13 = 24$ in each completion context), in contrast to the uniform state weight 2 across all 15 vertices of G_{32} and consistent with the fractional chromatic–clique gap $\chi_f(\Gamma(H_{\mathrm{YO}}^+)) = 35/11 > 3 = \omega(\Gamma(H_{\mathrm{YO}}^+))$.

S2.27. Hesse H_{37}: all states and separating profiles

The canonical states $s_1, \ldots, s_{153}$ are listed in Table S74. A two-valued coloring is a state with exactly one value one per context. Vertex numbers are those of Appendix S1.14 and Fig. 4: vertices $1, \ldots, 16$ form the core, and $17, \ldots, 37$ are completions. We list every state, a minimum separating family, and every vertex profile.

S2.27.1. State parametrization and all 27 branches

Choose $t = (t_0, t_1, t_2) \in \mathbb{Z}_3^3$ and $\varepsilon \in \{0, 1\}$. Put

$$s(M_{ak}) = \mathbf{1}_{k=t_a}, \qquad s(z) = \varepsilon, \tag{S9}$$

$$U(t) = \mathbb{Z}_3 \setminus \{t_0, t_1, t_2\}, \qquad W(t) = \mathbb{Z}_3 \setminus \{t_a - a : a \in \mathbb{Z}_3\}. \tag{S10}$$

For $\varepsilon = 0$, choose the values of P_k freely for $k \in U(t)$, with other P values zero. For $\varepsilon = 1$, all P values are zero. Independently, choose the values of Q_k freely for $k \in W(t)$, with other Q values zero. All completion values are forced:

$$s(A_k) = 1 - \varepsilon - s(P_k), \tag{S11}$$
$$s(B_{ak}) = 1 - s(M_{ak}) - s(P_k), \tag{S12}$$
$$s(C_{ak}) = 1 - s(M_{a,a+k}) - s(Q_k). \tag{S13}$$

Enumerate t lexicographically with $0 < 1 < 2$, then $\varepsilon = 1, 0$. Within each branch, list the available P bits in increasing k, followed by the available Q bits in increasing k, and order the resulting bit strings with 1 before 0. This rule fixes all state indices used in the text, tables, and verification data. Table S73 gives the index ranges of all 27 branches.

TABLE S73: All 27 state branches. The sets $U(t), W(t)$ identify the free core bits. The two choices of ε give $N_0 = 2^{|U|+|W|}$ and $N_1 = 2^{|W|}$ states; their index ranges refer to Table S74.

t	$U(t)$	$W(t)$	N_0	r for $\varepsilon = 0$	N_1	r for $\varepsilon = 1$
$(0,0,0)$	$\{1,2\}$	$\varnothing$	4	2–5	1	1
$(0,0,1)$	$\{2\}$	$\{1\}$	4	8–11	2	6–7
$(0,0,2)$	$\{1\}$	$\{1\}$	4	14–17	2	12–13
$(0,1,0)$	$\{2\}$	$\{2\}$	4	20–23	2	18–19
$(0,1,1)$	$\{2\}$	$\{1\}$	4	26–29	2	24–25
$(0,1,2)$	$\varnothing$	$\{1,2\}$	4	34–37	4	30–33
$(0,2,0)$	$\{1\}$	$\{2\}$	4	40–43	2	38–39
$(0,2,1)$	$\varnothing$	$\varnothing$	1	45	1	44
$(0,2,2)$	$\{1\}$	$\{2\}$	4	48–51	2	46–47
$(1,0,0)$	$\{2\}$	$\{0\}$	4	54–57	2	52–53
$(1,0,1)$	$\{2\}$	$\{0\}$	4	60–63	2	58–59
$(1,0,2)$	$\varnothing$	$\varnothing$	1	65	1	64
$(1,1,0)$	$\{2\}$	$\{2\}$	4	68–71	2	66–67
$(1,1,1)$	$\{0,2\}$	$\varnothing$	4	73–76	1	72
$(1,1,2)$	$\{0\}$	$\{2\}$	4	79–82	2	77–78
$(1,2,0)$	$\varnothing$	$\{0,2\}$	4	87–90	4	83–86
$(1,2,1)$	$\{0\}$	$\{0\}$	4	93–96	2	91–92
$(1,2,2)$	$\{0\}$	$\{2\}$	4	99–102	2	97–98
$(2,0,0)$	$\{1\}$	$\{0\}$	4	105–108	2	103–104
$(2,0,1)$	$\varnothing$	$\{0,1\}$	4	113–116	4	109–112
$(2,0,2)$	$\{1\}$	$\{1\}$	4	119–122	2	117–118
$(2,1,0)$	$\varnothing$	$\varnothing$	1	124	1	123
$(2,1,1)$	$\{0\}$	$\{1\}$	4	127–130	2	125–126
$(2,1,2)$	$\{0\}$	$\{1\}$	4	133–136	2	131–132
$(2,2,0)$	$\{1\}$	$\{0\}$	4	139–142	2	137–138
$(2,2,1)$	$\{0\}$	$\{0\}$	4	145–148	2	143–144
$(2,2,2)$	$\{0,1\}$	$\varnothing$	4	150–153	1	149
Total			99		54	

S2.27.2. Canonical two-valued colorings: all 153 states

Table S74 explicitly lists all 153 states. The last column is the complete binary vector in vertex order $1, \ldots, 37$, separated after the 16 core entries. Every row has context sum one in all 24 contexts. The support-size distribution is

m	12	13	14	15	16	18
$\#\{s : \lvert\operatorname{supp} s\rvert = m\}$	6	18	27	24	51	27

(S14)

TABLE S74: Canonical enumeration of all 153 two-valued states, in the canonical order defined above. The support is split into core and completion vertices. The last column is $(s_r(1), \ldots, s_r(16)\,;\, s_r(17), \ldots, s_r(37))$.

r	Core support	Completion support	$\lvert\operatorname{supp} s_r\rvert$	37-bit state vector
1	$\{1,2,5,8\}$	$\{21,22,24,25,27,28,30,31,32,33,35,37\}$	16	1100100100000000 0000110110110111110101
2	$\{2,5,8,12,13\}$	$\{17,30,31,32,33,35,37\}$	12	0100100100011000 1000000000000111110101
3	$\{2,5,8,12\}$	$\{17,19,22,25,28,30,31,32,33,35,37\}$	15	0100100100010000 1010010010011110101
4	$\{2,5,8,13\}$	$\{17,18,21,24,27,30,31,32,33,35,37\}$	15	0100100100001000 1100100100100111110101
5	$\{2,5,8\}$	$\{17,18,19,21,22,24,25,27,28,30,31,32,33,35,37\}$	18	0100100100000000 1110110110110111110101
6	$\{1,2,5,9,15\}$	$\{21,22,24,25,26,28,31,32,35\}$	14	1100100010000010 0000110111010011100100
7	$\{1,2,5,9\}$	$\{21,22,24,25,26,28,30,31,32,33,35,36\}$	16	1100100010000000 0000110111010111110110
8	$\{2,5,9,13,15\}$	$\{17,18,21,24,26,31,32,35\}$	13	0100100010001010 1100100101000011100100
9	$\{2,5,9,13\}$	$\{17,18,21,24,26,30,31,32,33,35,36\}$	15	0100100010001000 1100100101000011110110
10	$\{2,5,9,15\}$	$\{17,18,19,21,22,24,25,26,28,31,32,35\}$	16	0100100010000010 1110110111010011100100
11	$\{2,5,9\}$	$\{17,18,19,21,22,24,25,26,28,30,31,32,33,35,36\}$	18	0100100010000000 1110110111010111110110
12	$\{1,2,5,10,15\}$	$\{21,22,24,25,26,27,31,32,37\}$	14	1100100001000010 0000110111100011100001
13	$\{1,2,5,10\}$	$\{21,22,24,25,26,27,30,31,32,33,36,37\}$	16	1100100001000000 0000110111100111110011
14	$\{2,5,10,12,15\}$	$\{17,19,22,25,26,31,32,37\}$	13	0100100001010010 1010010011000011100001
15	$\{2,5,10,12\}$	$\{17,19,22,25,26,30,31,32,33,36,37\}$	15	0100100001010000 1010010011000111110011
16	$\{2,5,10,15\}$	$\{17,18,19,21,22,24,25,26,27,31,32,37\}$	16	0100100001000010 1110110111100011100001
17	$\{2,5,10\}$	$\{17,18,19,21,22,24,25,26,27,30,31,32,33,36,37\}$	18	0100100001000000 1110110111100111110011
18	$\{1,2,6,8,16\}$	$\{21,22,23,25,27,28,30,33,35\}$	14	1100010100000001 0000111010110101010100
19	$\{1,2,6,8\}$	$\{21,22,23,25,27,28,30,31,33,34,35,37\}$	16	1100010100000000 0000111010111011011101
20	$\{2,6,8,13,16\}$	$\{17,18,21,23,27,30,33,35\}$	13	0100010100001001 1100101000100101010100
21	$\{2,6,8,13\}$	$\{17,18,21,23,27,30,31,33,34,35,37\}$	15	0100010100001000 1100101000100111011101
22	$\{2,6,8,16\}$	$\{17,18,19,21,22,23,25,27,28,30,33,35\}$	16	0100010100000001 1110111010110101010100
23	$\{2,6,8\}$	$\{17,18,19,21,22,23,25,27,28,30,31,33,34,35,37\}$	18	0100010100000000 1110111010110111011101
24	$\{1,2,6,9,15\}$	$\{21,22,23,25,26,28,31,34,35\}$	14	1100010010000010 0000111010100100110 0
25	$\{1,2,6,9\}$	$\{21,22,23,25,26,28,30,31,33,34,35,36\}$	16	1100010010000000 0000111010110111011110
26	$\{2,6,9,13,15\}$	$\{17,18,21,23,26,31,34,35\}$	13	0100010010001010 1100101001000010011 00
27	$\{2,6,9,13\}$	$\{17,18,21,23,26,30,31,33,34,35,36\}$	15	0100010010001000 1100101001000111011110
28	$\{2,6,9,15\}$	$\{17,18,19,21,22,23,25,26,28,31,34,35\}$	16	0100010010000010 1110111010100100110 0
29	$\{2,6,9\}$	$\{17,18,19,21,22,23,25,26,28,30,31,33,34,35,36\}$	18	0100010010000000 1110111010110111011110
30	$\{1,2,6,10,15,16\}$	$\{21,22,23,25,26,27\}$	12	1100010001000011 0000111011100000000000
31	$\{1,2,6,10,15\}$	$\{21,22,23,25,26,27,31,34,37\}$	14	1100010001000010 0000111011100010010 01
32	$\{1,2,6,10,16\}$	$\{21,22,23,25,26,27,30,33,36\}$	14	1100010001000001 0000111011100100100 10
33	$\{1,2,6,10\}$	$\{21,22,23,25,26,27,30,31,33,34,36,37\}$	16	1100010001000000 0000111011100111011011
34	$\{2,6,10,15,16\}$	$\{17,18,19,21,22,23,25,26,27\}$	14	0100010001000011 1110111011100000000000
35	$\{2,6,10,15\}$	$\{17,18,19,21,22,23,25,26,27,31,34,37\}$	16	0100010001000010 1110111011100010010 01
36	$\{2,6,10,16\}$	$\{17,18,19,21,22,23,25,26,27,30,33,36\}$	16	0100010001000001 1110111011100100100 10
37	$\{2,6,10\}$	$\{17,18,19,21,22,23,25,26,27,30,31,33,34,36,37\}$	18	0100010001000000 1110111011100111011011
38	$\{1,2,7,8,16\}$	$\{21,22,23,24,27,28,30,32,35\}$	14	1100001100000001 0000111100110101010100
39	$\{1,2,7,8\}$	$\{21,22,23,24,27,28,30,31,32,34,35,37\}$	16	1100001100000000 0000111100111011101101
40	$\{2,7,8,12,16\}$	$\{17,19,22,23,28,30,32,35\}$	13	0100001100010001 1010011000010101010100
41	$\{2,7,8,12\}$	$\{17,19,22,23,28,30,31,32,34,35,37\}$	15	0100001100010000 1010011000010111011101
42	$\{2,7,8,16\}$	$\{17,18,19,21,22,23,24,27,28,30,32,35\}$	16	0100001100000001 1110111100110101010100
43	$\{2,7,8\}$	$\{17,18,19,21,22,23,24,27,28,30,31,32,34,35,37\}$	18	0100001100000000 1110111100111011101101
44	$\{1,2,7,9\}$	$\{21,22,23,24,26,28,30,31,32,34,35,36\}$	16	1100001010000000 0000111101010111011110
45	$\{2,7,9\}$	$\{17,18,19,21,22,23,24,26,28,30,31,32,34,35,36\}$	18	0100001010000000 1110111101010111011110
46	$\{1,2,7,10,16\}$	$\{21,22,23,24,26,27,30,32,36\}$	14	1100001001000001 0000111101100101001000
47	$\{1,2,7,10\}$	$\{21,22,23,24,26,27,30,31,32,34,36,37\}$	16	1100001001000000 0000111101100111011011
48	$\{2,7,10,12,16\}$	$\{17,19,22,23,26,30,32,36\}$	13	0100001001010001 1010011001000101000010
49	$\{2,7,10,12\}$	$\{17,19,22,23,26,30,31,32,34,36,37\}$	15	0100001001010000 1010011001000111011011
50	$\{2,7,10,16\}$	$\{17,18,19,21,22,23,24,26,27,30,32,36\}$	16	0100001001000001 1110111101100101001000
51	$\{2,7,10\}$	$\{17,18,19,21,22,23,24,26,27,30,31,32,34,36,37\}$	18	0100001001000000 1110111101100111011011
52	$\{1,3,5,8,14\}$	$\{20,22,24,25,27,28,31,33,37\}$	14	1010100100000100 0001010110110010100 01
53	$\{1,3,5,8\}$	$\{20,22,24,25,27,28,29,31,32,33,35,37\}$	16	1010100100000000 0001010110111011101 01
54	$\{3,5,8,13,14\}$	$\{17,18,20,24,27,31,33,37\}$	13	0010100100001100 1101000100100010100 01
55	$\{3,5,8,13\}$	$\{17,18,20,24,27,29,31,32,33,35,37\}$	15	0010100100001000 1101000100101011101 01
56	$\{3,5,8,14\}$	$\{17,18,19,20,22,24,25,27,28,31,33,37\}$	16	0010100100000100 1111010110110010100 01
57	$\{3,5,8\}$	$\{17,18,19,20,22,24,25,27,28,29,31,32,33,35,37\}$	18	0010100100000000 1111010110111011101 01
58	$\{1,3,5,9,14\}$	$\{20,22,24,25,26,28,31,33,36\}$	14	1010100010000100 0001010111010010100 10
59	$\{1,3,5,9\}$	$\{20,22,24,25,26,28,29,31,32,33,35,36\}$	16	1010100010000000 0001010111011011101 10
60	$\{3,5,9,13,14\}$	$\{17,18,20,24,26,31,33,36\}$	13	0010100010001100 1101000101000010100 10
61	$\{3,5,9,13\}$	$\{17,18,20,24,26,29,31,32,33,35,36\}$	15	0010100010001000 1101000101001011101 10
62	$\{3,5,9,14\}$	$\{17,18,19,20,22,24,25,26,28,31,33,36\}$	16	0010100010000100 1111010111010010100 10
63	$\{3,5,9\}$	$\{17,18,19,20,22,24,25,26,28,29,31,32,33,35,36\}$	18	0010100010000000 1111010111011011101 10
64	$\{1,3,5,10\}$	$\{20,22,24,25,26,27,29,31,32,33,36,37\}$	16	1010100001000000 0001010111101011101 11
65	$\{3,5,10\}$	$\{17,18,19,20,22,24,25,26,27,29,31,32,33,36,37\}$	18	0010100001000000 1111010111101011101 11
66	$\{1,3,6,8,16\}$	$\{20,22,23,25,27,28,29,33,35\}$	14	1010010100000001 0001011010110010100
67	$\{1,3,6,8\}$	$\{20,22,23,25,27,28,29,31,33,34,35,37\}$	16	1010010100000000 0001011010111011011 01

Continued on the next page.

TABLE S74: Canonical enumeration of all 153 two-valued states, in the canonical order defined above. The support is split into core and completion vertices. The last column is $(s_r(1), \ldots, s_r(16)\,;\, s_r(17), \ldots, s_r(37))$. (continued).

r	Core support	Completion support	$\lvert\operatorname{supp} s_r\rvert$	37-bit state vector
68	$\{3,6,8,13,16\}$	$\{17,18,20,23,27,29,33,35\}$	13	0010010100001001 1101001000101000010100
69	$\{3,6,8,13\}$	$\{17,18,20,23,27,29,31,33,34,35,37\}$	15	0010010100001000 1101001000101011101
70	$\{3,6,8,16\}$	$\{17,18,19,20,22,23,25,27,28,29,33,35\}$	16	0010010100000001 1111011010111000101000
71	$\{3,6,8\}$	$\{17,18,19,20,22,23,25,27,28,29,31,33,34,35,37\}$	18	0010010100000000 1111011010111010111101
72	$\{1,3,6,9\}$	$\{20,22,23,25,26,28,29,31,33,34,35,36\}$	16	1010010010000000 0001011011011010111110
73	$\{3,6,9,11,13\}$	$\{18,29,31,33,34,35,36\}$	12	0010010010101000 0100000000001010111110
74	$\{3,6,9,11\}$	$\{18,19,22,25,28,29,31,33,34,35,36\}$	15	0010010010100000 0110010010011010111110
75	$\{3,6,9,13\}$	$\{17,18,20,23,26,29,31,33,34,35,36\}$	15	0010010010001000 1101001000101011110
76	$\{3,6,9\}$	$\{17,18,19,20,22,23,25,26,28,29,31,33,34,35,36\}$	18	0010010010000000 1111011011011011110
77	$\{1,3,6,10,16\}$	$\{20,22,23,25,26,27,29,33,36\}$	14	1010010001000001 0001011011101000010010
78	$\{1,3,6,10\}$	$\{20,22,23,25,26,27,29,31,33,34,36,37\}$	16	1010010001000000 0001011011010101011011
79	$\{3,6,10,11,16\}$	$\{18,19,22,25,27,29,33,36\}$	13	0010010001100001 0110010010101000010010
80	$\{3,6,10,11\}$	$\{18,19,22,25,27,29,31,33,34,36,37\}$	15	0010010001100000 0110010010101011011
81	$\{3,6,10,16\}$	$\{17,18,19,20,22,23,25,26,27,29,33,36\}$	16	0010010001000001 1111011011101000010010
82	$\{3,6,10\}$	$\{17,18,19,20,22,23,25,26,27,29,31,33,34,36,37\}$	18	0010010001000000 1111011011101010111011
83	$\{1,3,7,8,14,16\}$	$\{20,22,23,24,27,28\}$	12	1010001100000101 0001011100110000000000
84	$\{1,3,7,8,14\}$	$\{20,22,23,24,27,28,31,34,37\}$	14	1010001100000100 0001011100110010010001
85	$\{1,3,7,8,16\}$	$\{20,22,23,24,27,28,29,32,35\}$	14	1010001100000001 0001011100111001001000
86	$\{1,3,7,8\}$	$\{20,22,23,24,27,28,29,31,32,34,35,37\}$	16	1010001100000000 0001011100111011011010
87	$\{3,7,8,14,16\}$	$\{17,18,19,20,22,23,24,27,28\}$	14	0010001100000101 1111011100110000000000
88	$\{3,7,8,14\}$	$\{17,18,19,20,22,23,24,27,28,31,34,37\}$	16	0010001100000100 1111011100110010010001
89	$\{3,7,8,16\}$	$\{17,18,19,20,22,23,24,27,28,29,32,35\}$	16	0010001100000001 1111011100111001001000
90	$\{3,7,8\}$	$\{17,18,19,20,22,23,24,27,28,29,31,32,34,35,37\}$	18	0010001100000000 1111011100111011011010
91	$\{1,3,7,9,14\}$	$\{20,22,23,24,26,28,31,34,36\}$	14	1010001010000100 0001011010100100101000
92	$\{1,3,7,9\}$	$\{20,22,23,24,26,28,29,31,32,34,35,36\}$	16	1010001010000000 0001011010110110111000
93	$\{3,7,9,11,14\}$	$\{18,19,22,24,28,31,34,36\}$	13	0010001010100100 0110010000100100101000
94	$\{3,7,9,11\}$	$\{18,19,22,24,28,29,31,32,34,35,36\}$	15	0010001010100000 0110010100011011011000
95	$\{3,7,9,14\}$	$\{17,18,19,20,22,23,24,26,28,31,34,36\}$	16	0010001010000100 1111011010010010001000
96	$\{3,7,9\}$	$\{17,18,19,20,22,23,24,26,28,29,31,32,34,35,36\}$	18	0010001010000000 1111011010110110111000
97	$\{1,3,7,10,16\}$	$\{20,22,23,24,26,27,29,32,36\}$	14	1010001001000001 0001011011010010001000
98	$\{1,3,7,10\}$	$\{20,22,23,24,26,27,29,31,32,34,36,37\}$	16	1010001001000000 0001011011010110111011
99	$\{3,7,10,11,16\}$	$\{18,19,22,24,27,29,32,36\}$	13	0010001001100001 0110010000100100100010
100	$\{3,7,10,11\}$	$\{18,19,22,24,27,29,31,32,34,36,37\}$	15	0010001001100000 0110010010101011011011
101	$\{3,7,10,16\}$	$\{17,18,19,20,22,23,24,26,27,29,32,36\}$	16	0010001001000001 1111011011010010001000
102	$\{3,7,10\}$	$\{17,18,19,20,22,23,24,26,27,29,31,32,34,36,37\}$	18	0010001001000000 1111011011010110111011
103	$\{1,4,5,8,14\}$	$\{20,21,24,25,27,28,30,33,37\}$	14	1001100100000100 0001100110110100100001
104	$\{1,4,5,8\}$	$\{20,21,24,25,27,28,29,30,32,33,35,37\}$	16	1001100100000000 0001100110111101101010
105	$\{4,5,8,12,14\}$	$\{17,19,20,25,28,30,33,37\}$	13	0001100100010100 1011000010010100100001
106	$\{4,5,8,12\}$	$\{17,19,20,25,28,29,30,32,33,35,37\}$	15	0001100100010000 1011000010011101101010
107	$\{4,5,8,14\}$	$\{17,18,19,20,21,24,25,27,28,30,33,37\}$	16	0001100100000100 1111100110110100100001
108	$\{4,5,8\}$	$\{17,18,19,20,21,24,25,27,28,29,30,32,33,35,37\}$	18	0001100100000000 1111100110111101101010
109	$\{1,4,5,9,14,15\}$	$\{20,21,24,25,26,28\}$	12	1001100010000110 0001100110100000000000
110	$\{1,4,5,9,14\}$	$\{20,21,24,25,26,28,30,33,36\}$	14	1001100010000100 0001100110110100100010
111	$\{1,4,5,9,15\}$	$\{20,21,24,25,26,28,29,32,35\}$	14	1001100010000010 0001100111011001001000
112	$\{1,4,5,9\}$	$\{20,21,24,25,26,28,29,30,32,33,35,36\}$	16	1001100010000000 0001100111011101101010
113	$\{4,5,9,14,15\}$	$\{17,18,19,20,21,24,25,26,28\}$	14	0001100010000110 1111100111010000000000
114	$\{4,5,9,14\}$	$\{17,18,19,20,21,24,25,26,28,30,33,36\}$	16	0001100010000100 1111100111010100100010
115	$\{4,5,9,15\}$	$\{17,18,19,20,21,24,25,26,28,29,32,35\}$	16	0001100010000010 1111100111011001001000
116	$\{4,5,9\}$	$\{17,18,19,20,21,24,25,26,28,29,30,32,33,35,36\}$	18	0001100010000000 1111100111011101101010
117	$\{1,4,5,10,15\}$	$\{20,21,24,25,26,27,29,32,37\}$	14	1001100001000010 0001100111010010001000
118	$\{1,4,5,10\}$	$\{20,21,24,25,26,27,29,30,32,33,36,37\}$	16	1001100001000000 0001100111011011011000
119	$\{4,5,10,12,15\}$	$\{17,19,20,25,26,29,32,37\}$	13	0001100001010010 1011000011001001000001
120	$\{4,5,10,12\}$	$\{17,19,20,25,26,29,30,32,33,36,37\}$	15	0001100001010000 1011000011001101101100
121	$\{4,5,10,15\}$	$\{17,18,19,20,21,24,25,26,27,29,32,37\}$	16	0001100001000010 1111100111010010001000
122	$\{4,5,10\}$	$\{17,18,19,20,21,24,25,26,27,29,30,32,33,36,37\}$	18	0001100001000000 1111100111011011011000
123	$\{1,4,6,8\}$	$\{20,21,23,25,27,28,29,30,33,34,35,37\}$	16	1001010100000000 0001101010111100111101
124	$\{4,6,8\}$	$\{17,18,19,20,21,23,25,27,28,29,30,33,34,35,37\}$	18	0001010100000000 1111101010111100111101
125	$\{1,4,6,9,15\}$	$\{20,21,23,25,26,28,29,34,35\}$	14	1001010010000010 0001101010110000110000
126	$\{1,4,6,9\}$	$\{20,21,23,25,26,28,29,30,33,34,35,36\}$	16	1001010010000000 0001101010111001111010
127	$\{4,6,9,11,15\}$	$\{18,19,21,25,28,29,34,35\}$	13	0001010010100010 0110100010011000011000
128	$\{4,6,9,11\}$	$\{18,19,21,25,28,29,30,33,34,35,36\}$	15	0001010010100000 0110100010011101111010
129	$\{4,6,9,15\}$	$\{17,18,19,20,21,23,25,26,28,29,34,35\}$	16	0001010010000010 1111101011011000011000
130	$\{4,6,9\}$	$\{17,18,19,20,21,23,25,26,28,29,30,33,34,35,36\}$	18	0001010010000000 1111101010110111001111
131	$\{1,4,6,10,15\}$	$\{20,21,23,25,26,27,29,34,37\}$	14	1001010001000010 0001101011010010001001
132	$\{1,4,6,10\}$	$\{20,21,23,25,26,27,29,30,33,34,36,37\}$	16	1001010001000000 0001101011011011011011
133	$\{4,6,10,11,15\}$	$\{18,19,21,25,27,29,34,37\}$	13	0001010001100010 0110100010101000010001

Continued on the next page.

TABLE S74: Canonical enumeration of all 153 two-valued states, in the canonical order defined above. The support is split into core and completion vertices. The last column is $(s_r(1), \ldots, s_r(16)\,;\, s_r(17), \ldots, s_r(37))$. (continued).

r	Core support	Completion support	$\lvert\operatorname{supp} s_r\rvert$	37-bit state vector
134	$\{4, 6, 10, 11\}$	$\{18, 19, 21, 25, 27, 29, 30, 33, 34, 36, 37\}$	15	0001010001100000 0110100010101100 11011
135	$\{4, 6, 10, 15\}$	$\{17, 18, 19, 20, 21, 23, 25, 26, 27, 29, 34, 37\}$	16	0001010001000010 1111101011101000 01001
136	$\{4, 6, 10\}$	$\{17, 18, 19, 20, 21, 23, 25, 26, 27, 29, 30, 33, 34, 36, 37\}$	18	0001010001000000 1111101011101100 11011
137	$\{1, 4, 7, 8, 14\}$	$\{20, 21, 23, 24, 27, 28, 30, 34, 37\}$	14	1001001100000100 0001101100110100 01001
138	$\{1, 4, 7, 8\}$	$\{20, 21, 23, 24, 27, 28, 29, 30, 32, 34, 35, 37\}$	16	1001001100000000 0001101100111110 10101
139	$\{4, 7, 8, 12, 14\}$	$\{17, 19, 20, 23, 28, 30, 34, 37\}$	13	0001001100010100 1011001000010100 01001
140	$\{4, 7, 8, 12\}$	$\{17, 19, 20, 23, 28, 29, 30, 32, 34, 35, 37\}$	15	0001001100010000 1011001000011101 01101
141	$\{4, 7, 8, 14\}$	$\{17, 18, 19, 20, 21, 23, 24, 27, 28, 30, 34, 37\}$	16	0001001100000100 1111101100110100 01001
142	$\{4, 7, 8\}$	$\{17, 18, 19, 20, 21, 23, 24, 27, 28, 29, 30, 32, 34, 35, 37\}$	18	0001001100000000 1111101100111101 01101
143	$\{1, 4, 7, 9, 14\}$	$\{20, 21, 23, 24, 26, 28, 30, 34, 36\}$	14	1001001010000100 0001101101010100 01010
144	$\{1, 4, 7, 9\}$	$\{20, 21, 23, 24, 26, 28, 29, 30, 32, 34, 35, 36\}$	16	1001001010000000 0001101101011101 01110
145	$\{4, 7, 9, 11, 14\}$	$\{18, 19, 21, 24, 28, 30, 34, 36\}$	13	0001001010100100 0110100100010100 01010
146	$\{4, 7, 9, 11\}$	$\{18, 19, 21, 24, 28, 29, 30, 32, 34, 35, 36\}$	15	0001001010100000 0110100100011101 01110
147	$\{4, 7, 9, 14\}$	$\{17, 18, 19, 20, 21, 23, 24, 26, 28, 30, 34, 36\}$	16	0001001010000100 1111101101010100 01010
148	$\{4, 7, 9\}$	$\{17, 18, 19, 20, 21, 23, 24, 26, 28, 29, 30, 32, 34, 35, 36\}$	18	0001001010000000 1111101101011101 01110
149	$\{1, 4, 7, 10\}$	$\{20, 21, 23, 24, 26, 27, 29, 30, 32, 34, 36, 37\}$	16	1001001001000000 0001101101011010 01011
150	$\{4, 7, 10, 11, 12\}$	$\{19, 29, 30, 32, 34, 36, 37\}$	12	0001001001110000 0010000000011010 10011
151	$\{4, 7, 10, 11\}$	$\{18, 19, 21, 24, 27, 29, 30, 32, 34, 36, 37\}$	15	0001001001100000 0110100100101101 01011
152	$\{4, 7, 10, 12\}$	$\{17, 19, 20, 23, 26, 29, 30, 32, 34, 36, 37\}$	15	0001001001010000 1011001001001101 01011
153	$\{4, 7, 10\}$	$\{17, 18, 19, 20, 21, 23, 24, 26, 27, 29, 30, 32, 34, 36, 37\}$	18	0001001001000000 1111101101101101 01011

S2.27.3. A minimum six-state unital, separating family

Define $\Omega_6 = \{\sigma_1, \ldots, \sigma_6\}$ in the following order:

$$(\sigma_1, \ldots, \sigma_6) = (s_3, s_{31}, s_{75}, s_{89}, s_{113}, s_{151}). \tag{S15}$$

Table S75 prints their complete supports. Table S76 gives all 37 profiles $(\sigma_1(i), \ldots, \sigma_6(i))$ and their active state sets. Every profile is nonzero and no two coincide, so Ω_6 is unital and separating. Since five states supply at most $2^5 - 1 = 31 < 37$ nonzero binary profiles, no smaller unital, separating family can exist.

TABLE S75: Complete supports and state vectors of the minimum unital, separating family Ω_6. The full state indices are those of Table S74.

State	Full index	Complete support	Weight	37-bit vector
σ_1	s_3	$\{2, 5, 8, 12, 17, 19, 22, 25, 28, 30, 31, 32, 33, 35, 37\}$	15	0100100100010000 1010010010011110101
σ_2	s_{31}	$\{1, 2, 6, 10, 15, 21, 22, 23, 25, 26, 27, 31, 34, 37\}$	14	1100010001000010 0000111011100010010 01
σ_3	s_{75}	$\{3, 6, 9, 13, 17, 18, 20, 23, 26, 29, 31, 33, 34, 35, 36\}$	15	0010010010001000 1101001001001011110
σ_4	s_{89}	$\{3, 7, 8, 16, 17, 18, 19, 20, 22, 23, 24, 27, 28, 29, 32, 35\}$	16	0010001100000001 1111011110011100100
σ_5	s_{113}	$\{4, 5, 9, 14, 15, 17, 18, 19, 20, 21, 24, 25, 26, 28\}$	14	0001100010000110 1111100111010000000
σ_6	s_{151}	$\{4, 7, 10, 11, 18, 19, 21, 24, 27, 29, 30, 32, 34, 36, 37\}$	15	0001001001100000 0110100100101101 01011

TABLE S76: All 37 vertex profiles on Ω_6, in two panels. Here $X_i^{(6)} = \{j : \sigma_j(i) = 1\}$, $W_6 = |X_i^{(6)}|$, and W_{153} is the weight across the full state space. The six-bit rows are nonzero and pairwise distinct.

i	Ray	$X_i^{(6)}$	Profile	W_6	W_{153}	i	Ray	$X_i^{(6)}$	Profile	W_6	W_{153}
1	z	$\{2\}$	010000	1	54	20	B_{00}	$\{3,4,5\}$	001110	3	86
2	M_{00}	$\{1,2\}$	110000	2	51	21	B_{01}	$\{2,5,6\}$	010011	3	86
3	M_{01}	$\{3,4\}$	001100	2	51	22	B_{02}	$\{1,2,4\}$	110100	3	86
4	M_{02}	$\{5,6\}$	000011	2	51	23	B_{10}	$\{2,3,4\}$	011100	3	86
5	M_{10}	$\{1,5\}$	100010	2	51	24	B_{11}	$\{4,5,6\}$	000111	3	86
6	M_{11}	$\{2,3\}$	011000	2	51	25	B_{12}	$\{1,2,5\}$	110010	3	86
7	M_{12}	$\{4,6\}$	000101	2	51	26	B_{20}	$\{2,3,5\}$	011010	3	86
8	M_{20}	$\{1,4\}$	100100	2	51	27	B_{21}	$\{2,4,6\}$	010101	3	86
9	M_{21}	$\{3,5\}$	001010	2	51	28	B_{22}	$\{1,4,5\}$	100110	3	86
10	M_{22}	$\{2,6\}$	010001	2	51	29	C_{00}	$\{3,4,6\}$	001101	3	76
11	P_0	$\{6\}$	000001	1	16	30	C_{01}	$\{1,6\}$	100001	2	76
12	P_1	$\{1\}$	100000	1	16	31	C_{02}	$\{1,2,3\}$	111000	3	76
13	P_2	$\{3\}$	001000	1	16	32	C_{10}	$\{1,4,6\}$	100101	3	76
14	Q_0	$\{5\}$	000010	1	26	33	C_{11}	$\{1,3\}$	101000	2	76
15	Q_1	$\{2,5\}$	010010	2	26	34	C_{12}	$\{2,3,6\}$	011001	3	76
16	Q_2	$\{4\}$	000100	1	26	35	C_{20}	$\{1,3,4\}$	101100	3	76
17	A_0	$\{1,3,4,5\}$	101110	4	83	36	C_{21}	$\{3,6\}$	001001	2	76
18	A_1	$\{3,4,5,6\}$	001111	4	83	37	C_{22}	$\{1,2,6\}$	110001	3	76
19	A_2	$\{1,4,5,6\}$	100111	4	83						

S2.27.4. All full 153-state vertex profiles

Table S77 prints the transpose of the complete state table: the full 153-bit profile of every vertex, in three consecutive blocks of 51 states. Thus it also enumerates the complete active state set $X_i = \{s_r : s_r(i) = 1\}$ for every vertex: a one in position r means exactly $s_r \in X_i$. The row weights are 54 for z, 51 for M, 16 for P, 26 for Q, 83 for A, 86 for B, and 76 for C.

TABLE S77: Complete 153-bit vertex profiles, with consecutive 51-bit blocks. Every row is nonzero and all 37 rows are distinct.

i	Ray	W_{153}	$s_1 \ldots s_{51}$	$s_{52} \ldots s_{102}$	$s_{103} \ldots s_{153}$
1	z	54	100001100001100001100001100001111000011000010110000	110000110000101100001000011000011110000110000110000	110000111000011000010110000110000110000110000010000
2	M_{00}	51	111111111111111111111111111111111111111111111111111	000000000000000000000000000000000000000000000000000	000000000000000000000000000000000000000000000000000
3	M_{01}	51	000000000000000000000000000000000000000000000000000	111111111111111111111111111111111111111111111111111	000000000000000000000000000000000000000000000000000
4	M_{02}	51	000000000000000000000000000000000000000000000000000	000000000000000000000000000000000000000000000000000	111111111111111111111111111111111111111111111111111
5	M_{10}	51	111111111111111100000000000000000000000000000000000	111111111111100000000000000000000000000000000000000	111111111111111111000000000000000000000000000000000
6	M_{11}	51	000000000000000011111111111111111111100000000000000	000000000000001111111111111111100000000000000000000	000000000000000000111111111111100000000000000000000
7	M_{12}	51	000000000000000000000000000000001111111111111111111	000000000000000000000000000000011111111111111111111	000000000000000000000000000000011111111111111111111
8	M_{20}	51	111110000000000011111100000000000001111110000000000	111111000000001111110000000000111111110000000000000	111111000000000000110000000000011111100000000000000
9	M_{21}	51	000001111100000000000111111000000000000011000000000	000000111111000000011111000000000000111111000000000	000000111111100000001111110000000000011111100000000
10	M_{22}	51	000000000001111110000000000011111111000000011111111	000000000001100000000001111110000000000000111111111	000000000000111111000000011111100000000011111111111
11	P_0	16	000000000000000000000000000000000000000000000000000	000000000000000000000011000011000000000011000011000	000000000000000000000011000011000000000011000011000
12	P_1	16	011000000000110000000000000000000000011000000110000	000000000000000000000000000000000000000000000000000	001100000000000110000000000000000011000000001010000
13	P_2	16	010100011000000000110000110000000000000000000000000	001100001100000011000101000000000000000000000000000	000000000000000000000000000000000000000000000000000
14	Q_0	26	000000000000000000000000000000000000000000000000000	101010101010000000000000000000011001100101010000000	101010110011000000000000000000000010101010101000000
15	Q_1	26	000001010101010100000001010101100110000000000000000	000000000000000000000000000000000000000000000000000	000000101010101010100010101010101000000000000000000
16	Q_2	26	000000000000000010101000000010101010101000101010000	000000000000001010100000010101010101000000101010000	000000000000000000000000000000000000000000000000000
17	A_0	83	011110011110011110011110011110000111100111101001111	001111001111010011110001100001100001111000011000011	001111000011110011110100001100001100111100001100011
18	A_1	83	000110011100001100111100111100001111000011010000111	001111001111010011110111100111000011100111100111111	000011000011110000110100111001111000011001111001101
19	A_2	83	001010000110011110000110000110000111100111010011111	000011000011010000110010100111000011110011110011111	001111000011110011110100111001111000011001111011111
20	B_{00}	86	000000000000000000000000000000000000000000000000000	111111111111111111111111001111001111111111111001110011	111111111111111111111111001111001111111111111001110011
21	B_{01}	86	100111111111100111111111111111111111111001111110011	000000000000000000000000000000000000000000000000000000	110011111111111100111111111111111111001111111110101
22	B_{02}	86	101011100111111111100111100111111111111111111111111	110011100111111001110101111111111111111111111111111	000000000000000000000000000000000000000000000000000000
23	B_{10}	86	000000000000000011111111111111111111111111111111111	000000000000001111111001111001111111111001110011111	000000000000000000000011100111100111111111001110011
24	B_{11}	86	100111111111100110000000000000000000011001111110011	111111111111110000000000000001111111111111111111111	110011111111111100110000000000000110011111111101011
25	B_{12}	86	101011100111111111100111100111111111100000000000000	110011100111111001110101111111000000000000000000000	111111111111111111111111111111111111100000000000000000
26	B_{20}	86	000001111111111100000011111111111100000011111111111	000000111111110000001001110011000000011001110011	000000111111111111001100111100110000001100110011
27	B_{21}	86	100110000001100111111100000011111111100110011011111	111111000000111111110000011111111111110000011111111	110011000000011001111000000111111100110000001010111
28	B_{22}	86	101011100110000011001111001100000000111111110000000	110011100110011001110101000001111111111111000000000	111111111111100000011111111000000111111111111000000
29	C_{00}	76	000000000000000000000000000000000000000000000000000	010101010101111111111111111111111111001100110101011111	010101001100111111111111111111111111101010101011111111
30	C_{01}	76	111110101010101011111110101010011001111111111111111	000000000000000000000000000000000000000000000000000000	111111010101010101011101010101010111111111111111111
31	C_{02}	76	111111111111111110101011111110101010101010111010101	111111111111110101011111010101010101011111111010101	000000000000000000000000000000000000000000000000000000
32	C_{10}	76	111111111111111100000000000000000000011111111111111	010101010101100000000000000000000000000000000000000	010101001100111111111100000000000000000000000000000
33	C_{11}	76	111110101010101011111110101010011001100000000000000	111111111111111111111111111111111000000000000000000	111111010101010101011101010101010000000000000000000
34	C_{12}	76	000000000000000010101111111010101010101011010101	000000000000001010111111010101010101011111111010101	000000000000000000011111111111111111111111111111111
35	C_{20}	76	111111111110000011111111111100000000111111110000000	010101010101001111111111000000001100110101010000000	010101001100110000001111111100000001010101010100000
36	C_{21}	76	000000101010101000000010101001100110000001111111111	000000111111110000001111111111000000001111111111111	000000010101010101010001010101010100000011111111111
37	C_{22}	76	111110000001111110101010000000101010101010100010101	111111000000110101010000001010101010100000010101	111111000000001111111100000011111111111100000011111

S2.27.5. The six-state double cover

The fractional-coloring certificate uses a second six-state family,

$$\mathcal{R}_6 = \{s_2, s_{30}, s_{73}, s_{99}, s_{109}, s_{139}\}. \tag{S16}$$

Table S78 specifies these states in the branch parametrization. Each of the 37 vertices occurs in exactly two of their supports. Thus weights 1/2 give a fractional coloring of total weight three, and uniform weights 1/6 reproduce the maximally mixed probabilities 1/3 for every ray. The family Ω_6 in Eq. (S15) supplies the separating certificate; $\mathcal{R}_6$ supplies the double-cover certificate.

TABLE S78: Explicit six-state double cover $\mathcal{R}_6$. Omitted P, Q values are zero; completion values follow from the state equations.

j	State	t	ε	Active P subscripts	Active Q subscripts
1	s_2	$(0,0,0)$	0	$\{1,2\}$	$\varnothing$
2	s_{30}	$(0,1,2)$	1	$\varnothing$	$\{1,2\}$
3	s_{73}	$(1,1,1)$	0	$\{0,2\}$	$\varnothing$
4	s_{99}	$(1,2,2)$	0	$\{0\}$	$\{2\}$
5	s_{109}	$(2,0,1)$	1	$\varnothing$	$\{0,1\}$
6	s_{139}	$(2,2,0)$	0	$\{1\}$	$\{0\}$

S2.27.6. Order determination and exact verification

The companion file `sources/hesse/hesse37-canonical-evaluations.txt` records the 24 numbered contexts and all 153 binary state vectors. The parametrization in this appendix defines their canonical order independently of any external program.

Extend the canonical states to the 76 events of $L(H_{37})$ as in Sec. V A. If $X_x = \{r : s_r(x) = 1\}$, then the full profiles satisfy

$$X_x \subseteq X_y \quad \Longleftrightarrow \quad x \leq y. \tag{S17}$$

The verifier checks all $76^2 = 5{,}776$ ordered event pairs against the incidence order in Eq. (S27). Whenever $x \not\leq y$, the companion file `sources/hesse/hesse37-order-witnesses.csv` gives the least canonical index r with $s_r(x) = 1$ and $s_r(y) = 0$. Its event labels `a1`,...,`a37` denote atoms and `c1`,...,`c37` their complements; `0` and `1` denote the bounds. Every one of the 594 distinct nonorthogonal atom pairs also has a joint-one state. Thus the full family is order determining; the six-state family Ω_6 supplies the minimum unital, separating vertex representation in Eq. (S26). The distinct minima for event separation and order determination are certified in Appendix S3.3.

The verifier `hesse37-verify.py` uses exact rational arithmetic in $\mathbb{Q}(\zeta)$, with no numerical tolerance or external solver. It checks distinct rays, all 666 Hermitian inner products, the 24 bases, all graph triangles, linearity, the chromatic number and core criticality, all 153 states, the minimum separating family, both profile tables, the maximum squared ray overlap, the six-state double cover, and all event-order comparisons. It also compares the reconstructed state vectors with the archived enumeration. The accompanying machine-readable data give the same numbering and context ordering as the figure and these appendices. From the extracted joint project directory, run

```
python verification/verify-integrated.py
```

S2.28. Four-dimensional $H_{\mathrm{YO}}^{(4)}$ and H_{65}: states and profiles

This appendix provides the complete enumeration of all two-valued states on the four-dimensional examples $H_{\mathrm{YO}}^{(4)}$ and H_{65}: the 25 two-valued states of the non-linear warm-up hypergraph $H_{\mathrm{YO}}^{(4)}$ (Appendix S2.28.1), and the complete branch-by-branch parametrization of all $24 \times 2^{14} = 393{,}216$ two-valued states—together with the 48 canonical unital and separating states $s_{\alpha,\varepsilon}$ and the 65 vertex state-profiles—of the canonical linear hypergraph H_{65} (Appendix S2.28.2).

S2.28.1. Complete enumeration of the 25 two-valued states on $H_{\mathrm{YO}}^{(4)}$

Table S79 lists all 25 two-valued states $s_1, \dots, s_{25} : V_{26} \to \{0,1\}$ on $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$. States $s_1, \dots, s_{24}$ have $s_r(26) = 0$ and correspond to the 24 two-valued states of the completed Yu–Oh hypergraph H_{YO}^{+} in $\mathbb{R}^3$ (grouped into 12 pairs of states sharing the same support on the three coordinate contexts $\{1,2,3\}$, $\{4,5,6\}$, $\{7,8,9\}$), while s_{25} is the unique apex state with support $\{26\}$. Table S80 displays the transposed 26×25 vertex-by-state binary matrix, listing for each vertex $i \in \{1, \dots, 26\}$ its state weight $\sum_{r=1}^{25} s_r(i) \in \{1, 3, 8, 13\}$, the set of states $A_i = \{s_r : s_r(i) = 1\}$, and its 25-bit profile $(s_1(i), \dots, s_{25}(i))$.

Inspection of Table S80 highlights three structural properties of the state space $\mathcal{S}(H_{\mathrm{YO}}^{(4)})$:

1. *Unitality and atom separation:* Every row of Table S80 has positive weight (1, 3, 8, or 13) and all 26 binary rows are distinct ($A_i \neq A_j$ for all $1 \le i < j \le 26$), so $\mathcal{S}(H_{\mathrm{YO}}^{(4)})$ is nonempty, unital, and separating on atoms.

2. *Exclusion of simultaneous* 1*s on the non-orthogonal triad atoms* $\{10, 11, 12, 13\}$*:* Although the four Yu–Oh triad rays $v_{10}, v_{11}, v_{12}, v_{13}$ are pairwise non-orthogonal ($v_i \cdot v_j = \pm 1 \neq 0$ for $10 \le i < j \le 13$), their active state sets in Table S80 are pairwise disjoint:

$$\begin{aligned} A_{10} &= \{s_{13}, s_{19}, s_{23}\}, \quad A_{11} = \{s_5, s_7, s_{15}\}, \\ A_{12} &= \{s_1, s_3, s_9\}, \qquad A_{13} = \{s_{11}, s_{17}, s_{21}\}. \end{aligned} \tag{S18}$$

   Consequently, $\sum_{i=10}^{13} s(i) \le 1$ holds for every $s \in \mathcal{S}(H_{\mathrm{YO}}^{(4)})$, and no two-valued state assigns $(1,1)$ to any of the $\binom{4}{2} = 6$ non-orthogonal pairs $\{10,11\}$, $\{10,12\}$, $\{10,13\}$, $\{11,12\}$, $\{11,13\}$, $\{12,13\}$.

3. *Failure of order-determinacy:* Even though $A_i \neq A_j$ for all $i \neq j$, the state space $\mathcal{S}(H_{\mathrm{YO}}^{(4)})$ is not order-determining on atoms or compound propositions [3]. In particular, each triad atom $t \in \{10, 11, 12, 13\}$ (*weight* 3) is strictly dominated across all 25 states ($A_t \subsetneq A_w$, i.e., $s(t) \le s(w)$ for all $s \in \mathcal{S}(H_{\mathrm{YO}}^{(4)})$) by three non-orthogonal completion atoms $w \in \{14, \dots, 25\}$ (*weight* 13):

$$\begin{aligned} A_{10} &\subsetneq A_{16},\, A_{20},\, A_{24}, \quad A_{11} \subsetneq A_{18},\, A_{21},\, A_{25}, \\ A_{12} &\subsetneq A_{14},\, A_{22},\, A_{23}, \quad A_{13} \subsetneq A_{15},\, A_{17},\, A_{19}, \end{aligned} \tag{S19}$$

   together with compound-event inclusions such as $A_{12} \subsetneq A_3 \dot\cup A_{14}$, $A_{11} \subsetneq A_3 \dot\cup A_{25}$, and their unions with the apex set $A_{26} = \{s_{25}\}$.

TABLE S79: Complete enumeration of the 25 two-valued states $s_1, \dots, s_{25} : V_{26} \to \{0,1\}$ on the non-linear warm-up hypergraph $H_{\mathrm{YO}}^{(4)} = (V_{26}, E_{16}^{(4)})$, showing for each state its support on the 13-ray Yu–Oh core $\{1, \dots, 13\}$, its forced support on $\{14, \dots, 26\}$, its total support size $|S_r|$, and its 26-bit binary vector $(s_r(1 \dots 13)\, s_r(14 \dots 25)\, s_r(26))$.

State	Core support	Completion/apex support	$\lvert S_r\rvert$	26-bit state vector
s_1	$\{1,4,8,12\}$	$\{14,15,21,22,23\}$	9	1001000100010 110000011100 0
s_2	$\{1,4,8\}$	$\{14,15,16,17,21,22,23,25\}$	11	1001000100000 111100011101 0
s_3	$\{1,5,7,12\}$	$\{14,18,22,23,24\}$	9	1000101000010 100010001110 0
s_4	$\{1,5,7\}$	$\{14,16,17,18,22,23,24,25\}$	11	1000101000000 101110001111 0
s_5	$\{1,5,9,11\}$	$\{17,18,21,22,25\}$	9	1000100010100 000110011001 0
s_6	$\{1,5,9\}$	$\{14,15,17,18,21,22,24,25\}$	11	1000100010000 110110011011 0
s_7	$\{1,6,8,11\}$	$\{16,18,21,23,25\}$	9	1000010100100 001010010101 0
s_8	$\{1,6,8\}$	$\{14,15,16,18,21,23,24,25\}$	11	1000010100000 111010010111 0
s_9	$\{2,4,7,12\}$	$\{14,19,20,22,23\}$	9	0101001000010 100001101100 0
s_{10}	$\{2,4,7\}$	$\{14,16,17,19,20,22,23,25\}$	11	0101001000000 101101101101 0
s_{11}	$\{2,4,9,13\}$	$\{14,15,17,19,25\}$	9	0101000010001 110101000001 0
s_{12}	$\{2,4,9\}$	$\{14,15,17,19,20,21,22,25\}$	11	0101000010000 110101111001 0
s_{13}	$\{2,6,7,10\}$	$\{14,16,20,24,25\}$	9	0100011001000 101000100011 0
s_{14}	$\{2,6,7\}$	$\{14,16,18,19,20,23,24,25\}$	11	0100011000000 101011100111 0
s_{15}	$\{2,6,9,11\}$	$\{18,19,20,21,25\}$	9	0100010010100 000011110001 0

Continued on the next page.

TABLE S79: All 25 states of $H_{\mathrm{YO}}^{(4)}$ (continued).

State	Core support	Completion/apex support	$\lvert S_r\rvert$	26-bit state vector
s_{16}	$\{2,6,9\}$	$\{14,15,18,19,20,21,24,25\}$	11	0100010010000 1100111100110
s_{17}	$\{3,4,8,13\}$	$\{15,16,17,19,23\}$	9	0011000100001 0111010001000
s_{18}	$\{3,4,8\}$	$\{15,16,17,19,20,21,22,23\}$	11	0011000100000 0111011111000
s_{19}	$\{3,5,7,10\}$	$\{16,17,20,22,24\}$	9	0010101001000 0011001010100
s_{20}	$\{3,5,7\}$	$\{16,17,18,19,20,22,23,24\}$	11	0010101000000 0011111011100
s_{21}	$\{3,5,9,13\}$	$\{15,17,18,19,24\}$	9	0010100010001 0101110000100
s_{22}	$\{3,5,9\}$	$\{15,17,18,19,20,21,22,24\}$	11	0010100010000 0101111110100
s_{23}	$\{3,6,8,10\}$	$\{15,16,20,21,24\}$	9	0010010101000 0110001100100
s_{24}	$\{3,6,8\}$	$\{15,16,18,19,20,21,23,24\}$	11	0010010100000 0110111101100
s_{25}	$\emptyset$	$\{26\}$	1	0000000000000 0000000000001

TABLE S80: Vertex state profiles across the 25 two-valued states $s_1,\dots,s_{25}$ of $H_{\mathrm{YO}}^{(4)}$, showing for each vertex $i \in \{1,\dots,26\}$ its state weight $\sum_{r=1}^{25} s_r(i)$, the active state indices $\{r : s_r(i) = 1\}$, and the 25-bit profile $(s_{1\dots12}(i)\, s_{13\dots24}(i)\, s_{25}(i))$. All 26 rows are nonzero and pairwise distinct.

Vertex i	Weight $\sum_r s_r(i)$	Active states $\{r \in \{1,\dots,25\} : s_r(i) = 1\}$	25-bit profile $(s_{1\dots12}(i)\, s_{13\dots24}(i)\, s_{25}(i))$
1	8	$\{1,2,3,4,5,6,7,8\}$	1111111100000 0000000000000
2	8	$\{9,10,11,12,13,14,15,16\}$	0000000011110 1111000000000
3	8	$\{17,18,19,20,21,22,23,24\}$	0000000000000 0000111111110
4	8	$\{1,2,9,10,11,12,17,18\}$	1100000011110 0000110000000
5	8	$\{3,4,5,6,19,20,21,22\}$	0011110000000 0000001111000
6	8	$\{7,8,13,14,15,16,23,24\}$	0000001100000 1111000000110
7	8	$\{3,4,9,10,13,14,19,20\}$	0011000011000 1100001100000
8	8	$\{1,2,7,8,17,18,23,24\}$	1100001100000 0000110000110
9	8	$\{5,6,11,12,15,16,21,22\}$	0000110000110 0011000011000
10	3	$\{13,19,23\}$	0000000000000 1000001000100
11	3	$\{5,7,15\}$	0000101000000 0010000000000
12	3	$\{1,3,9\}$	1010000010000 0000000000000
13	3	$\{11,17,21\}$	0000000000100 0000100010000
14	13	$\{1,2,3,4,6,8,9,10,11,12,13,14,16\}$	1111010111110 1101000000000
15	13	$\{1,2,6,8,11,12,16,17,18,21,22,23,24\}$	1100010100110 0001110011110
16	13	$\{2,4,7,8,10,13,14,17,18,19,20,23,24\}$	0101001101000 1100111100110
17	13	$\{2,4,5,6,10,11,12,17,18,19,20,21,22\}$	0101110001110 0000111111000
18	13	$\{3,4,5,6,7,8,14,15,16,20,21,22,24\}$	0011111100000 0111000111010
19	13	$\{9,10,11,12,14,15,16,17,18,20,21,22,24\}$	0000000011110 0111110111010
20	13	$\{9,10,12,13,14,15,16,18,19,20,22,23,24\}$	0000000011010 1111011101110
21	13	$\{1,2,5,6,7,8,12,15,16,18,22,23,24\}$	1100111100010 0011010001110
22	13	$\{1,2,3,4,5,6,9,10,12,18,19,20,22\}$	1111110011010 0000011101000
23	13	$\{1,2,3,4,7,8,9,10,14,17,18,20,24\}$	1111001111000 1001101000100
24	13	$\{3,4,6,8,13,14,16,19,20,21,22,23,24\}$	0011010100001 1010011111110
25	13	$\{2,4,5,6,7,8,10,11,12,13,14,15,16\}$	0101111101111 1111000000000
26	1	$\{25\}$	0000000000000 0000000000001

S2.28.2. Complete parametrization of all 393,216 *two-valued states and the* 48 *canonical separating states on* H_{65}

Every two-valued state $s \in \mathcal{S}(H_{65})$ on the canonical linear hypergraph $H_{65} = (V_{65}, E_{29})$ is uniquely indexed by a pair $(\alpha, \boldsymbol{\varepsilon})$ consisting of:

1. a branch index $\alpha \in \{1,\dots,24\}$ selecting one of the 24 four-element core state supports $S_\alpha \subset V_{\mathrm{core}} = \{1,\dots,17\}$; and
2. a 14-bit binary vector $\boldsymbol{\varepsilon} = (\varepsilon_m)_{m\in\mathcal{F}_\alpha} \in \{0,1\}^{14}$ indexed by the 14 *free completion contexts*

$$\mathcal{F}_\alpha = \big\{m \in \{1,\dots,24\} : S_\alpha \cap \{u_m, v_m\} = \emptyset\big\},$$

whose complement is the set of 10 *hit completion contexts* $\mathcal{H}_\alpha = \{1,\dots,24\} \setminus \mathcal{F}_\alpha$.

Explicitly, the state $s_{\alpha,\boldsymbol{\varepsilon}} : V_{65} \to \{0,1\}$ takes values

$$\begin{aligned} s_{\alpha,\boldsymbol{\varepsilon}}(i) &= \mathbf{1}_{i\in S_\alpha} \qquad (1 \le i \le 17), \\ \big(s_{\alpha,\boldsymbol{\varepsilon}}(2m+16),\, s_{\alpha,\boldsymbol{\varepsilon}}(2m+17)\big) &= \begin{cases} (0,0), & m \in \mathcal{H}_\alpha, \\ (1-\varepsilon_m,\, \varepsilon_m), & m \in \mathcal{F}_\alpha. \end{cases} \end{aligned} \tag{S20}$$

Because $|\mathcal{F}_\alpha| = 14$ for every $\alpha \in \{1,\dots,24\}$, each branch α yields $2^{14} = 16{,}384$ states $s_{\alpha,\boldsymbol{\varepsilon}}$, giving $|\mathcal{S}(H_{65})| = 24 \times 2^{14} = 393{,}216$ two-valued states in total. Setting all 14 free bits equal to $\varepsilon \in \{0,1\}$ ($\varepsilon_m = \varepsilon$ for all $m \in \mathcal{F}_\alpha$) yields the 48 *canonical states* $s_{\alpha,0}, s_{\alpha,1}$ of Eq. (S29), which have support size $|S_\alpha| + 14 = 18$:

$$\begin{aligned} \mathrm{supp}(s_{\alpha,0}) &= S_\alpha \cup \big\{2m+16 : m \in \mathcal{F}_\alpha\big\}, \\ \mathrm{supp}(s_{\alpha,1}) &= S_\alpha \cup \big\{2m+17 : m \in \mathcal{F}_\alpha\big\}. \end{aligned} \tag{S21}$$

Table S81 enumerates all 24 branches $\alpha = 1,\dots,24$, specifying S_α, the 10 hit context indices $\mathcal{H}_\alpha \subset \{1,\dots,24\}$, the 14 free context indices $\mathcal{F}_\alpha \subset \{1,\dots,24\}$, and the 24-bit free-context indicator string. Table S82 lists the resulting state profiles and weights for all 65 vertices of H_{65} across both the 48 canonical states $\{s_{\alpha,\varepsilon}\}$ and the full space of 393,216 states: across the 48 canonical states $(s_{1,0}, s_{1,1}, \dots, s_{24,0}, s_{24,1})$, each '1' in row i of Table S82 expands to '(1,1)' for core vertices $i \in \{1,\dots,17\}$, to '(1,0)' for first completion vertices $w_{m,1} = 2m+16$, and to '(0,1)' for second completion vertices $w_{m,2} = 2m+17$ (and each '0' expands to '(0,0)'), making all 65 canonical 48-bit vertex profiles nonzero and pairwise distinct by inspection.

TABLE S81: Complete branch-by-branch parametrization of the $24 \times 2^{14} = 393{,}216$ two-valued states $s_{\alpha,\boldsymbol{\varepsilon}}$ and the 48 canonical states $s_{\alpha,0}, s_{\alpha,1}$ on H_{65} via Eqs. (S20)–(S21). For each $\alpha \in \{1,\dots,24\}$, the table lists the 4-element core support S_α, the 10 hit completion context indices $m \in \mathcal{H}_\alpha$ where $(s(2m+16), s(2m+17)) = (0,0)$ is forced on E_m, the 14 free completion context indices $m \in \mathcal{F}_\alpha$ where $(s(2m+16), s(2m+17)) = (1-\varepsilon_m, \varepsilon_m)$ with $\varepsilon_m \in \{0,1\}$ on E_m, and the 24-bit indicator string $(\mathbf{1}_{1\in\mathcal{F}_\alpha} \dots \mathbf{1}_{12\in\mathcal{F}_\alpha}\, \mathbf{1}_{13\in\mathcal{F}_\alpha} \dots \mathbf{1}_{24\in\mathcal{F}_\alpha})$.

Branch	Core support S_α	Hit contexts $\mathcal{H}_\alpha$	Free contexts $\mathcal{F}_\alpha$	Free indicator $m = 1\dots24$
S_1	{1,5,9,16}	{3,7,9,10,11,13,19,21,22,23}	{1,2,4,5,6,8,12,14,15,16,17,18,20,24}	110111010001 011111010001
S_2	{1,5,10,14}	{3,5,6,7,11,13,17,18,19,23}	{1,2,4,8,9,10,12,14,15,16,20,21,22,24}	110100011101 011100011101
S_3	{1,6,9,16}	{3,7,9,10,11,15,17,21,22,24}	{1,2,4,5,6,8,12,13,14,16,18,19,20,23}	110111010001 110101110010
S_4	{1,6,11,12}	{1,2,3,7,11,13,14,15,17,24}	{4,5,6,8,9,10,12,16,18,19,20,21,22,23}	000111011101 000101111110
S_5	{1,7,10,14}	{3,5,6,7,11,16,17,18,20,21}	{1,2,4,8,9,10,12,13,14,15,19,22,23,24}	110100011101 111000100111
S_6	{1,7,11,12}	{1,2,3,7,11,13,14,16,20,21}	{4,5,6,8,9,10,12,15,17,18,19,22,23,24}	000111011101 001011100111
S_7	{2,5,9,16}	{1,5,9,10,12,13,19,21,22,23}	{2,3,4,6,7,8,11,14,15,16,17,18,20,24}	011101110010 011111010001
S_8	{2,5,11,13}	{1,3,4,5,12,13,15,16,19,23}	{2,6,7,8,9,10,11,14,17,18,20,21,22,24}	010001111110 010011011101
S_9	{2,6,9,16}	{1,5,9,10,12,15,17,21,22,24}	{2,3,4,6,7,8,11,13,14,16,18,19,20,23}	011101110010 110101110010
S_{10}	{2,6,10,15}	{1,5,7,8,12,15,17,19,20,24}	{2,3,4,6,9,10,11,13,14,16,18,21,22,23}	011101001110 110101001110
S_{11}	{2,8,10,15}	{1,5,7,8,12,14,18,19,20,22}	{2,3,4,6,9,10,11,13,15,16,17,21,23,24}	011101001110 101110001011
S_{12}	{2,8,11,13}	{1,3,4,5,12,14,15,16,18,22}	{2,6,7,8,9,10,11,13,17,19,20,21,23,24}	010001111110 100010111011
S_{13}	{3,5,10,14}	{2,5,6,8,9,13,17,18,19,23}	{1,3,4,7,10,11,12,14,15,16,20,21,22,24}	101100100111 011100011101
S_{14}	{3,5,11,13}	{2,3,4,8,9,13,15,16,19,23}	{1,5,6,7,10,11,12,14,17,18,20,21,22,24}	100011100111 010011011101
S_{15}	{3,7,9,17}	{2,8,9,11,12,16,20,21,23,24}	{1,3,4,5,6,7,10,13,14,15,17,18,19,22}	101111100100 111011100100
S_{16}	{3,7,10,14}	{2,5,6,8,9,16,17,18,20,21}	{1,3,4,7,10,11,12,13,14,15,19,22,23,24}	101100100111 111000100111
S_{17}	{3,8,9,17}	{2,8,9,11,12,14,18,22,23,24}	{1,3,4,5,6,7,10,13,15,16,17,19,20,21}	101111100100 101110111000
S_{18}	{3,8,11,13}	{2,3,4,8,9,14,15,16,18,22}	{1,5,6,7,10,11,12,13,17,19,20,21,23,24}	100011100111 100010111011
S_{19}	{4,6,10,15}	{4,6,7,8,10,15,17,19,20,24}	{1,2,3,5,9,11,12,13,14,16,18,21,22,23}	111010001011 110101001110
S_{20}	{4,6,11,12}	{1,2,4,6,10,13,14,15,17,24}	{3,5,7,8,9,11,12,16,18,19,20,21,22,23}	001010111011 000101111110
S_{21}	{4,7,9,17}	{4,6,10,11,12,16,20,21,23,24}	{1,2,3,5,7,8,9,13,14,15,17,18,19,22}	111010111000 111011100100
S_{22}	{4,7,11,12}	{1,2,4,6,10,13,14,16,20,21}	{3,5,7,8,9,11,12,15,17,18,19,22,23,24}	001010111011 001011100111
S_{23}	{4,8,9,17}	{4,6,10,11,12,14,18,22,23,24}	{1,2,3,5,7,8,9,13,15,16,17,19,20,21}	111010111000 101110111000
S_{24}	{4,8,10,15}	{4,6,7,8,10,14,18,19,20,22}	{1,2,3,5,9,11,12,13,15,16,17,21,23,24}	111010001011 101110001011

TABLE S82: Vertex state profiles and weights for all 65 vertices of H_{65}, showing for each core vertex $i \in \{1,\dots,17\}$ and each completion pair $(w_{m,1}, w_{m,2}) = (2m+16, 2m+17)$ $(m = 1,\dots,24)$ the number of active branches W_{24}, the weight $W_{48} = \sum_{\alpha,\varepsilon} s_{\alpha,\varepsilon}(i)$ across the 48 canonical states, the weight $W_{393,216} = \sum_{s\in\mathcal{S}(H_{65})} s(i)$ across all 393,216 states, the active branch indices $\alpha \in \{1,\dots,24\}$ (equal to $Z(e_m)$ for $(w_{m,1}, w_{m,2})$), and the 24-bit branch profile across $S_1,\dots,S_{24}$ (in which each `1` expands to `11` for core vertices i, to `10` for $w_{m,1}$, and to `01` for $w_{m,2}$ across the 48 canonical states).

Vertex / pair	W_{24}	W_{48}	$W_{393,216}$	Active branches α	24-bit branch profile
Core 1	6	12	98,304	$\{1,2,3,4,5,6\}$	11111100 00000000 00000000
Core 2	6	12	98,304	$\{7,8,9,10,11,12\}$	00000011 11110000 00000000
Core 3	6	12	98,304	$\{13,14,15,16,17,18\}$	00000000 00001111 11000000
Core 4	6	12	98,304	$\{19,20,21,22,23,24\}$	00000000 00000000 00111111
Core 5	6	12	98,304	$\{1,2,7,8,13,14\}$	11000011 00001100 00000000
Core 6	6	12	98,304	$\{3,4,9,10,19,20\}$	00110000 11000000 00110000
Core 7	6	12	98,304	$\{5,6,15,16,21,22\}$	00001100 00000011 00001100
Core 8	6	12	98,304	$\{11,12,17,18,23,24\}$	00000000 00110000 11000011
Core 9	8	16	131,072	$\{1,3,7,9,15,17,21,23\}$	10100010 10000010 10001010
Core 10	8	16	131,072	$\{2,5,10,11,13,16,19,24\}$	01001000 01101001 00100001
Core 11	8	16	131,072	$\{4,6,8,12,14,18,20,22\}$	00010101 00010100 01010100
Core 12	4	8	65,536	$\{4,6,20,22\}$	00010100 00000000 00010100
Core 13	4	8	65,536	$\{8,12,14,18\}$	00000001 00010100 01000000
Core 14	4	8	65,536	$\{2,5,13,16\}$	01001000 00001001 00000000
Core 15	4	8	65,536	$\{10,11,19,24\}$	00000000 01100000 00100001
Core 16	4	8	65,536	$\{1,3,7,9\}$	10100010 10000000 00000000
Core 17	4	8	65,536	$\{15,17,21,23\}$	00000000 00000010 10001010
E_1 : $(18,19)$	14	14	114,688	$\{1,2,3,5,13,14,15,16,17,18,19,21,23,24\}$	11101000 00001111 11101011
E_2 : $(20,21)$	14	14	114,688	$\{1,2,3,5,7,8,9,10,11,12,19,21,23,24\}$	11101011 11110000 00101011
E_3 : $(22,23)$	14	14	114,688	$\{7,9,10,11,13,15,16,17,19,20,21,22,23,24\}$	00000010 11101011 10111111
E_4 : $(24,25)$	14	14	114,688	$\{1,2,3,4,5,6,7,9,10,11,13,15,16,17\}$	11111110 11101011 10000000
E_5 : $(26,27)$	14	14	114,688	$\{1,3,4,6,14,15,17,18,19,20,21,22,23,24\}$	10110100 00000110 11111111
E_6 : $(28,29)$	14	14	114,688	$\{1,3,4,6,7,8,9,10,11,12,14,15,17,18\}$	10110111 11110110 11000000
E_7 : $(30,31)$	14	14	114,688	$\{7,8,9,12,13,14,15,16,17,18,20,21,22,23\}$	00000011 10011111 11011110
E_8 : $(32,33)$	14	14	114,688	$\{1,2,3,4,5,6,7,8,9,12,20,21,22,23\}$	11111111 10010000 00011110
E_9 : $(34,35)$	14	14	114,688	$\{2,4,5,6,8,10,11,12,19,20,21,22,23,24\}$	01011101 01110000 00111111
E_{10} : $(36,37)$	14	14	114,688	$\{2,4,5,6,8,10,11,12,13,14,15,16,17,18\}$	01011101 01111111 11000000
E_{11} : $(38,39)$	14	14	114,688	$\{7,8,9,10,11,12,13,14,16,18,19,20,22,24\}$	00000011 11111101 01110101
E_{12} : $(40,41)$	14	14	114,688	$\{1,2,3,4,5,6,13,14,16,18,19,20,22,24\}$	11111100 00001101 01110101
E_{13} : $(42,43)$	14	14	114,688	$\{3,5,9,10,11,12,15,16,17,18,19,21,23,24\}$	00101000 11110011 11101011
E_{14} : $(44,45)$	14	14	114,688	$\{1,2,3,5,7,8,9,10,13,14,15,16,19,21\}$	11101011 11001111 00101000
E_{15} : $(46,47)$	14	14	114,688	$\{1,2,5,6,7,11,13,15,16,17,21,22,23,24\}$	11001110 00101011 10001111
E_{16} : $(48,49)$	14	14	114,688	$\{1,2,3,4,7,9,10,11,13,17,19,20,23,24\}$	11110010 11101000 10110011
E_{17} : $(50,51)$	14	14	114,688	$\{1,6,7,8,11,12,14,15,17,18,21,22,23,24\}$	10000111 00110110 11001111
E_{18} : $(52,53)$	14	14	114,688	$\{1,3,4,6,7,8,9,10,14,15,19,20,21,22\}$	10110111 11000110 00111100
E_{19} : $(54,55)$	14	14	114,688	$\{3,4,5,6,9,12,15,16,17,18,20,21,22,23\}$	00111100 10010011 11011110
E_{20} : $(56,57)$	14	14	114,688	$\{1,2,3,4,7,8,9,12,13,14,17,18,20,23\}$	11110011 10011100 11010010
E_{21} : $(58,59)$	14	14	114,688	$\{2,4,8,10,11,12,13,14,17,18,19,20,23,24\}$	01010001 01111100 11110011
E_{22} : $(60,61)$	14	14	114,688	$\{2,4,5,6,8,10,13,14,15,16,19,20,21,22\}$	01011101 01001111 00111100
E_{23} : $(62,63)$	14	14	114,688	$\{3,4,5,6,9,10,11,12,16,18,19,20,22,24\}$	00111100 11110001 01110101
E_{24} : $(64,65)$	14	14	114,688	$\{1,2,5,6,7,8,11,12,13,14,16,18,22,24\}$	11001111 00111101 01000101

S2.29. Parent-state and deletion certificates

The 49-ray, 87-ray, 40-ray, and 72-ray parents have no context states. The two smaller 68-ray parents are treated in Appendix S2.30.1. The first and last two have the independently verified complete bases printed in Appendix S1.16; the 87-ray parent contains the known 69-ray KS lineage. The 70-ray single-apex lift of the completed Cabello parent instead has the unique state supported at its apex.

For the real Yu–Oh parent, the intermediate 33 rays admit 192 assignments to the twenty bases alone and none after its sixteen additional pair exclusions. The sixteen missing pairs are uniquely completed by rays 34–49. The x-axis-plane extension by rays 26–29 alone has 72 basis assignments; all 24 original states extend. Including its eight additional pair exclusions leaves eight extensions, all with $s(2) = 1$. The z-axis-plane case is identical, with $s(5) = 1$. The two added bases on either axis each exclude twelve of the sixteen original axis-zero states when their further pair exclusions are included; eight original states are excluded by both, leaving none. Thus the axis forcing uses the

complete orthogonality constraints, not the added bases alone.

TABLE S83. State counts after deleting each single context of the real 49-ray parent. Every count is positive. Context numbering is Table S21.

Deleted context	States	Deleted context	States	Deleted context	States
1	12	13	2	25	2
2	4	14	4	26	2
3	4	15	4	27	2
4	32	16	16	28	2
5	2	17	4	29	2
6	2	18	4	30	2
7	4	19	4	31	2
8	4	20	4	32	2
9	12	21	2	33	2
10	4	22	2	34	2
11	4	23	2	35	2
12	2	24	2	36	2

All complete states of H_{65} are additionally supplied as `data/h65-all-states.csv.gz`. Each row gives its branch index, the fourteen-bit choice index within that branch, and its 65-bit state vector in vertex order. There are exactly $24 \times 2^{14} = 393{,}216$ data rows. This supplements the complete parametrization and the printed 48-state family; it does not assert that 48 is minimal. The 153 Hesse states, their event-order witnesses, and all original printed profiles remain in the accompanying Hesse data.

S2.30. States of the Cabello restrictions and completions

S2.30.1. States of the vertex-deleted 45-ray completion

The nine-context $C_{18,9}$, its eighteen-context faithful completion, and both 68-ray parents have no context states by their retained parity proofs. For the 45-ray construction, use exactly the local labels in Tables S31 and S32. Table S84 lists all twelve admissible core supports. For a support S_α, let F_α be the bridge indices missed by S_α. On every hit bridge both fresh values are zero; on each missed bridge independently choose exactly one of its two fresh vertices. This enumerates every context state once.

TABLE S84: All twelve core supports; $m = |F_\alpha|$. The final column specifies extendibility under all geometric pair exclusions.

α	S_α	m	Geometric extension
1	$\{1,6,11,12\}$	8	yes
2	$\{1,10,14,18\}$	8	yes
3	$\{3,7,10,14\}$	8	yes
4	$\{3,8,11,13\}$	8	yes
5	$\{3,13,14,17\}$	8	no
6	$\{2,6,10,15\}$	8	yes
7	$\{6,12,15,16\}$	8	no
8	$\{8,10,15,18\}$	8	yes
9	$\{9,16,18,19\}$	9	no
10	$\{9,17,18,19\}$	9	no
11	$\{2,11,13,19\}$	8	yes
12	$\{7,11,12,19\}$	8	yes

TABLE S85: The fourteen bridge pairs, fresh labels, and zero-profiles; each Z_m is the set of missed-bridge branch indices.

m	Bridge	Fresh pair	Z_m
1	$\{1,2\}$	$\{21,22\}$	$\{3,4,5,7,8,9,10,12\}$
2	$\{1,13\}$	$\{23,24\}$	$\{3,6,7,8,9,10,12\}$
3	$\{1,17\}$	$\{25,26\}$	$\{3,4,6,7,8,9,11,12\}$
4	$\{2,14\}$	$\{27,28\}$	$\{1,4,7,8,9,10,12\}$
5	$\{2,17\}$	$\{29,30\}$	$\{1,2,3,4,7,8,9,12\}$
6	$\{3,16\}$	$\{31,32\}$	$\{1,2,6,8,10,11,12\}$
7	$\{6,17\}$	$\{33,34\}$	$\{2,3,4,8,9,11,12\}$
8	$\{7,8\}$	$\{35,36\}$	$\{1,2,5,6,7,9,10,11\}$
9	$\{7,15\}$	$\{37,38\}$	$\{1,2,4,5,9,10,11\}$
10	$\{7,16\}$	$\{39,40\}$	$\{1,2,4,5,6,8,10,11\}$
11	$\{8,12\}$	$\{41,42\}$	$\{2,3,5,6,9,10,11\}$
12	$\{8,16\}$	$\{43,44\}$	$\{1,2,3,5,6,10,11,12\}$
13	$\{10,19\}$	$\{45,46\}$	$\{1,4,5,7\}$
14	$\{11,18\}$	$\{47,48\}$	$\{3,5,6,7\}$

Taking the first fresh vertex on every missed bridge gives $s_{\alpha,0}$, and taking the second gives $s_{\alpha,1}$. Table S86 explicitly enumerates all 24 canonical context states. Their profiles are nonzero and pairwise distinct on all 45 vertices. The twelve branch supports and fourteen distinct nonempty zero-profiles verify this by Lemma 21(3).

TABLE S86: All 24 canonical context-state supports of the 45-ray example. Some fail geometric pair exclusions.

α	ε	Support
1	0	$\{1,6,11,12,27,29,31,35,37,39,43,45\}$
1	1	$\{1,6,11,12,28,30,32,36,38,40,44,46\}$
2	0	$\{1,10,14,18,29,31,33,35,37,39,41,43\}$
2	1	$\{1,10,14,18,30,32,34,36,38,40,42,44\}$
3	0	$\{3,7,10,14,21,23,25,29,33,41,43,47\}$
3	1	$\{3,7,10,14,22,24,26,30,34,42,44,48\}$
4	0	$\{3,8,11,13,21,25,27,29,33,37,39,45\}$
4	1	$\{3,8,11,13,22,26,28,30,34,38,40,46\}$
5	0	$\{3,13,14,17,21,35,37,39,41,43,45,47\}$
5	1	$\{3,13,14,17,22,36,38,40,42,44,46,48\}$
6	0	$\{2,6,10,15,23,25,31,35,39,41,43,47\}$
6	1	$\{2,6,10,15,24,26,32,36,40,42,44,48\}$
7	0	$\{6,12,15,16,21,23,25,27,29,35,45,47\}$
7	1	$\{6,12,15,16,22,24,26,28,30,36,46,48\}$
8	0	$\{8,10,15,18,21,23,25,27,29,31,33,39\}$
8	1	$\{8,10,15,18,22,24,26,28,30,32,34,40\}$
9	0	$\{9,16,18,19,21,23,25,27,29,33,35,37,41\}$
9	1	$\{9,16,18,19,22,24,26,28,30,34,36,38,42\}$
10	0	$\{9,17,18,19,21,23,27,31,35,37,39,41,43\}$
10	1	$\{9,17,18,19,22,24,28,32,36,38,40,42,44\}$
11	0	$\{2,11,13,19,25,31,33,35,37,39,41,43\}$
11	1	$\{2,11,13,19,26,32,34,36,38,40,42,44\}$
12	0	$\{7,11,12,19,21,23,25,27,29,31,33,43\}$
12	1	$\{7,11,12,19,22,24,26,28,30,32,34,44\}$

Ten branches miss eight pairs and two miss nine. Therefore

$$|\mathcal{S}_{\mathrm{ctx}}| = 10\,2^8 + 2\,2^9 = 3{,}584, \qquad |\mathcal{S}_{\mathrm{geom}}| = 8\,2^8 = 2048. \tag{S22}$$

The geometric branches are $1, 2, 3, 4, 6, 8, 11, 12$; arbitrary fresh choices within each of these branches remain geometrically compatible. The other branches are excluded by the four forced pairs. Both full enumerations are supplied as compressed CSV data, in addition to this complete finite parametrization. Every geometric state vanishes on 9, 16, and 17, so this family is neither unital nor separating.

For an explicit counterexample, the context state supported at

$$\{6, 12, 15, 16, 21, 23, 25, 27, 29, 35, 45, 47\}$$

selects both 16 and 21, although $v_{16} \cdot v_{21} = 0$.

S2.30.2. Deleting one of the nine Cabello contexts

Number the nine contexts as the first nine rows of Table S29. For every deletion there are 26 context states, six geometric states, and a proper four-coloring of the remaining context graph. The color strings below use the ordered ray list $(1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20)$. The geometric states vanish precisely on the omitted context.

TABLE S87: All single-context deletion counts and four-coloring witnesses for $C_{18,9}$.

e	Omitted context	$\lvert\mathcal{S}_{\mathrm{ctx}}\rvert$	$\lvert\mathcal{S}_{\mathrm{geom}}\rvert$	Four-color string
1	$\{9, 10, 12, 13\}$	26	6	012102203020112333
2	$\{9, 11, 14, 15\}$	26	6	012021231012102333
3	$\{10, 11, 16, 17\}$	26	6	012021320012102333
4	$\{2, 3, 12, 18\}$	26	6	013103302120113322
5	$\{1, 3, 15, 19\}$	26	6	103130320102113232
6	$\{1, 2, 17, 20\}$	26	6	330130230103112222
7	$\{6, 7, 13, 18\}$	26	6	012221213030102333
8	$\{6, 8, 14, 19\}$	26	6	012202230013112333
9	$\{7, 8, 16, 20\}$	26	6	012021320012113332

For deletion of context 9, $\{7, 8, 16, 20\}$, Table S88 explicitly enumerates all 26 supports; exactly the six marked rows satisfy every geometric pair exclusion. All nine full deletion-state lists accompany the data.

TABLE S88: All 26 states after deleting context 9; geometric compatibility is marked explicitly.

j	Support	Geometric
1	$\{1, 6, 11, 12\}$	yes
2	$\{1, 7, 8, 11, 12\}$	no
3	$\{1, 7, 12, 14, 16\}$	no
4	$\{1, 8, 9, 16, 18\}$	no
5	$\{1, 10, 14, 18\}$	yes
6	$\{2, 6, 10, 15\}$	yes
7	$\{2, 7, 8, 10, 15\}$	no
8	$\{2, 8, 13, 15, 16\}$	no
9	$\{2, 7, 9, 16, 19\}$	no
10	$\{2, 11, 13, 19\}$	yes
11	$\{3, 6, 9, 17\}$	no
12	$\{3, 7, 8, 9, 17\}$	no
13	$\{3, 13, 14, 17\}$	yes
14	$\{6, 12, 15, 17\}$	no
15	$\{7, 8, 12, 15, 17\}$	no
16	$\{9, 17, 18, 19\}$	yes
17	$\{3, 6, 9, 16, 20\}$	no
18	$\{3, 7, 8, 9, 16, 20\}$	no
19	$\{3, 7, 10, 14, 20\}$	no
20	$\{3, 8, 11, 13, 20\}$	no
21	$\{3, 13, 14, 16, 20\}$	no
22	$\{6, 12, 15, 16, 20\}$	no
23	$\{7, 11, 12, 19, 20\}$	no
24	$\{7, 8, 12, 15, 16, 20\}$	no
25	$\{9, 16, 18, 19, 20\}$	no
26	$\{8, 10, 15, 18, 20\}$	no

S2.30.3. All single-ray deletions and the alternative parity subsystem

The next table records the two span failures and all forced-zero core rays after geometric pair exclusion for every possible single-ray deletion. An entry $\{u, v\} : z$ means $v_z \in \mathrm{span}(v_u, v_v)$. Every row has twelve compatible core states, 3,584 abstract completion states, and 2,048 geometric completion states, with exactly four forced extra pairs and no other extra orthogonality in the supplied realization. The complete core supports and 24 canonical states for every row are also supplied in the data.

TABLE S89: All eighteen vertex-deletion checks; the completed geometric families fail unitality and separation in every case.

Deleted ray	First span failure	Second span failure	Forced-zero core rays
1	$\{15, 19\} : 2$	$\{17, 20\} : 3$	$\{2, 3, 13\}$
2	$\{12, 18\} : 1$	$\{17, 20\} : 3$	$\{1, 3, 14\}$
3	$\{12, 18\} : 1$	$\{15, 19\} : 2$	$\{1, 2, 16\}$
6	$\{13, 18\} : 8$	$\{14, 19\} : 7$	$\{7, 8, 17\}$
7	$\{13, 18\} : 8$	$\{16, 20\} : 6$	$\{6, 8, 15\}$
8	$\{14, 19\} : 7$	$\{16, 20\} : 6$	$\{6, 7, 12\}$
9	$\{12, 13\} : 11$	$\{14, 15\} : 10$	$\{10, 11, 20\}$
10	$\{12, 13\} : 11$	$\{16, 17\} : 9$	$\{9, 11, 19\}$
11	$\{14, 15\} : 10$	$\{16, 17\} : 9$	$\{9, 10, 18\}$
12	$\{2, 3\} : 13$	$\{9, 10\} : 18$	$\{8, 13, 18\}$
13	$\{6, 7\} : 12$	$\{9, 10\} : 18$	$\{1, 12, 18\}$
14	$\{6, 8\} : 15$	$\{9, 11\} : 19$	$\{2, 15, 19\}$
15	$\{1, 3\} : 14$	$\{9, 11\} : 19$	$\{7, 14, 19\}$
16	$\{7, 8\} : 17$	$\{10, 11\} : 20$	$\{3, 17, 20\}$
17	$\{1, 2\} : 16$	$\{10, 11\} : 20$	$\{6, 16, 20\}$
18	$\{2, 3\} : 13$	$\{6, 7\} : 12$	$\{11, 12, 13\}$
19	$\{1, 3\} : 14$	$\{6, 8\} : 15$	$\{10, 14, 15\}$
20	$\{1, 2\} : 16$	$\{7, 8\} : 17$	$\{9, 16, 17\}$

For the alternative subsystem, take parent variant 2 of Table S35 and retain vertices 1–17 and 66–68. It is stateless. Each ray deletion instead leaves the positive geometric state count below; restriction of any such state shows that no eighteen-ray subset can be KS.

TABLE S90: Geometric state counts after every single-ray deletion of the twenty-ray parity subsystem.

Deleted ray	States	Deleted ray	States
1	8	2	8
3	8	4	28
5	28	6	8
7	8	8	8
9	12	10	12
11	12	12	10
13	10	14	10
15	10	16	10
17	10	66	8
67	8	68	8

S3. SUPPLEMENTARY PROOFS, REDUCTIONS, AND EXACT VERIFICATION CERTIFICATES

S3.1. Formulas used by the finite certificates

The following formulas restore the explicit conventions used by the later certificates. State indices are local to the corresponding complete table. For Hesse, the six-state vertex-separating family, its seven-state event-separating extension, and the eighteen-state order-determining family are respectively

$$\Omega_6 = \{s_3, s_{31}, s_{75}, s_{89}, s_{113}, s_{151}\}. \tag{S23}$$

$$\Omega_7 = \Omega_6 \cup \{s_{84}\} \tag{S24}$$

$$\begin{aligned}\Omega_{18} = \{&s_2, s_8, s_{20}, s_{30}, s_{40}, s_{45},\\ &s_{54}, s_{64}, s_{73}, s_{79}, s_{83}, s_{93},\\ &s_{105}, s_{109}, s_{119}, s_{123}, s_{127}, s_{150}\}.\end{aligned} \tag{S25}$$

The members of Ω_6 are denoted $\sigma_1, \dots, \sigma_6$ in the displayed order. Their atom profiles and the nontrivial atom–coatom order are

$$\kappa(i) = (\sigma_1(i), \dots, \sigma_6(i)). \tag{S26}$$

$$a_i \le a_j^{\perp} \quad \Longleftrightarrow \quad \{i, j\} \in E(\Gamma(H_{37})). \tag{S27}$$

For H_{65}, the following core coloring extends to the completed hypergraph. A base-compatible independent support S_α is extended canonically using $\varepsilon \in \{0, 1\}$. In the plane formula, η_A, η_B are an integer orthogonal basis and $n_A = \|\eta_A\|^2$, $n_B = \|\eta_B\|^2$.

$$\begin{aligned}(c(1), \dots, c(17)) = (&0, 1, 2, 3, 0, 1, 2, 4, 0, 1, 3,\\ &3, 4, 2, 1, 0, 2).\end{aligned} \tag{S28}$$

$$\begin{aligned}&\big(s_{\alpha,\varepsilon}(w_{m,1}),\ s_{\alpha,\varepsilon}(w_{m,2})\big)\\ &\quad = \begin{cases}(0, 0), & |S_\alpha \cap \{u_m, v_m\}| = 1,\\ (1-\varepsilon,\ \varepsilon), & S_\alpha \cap \{u_m, v_m\} = \emptyset.\end{cases}\end{aligned} \tag{S29}$$

$$\begin{aligned}w_{m,1}(a, b) &= a\,\eta_A + b\,\eta_B,\\ w_{m,2}(a, b) &= -n_B b\,\eta_A + n_A a\,\eta_B\end{aligned} \tag{S30}$$

The eight rays adjoined to completed Yu–Oh and the nine uncovered pairs of the Cabello subsystem are fixed by the following local labels. The last display records the completed historical Γ_1 TITS incidence.

$$\begin{aligned}&(0, 1, 2), (0, 2, -1), (0, 1, -2), (0, 2, 1),\\ &(1, 2, 0), (2, -1, 0), (1, -2, 0), (2, 1, 0).\end{aligned} \tag{S31}$$

$$\begin{aligned}&\{1, 13\}, \{2, 14\}, \{3, 16\}, \{6, 17\}, \{7, 15\},\\ &\{8, 12\}, \{9, c\}, \{10, b\}, \{11, a\}.\end{aligned} \tag{S32}$$

$$E(I_{16,9}) = E(B_{13,7}) \cup \{\{4, 11, 24\}, \{4, 10, 18\}\}. \tag{S33}$$

a. Fractional value of the faithful Cabello completion. The geometric orthogonality graph of $C_{18,9}$ has independence number four and exactly 54 independent four-sets, with each of its 18 rays belonging to twelve of them. Assigning weight 1/12 to each set gives a fractional cover of total weight 9/2, matching 18/4. Fresh pair completion preserves this value by the main completion proposition, proving $\chi_f(C_{36,18}) = 9/2$ as recorded in the main comparison. The complete 54-set certificate is in the electronic audit data.

S3.2. Snark obstructions and quantitative robustness bounds

S3.2.1. Cubic graphs, Blanuša snarks, and Isaacs dot products

For a finite simple cubic graph X, let $H(X)$ be the context hypergraph whose vertices are the edges of X and whose contexts are the three edges incident with each vertex of X. Then $\Gamma(H(X)) = L(X)$ and $\chi(H(X)) = \chi'(X)$. An orthogonal ray assignment for $H(X)$ in $\mathbb{F}^3$ is an assignment of nonzero vectors to the edges of X such that incident edges are orthogonal.

Silva and Ibiapina [4, Question 6] asked whether an orthogonal vector edge coloring in $\mathbb{R}^3$ can exist when $\chi'(X) = 4$. Theorem 11 excludes the Petersen graph P even over $\mathbb{C}$. Using the multipole forcing law of Theorem 11(1), we can extend this exclusion to the two 18-vertex Blanuša snarks [5] and the infinite family of Isaacs dot-product snarks [6].

Recall that for a cubic graph G with two nonadjacent edges $e_1 = \{x_1, y_1\}$ and $e_2 = \{x_2, y_2\}$, the Isaacs dot product $G \cdot P$ is formed by removing e_1, e_2 from G, removing two adjacent vertices u, v from the Petersen graph P (leaving the 4-pole $P \setminus \{u, v\}$ whose dual hypergraph is precisely H_8), and attaching the two dangling edges $\{2, 8\}$ of u to $\{x_1, y_1\}$ and the two dangling edges $\{13, 14\}$ of v to $\{x_2, y_2\}$; the two distinct choices of nonadjacent edges in $G = P$ produce the two Blanuša snarks B_1 and B_2 [5–7].

Corollary 1 (Obstruction for Blanuša and dot-product snarks)**.** *If a cubic graph G admits no orthogonal vector edge coloring in $\mathbb{C}^3$, then neither does any Isaacs dot product $G \cdot P$. In particular, both 18-vertex Blanuša snarks $B_1, B_2 = P \cdot P$ and all iterated dot products $X = P \cdot P \cdots P$ satisfy $\chi(H(X)) = \chi'(X) = 4 > 3 = \omega(L(X))$ and admit no orthogonal vector edge coloring in $\mathbb{C}^3$.*

Proof. In any orthogonal vector edge coloring of $G \cdot P$ in $\mathbb{C}^3$, the restriction to the embedded copy of $H_8 = H(P \setminus \{u, v\})$ satisfies $[v_2] = [v_8]$ and $[v_{13}] = [v_{14}]$ by Theorem 11(1). Assigning the common ray $[v_2] = [v_8]$ to $e_1 = \{x_1, y_1\}$ and the common ray $[v_{13}] = [v_{14}]$ to $e_2 = \{x_2, y_2\}$ preserves orthogonality at x_1, y_1, x_2, y_2 and therefore yields a valid orthogonal vector edge coloring of G in $\mathbb{C}^3$. Since P admits no such coloring by Theorem 11, induction excludes B_1, B_2 and all iterated dot products with P. □

Generalized Petersen graphs [8] and flower snarks provide further families for comparing real, complex, injective, and edge-faithful realizability.

S3.2.2. Explicit analytic lower bound and tetrahedral tight-frame upper bounds for Δ_3

Let A be the 30 edges of $\Gamma(G_{32})$ and define the minimum total squared orthogonality error in $\mathbb{C}^3$:

$$\Delta_3 = \min_{\substack{v_i \in \mathbb{C}^3 \\ \|v_i\|=1 \ (1 \le i \le 15)}} \sum_{\{i,j\} \in A} |\langle v_i, v_j \rangle|^2. \tag{S34}$$

By compactness of $(S^5)^{15}$ and Theorem 11, $\Delta_3 > 0$. We now establish both a certified rational analytic lower bound and explicit tetrahedral tight-frame upper bounds.

Proposition 2 (Quantitative bounds on Δ_3)**.** *Every assignment of unit vectors $v_1, \ldots, v_{15} \in \mathbb{C}^3$ satisfies*

$$\varepsilon := \max_{\{i,j\} \in A} |\langle v_i, v_j \rangle| > \frac{1}{120}, \tag{S35}$$

and the minimum total squared error in Eq. (S34) satisfies

$$\begin{aligned} \frac{1}{14400} < \Delta_3 &\le \min_{a^2+b^2+c^2=1} F(a, b, c) \\ &\le \frac{9275914990340646}{10001032226635921} < 0.927496 < \frac{18}{19}, \end{aligned} \tag{S36}$$

where

$$F(a,b,c) = 12a^2 + 12(ab+bc-ca)^2 + 6(b^2-a^2-c^2)^2. \tag{S37}$$

Proof. To prove the lower bound (S35), let $\varepsilon = \max_{\{i,j\}\in A} |\langle v_i, v_j\rangle|$. For any context $e = \{x,y,z\}$, the Gram matrix of $\{v_x, v_y, v_z\}$ has diagonal entries 1 and off-diagonal entries of modulus at most ε, so Gershgorin's circle theorem implies $\|I_3 - (v_x v_x^\dagger + v_y v_y^\dagger + v_z v_z^\dagger)\|_{\mathrm{op}} \le 2\varepsilon$. Hence for any unit vectors $u, w \in \mathbb{C}^3$,

$$\Big|1 - \big(|\langle u,x\rangle|^2 + |\langle u,y\rangle|^2 + |\langle u,z\rangle|^2\big)\Big| \le 2\varepsilon, \tag{S38}$$

and if $\{u,x\}, \{w,z\} \in A$, then

$$\big|\langle u,w\rangle - \langle u,y\rangle\langle y,w\rangle\big| \le 4\varepsilon. \tag{S39}$$

In $\{5,6,7\}$ and $\{9,10,11\}$, the vectors $w_7 = v_{12} - v_5\langle v_5, v_{12}\rangle$ and $w_9 = v_4 - v_{11}\langle v_{11}, v_4\rangle$ satisfy $\|w_7 - v_7\langle v_7, v_{12}\rangle\| \le 3\varepsilon$ and $\|w_9 - v_9\langle v_9, v_4\rangle\| \le 3\varepsilon$, so $|\langle w_7, w_9\rangle| \le 7\varepsilon$. Expanding $\langle w_7, w_9\rangle$ and using $|\langle v_{12}, v_{11}\rangle| \le \varepsilon$, $|\langle v_5, v_4\rangle| \le \varepsilon$, and Eq. (S39) on $\{1,2,3\}$ for $x \in \{v_{11}, v_{12}\}$, $y \in \{v_4, v_5\}$ yields

$$\big|\langle v_{12}, v_2\rangle\langle v_2, v_4\rangle\big| \le (7+2+4+12)\varepsilon = 25\varepsilon.$$

By reflection symmetry, we may assume $|\langle v_2, v_4\rangle| \le 5\sqrt{\varepsilon}$. Applying Eq. (S38) to v_4 in $\{1,2,3\}$ and to v_2 in $\{3,4,5\}$ gives $|\langle v_1, v_4\rangle|^2 \ge 1 - 27\varepsilon - \varepsilon^2$ and $|\langle v_2, v_5\rangle|^2 \ge 1 - 27\varepsilon - \varepsilon^2$, so $\min_\theta \|v_1 - e^{i\theta} v_4\| \le \sqrt{54\varepsilon + 2\varepsilon^2}$ and $\min_\theta \|v_2 - e^{i\theta} v_5\| \le \sqrt{54\varepsilon + 2\varepsilon^2}$. Since $|\langle v_4, v_{10}\rangle| \le \varepsilon$ and $|\langle v_5, v_7\rangle| \le \varepsilon$, Eq. (S38) applied to v_1 in $\{9,10,11\}$ and to v_2 in $\{7,8,9\}$ gives

$$|\langle v_i, v_9\rangle|^2 \ge 1 - 2\varepsilon - \varepsilon^2 - \big(\varepsilon + \sqrt{54\varepsilon + 2\varepsilon^2}\big)^2 \qquad (i = 1,2).$$

Comparing the sum of these two inequalities with $|\langle v_1, v_9\rangle|^2 + |\langle v_2, v_9\rangle|^2 \le 1 + 2\varepsilon$ from $\{1,2,3\}$ gives

$$1 \le 6\varepsilon + 2\varepsilon^2 + 2\big(\varepsilon + \sqrt{54\varepsilon + 2\varepsilon^2}\big)^2.$$

The right-hand side is strictly increasing for $\varepsilon > 0$, and at $\varepsilon = 1/120$ we have $54\varepsilon + 2\varepsilon^2 = 3241/7200 < (27/40)^2$, so the right-hand side is strictly bounded above by

$$\frac{6}{120} + \frac{2}{120^2} + 2\left(\frac{1}{120} + \frac{27}{40}\right)^2 = \frac{1417}{1440} < 1,$$

certifying $\varepsilon > 1/120$ and $\Delta_3 \ge \varepsilon^2 > 1/14400$.

For the upper bounds, set $v_4 = e_1$, $v_8 = e_2$, $v_{12} = e_3$, and for any unit vector $(a,b,c) \in S^2$, define the remaining 12 unit vectors as representatives of the tetrahedral A_4-orbit of rays:

$$\begin{aligned}
v_1 &= (-b,-c,-a)^T, & v_2 &= (c,-a,b)^T,\\
v_3 &= (-a,-b,c)^T, & v_5 &= (-a,b,c)^T,\\
v_6 &= (-b,c,-a)^T, & v_7 &= (c,a,b)^T,\\
v_9 &= (-c,-a,b)^T, & v_{10} &= (-a,b,-c)^T,\\
v_{11} &= (b,-c,-a)^T, & v_{13} &= (a,b,c)^T,\\
v_{14} &= (b,c,-a)^T, & v_{15} &= (-c,a,b)^T.
\end{aligned} \tag{S40}$$

These 15 vectors form a tight frame $\sum_{i=1}^{15} v_i v_i^T = 5I_3$, and the sum of squared overlaps on the 30 edges of $\Gamma(G_{32})$ equals $F(a,b,c)$ in Eq. (S37). Evaluating Eq. (S40) at the normalized integer direction $(a,b,c) = (1,3,-3)/\sqrt{19}$ yields rational squared overlaps with $\Delta_3 \le 342/361 = 18/19 \approx 0.947368$ and $\max_{\{i,j\}\in A} |\langle v_i, v_j\rangle|^2 = 1/19$; evaluating at the refined normalized integer direction $(a,b,c) = (2332, 7101, -6644)/\sqrt{100005161}$ yields the certified rational bound in Eq. (S36). □

Numerical minimization of F on S^2 (as well as unrestricted local searches over $(S^5)^{15}$) finds an attained candidate minimum near 0.927495724691 at $(a,b,c) \approx (0.23319713, 0.71005263, -0.66441280)$; certifying whether the unrestricted global minimum Δ_3 coincides with $\min_{S^2} F$ remains an open optimization problem. Choosing instead the rational unit vector $(a,b,c) = \frac{1}{3}(1,2,-2)$ in Eq. (S40) makes $ab + bc - ca = 0$, so four of the ten contexts of G_{32} ($\{1,2,3\}, \{5,6,7\}, \{9,10,11\}, \{13,14,15\}$) become exact orthonormal bases of $\mathbb{R}^3$, all 15 rays are distinct, and $\Delta_3 \le 38/27$ with $\max_{\{i,j\}\in A} |\langle v_i, v_j\rangle| = 1/3$.

S3.3. Hesse H_{37}: minimum state families

All state indices in this appendix use the canonical enumeration of Appendix S2.27. The minima for unital vertex separation and event separation have the counting lower bounds $2^5 - 1 < 37$ and $2^6 < 76$. The families in Eqs. (S23) and (S24) attain those bounds: their profiles give respectively 37 distinct nonzero atom profiles and 76 distinct event profiles. The seven-state family still has 462 ordered event pairs for which inclusion of the profiles does not imply the lattice order. The family in Eq. (S25) has no such pair and hence gives the upper bound eighteen for order determination. This appendix supplies a computer-assisted proof of the matching lower bound.

S3.3.1. Reduction to families of seventeen states

Let $D = \{P_0, P_1, P_2, Q_0, Q_1, Q_2\}$. For each unordered pair $\{u, v\} \subset D$, order determination requires a state with $s(u) = s(v) = 1$: the pair is nonorthogonal, so $u \not\leq v^\perp$. Every state takes value one on at most two members of D. Consequently the fifteen pairs require fifteen different states. Table S91 gives their complete witness classes. These classes are disjoint, their union T has 27 states, and a choice of one witness from each class has $1^3 2^{12} = 4{,}096$ possibilities.

TABLE S91: Complete joint-one witness classes for the fifteen pairs in D. Each entry lists all canonical indices r for which $s_r(u) = s_r(v) = 1$.

Pair $\{u, v\}$	Witness indices r
$\{P_0, P_1\}$	150
$\{P_0, P_2\}$	73
$\{P_0, Q_0\}$	93, 145
$\{P_0, Q_1\}$	127, 133
$\{P_0, Q_2\}$	79, 99
$\{P_1, P_2\}$	2
$\{P_1, Q_0\}$	105, 139
$\{P_1, Q_1\}$	14, 119
$\{P_1, Q_2\}$	40, 48
$\{P_2, Q_0\}$	54, 60
$\{P_2, Q_1\}$	8, 26
$\{P_2, Q_2\}$	20, 68
$\{Q_0, Q_1\}$	109, 113
$\{Q_0, Q_2\}$	83, 87
$\{Q_1, Q_2\}$	30, 34

The nine nonorthogonal pairs $\{z, C_{ak}\}$ require further joint-one witnesses. No state in T witnesses any of them. Indeed, if a state activates a P vertex then $s(z) = 0$. If it instead activates two Q vertices, the state equations force the selected M vertices into the remaining Q column, and all nine C values are zero. Any state outside T witnesses at most six of the nine pairs: for $s(z) = 1$, the three selected M vertices force at least three C values to vanish. Thus at least two states outside T are necessary.

It follows that an order-determining family of at most seventeen states would have exactly seventeen members: one from each of the fifteen classes in Table S91, and two outside T that together witness all nine $\{z, C_{ak}\}$ pairs. Enumerating the $\binom{126}{2}$ unordered pairs outside T leaves exactly 132 with this property. The reduction therefore leaves precisely

$$4{,}096 \cdot 132 = 540{,}672 \tag{S41}$$

candidate families. The nine pairs alone do not require three extra states; for example, s_1 and s_{72} jointly witness all nine. The remaining order requirements must also be checked.

S3.3.2. Exhaustive order certificate

For each of the 5,407 ordered event pairs $x \not\leq y$, define its nonempty witness set

$$W_{xy} = \{r \in \{1, \ldots, 153\} : s_r(x) = 1,\ s_r(y) = 0\}. \tag{S42}$$

A family is order determining exactly when its index set meets every W_{xy}. Repeated sets and proper supersets can be discarded, because meeting an inclusion-minimal set also meets every set containing it. There are exactly 81

inclusion-minimal witness sets. The exhaustive check forms each of the 540,672 families in Eq. (S41) and counts the minimal witness sets that it meets. The largest count is 75, strictly below 81. Hence no family of seventeen states is order determining. Together with Ω_{18}, this proves the minimum eighteen.

The accompanying Hesse extension checker, invoked by the joint entry point, reconstructs the contexts and all 153 states from their symbolic parametrization, checks their agreement with `hesse37-data.json`, and performs this finite search using integer bit sets and the Python standard library. No numerical tolerance, optimization solver, or unreported search cutoff is used. It also verifies the printed seven- and eighteen-state families and every entry of Table S91. From the extracted joint project directory, run the joint entry point, without Python's optimization option:

```
python verification/verify-integrated.py
```

The generated file `verification/_work/hesse/hesse37-extensions-certificate.json` records all 81 minimal witness sets, a representative event pair for each, all 132 possible extra pairs, the coverage-count distribution over the 540,672 candidates, and all event profiles for Ω_6, Ω_7, and Ω_{18}. The generated file `verification/_work/hesse/hesse37-order18-witnesses.csv` records a witness in Ω_{18} for every $x \not\leq y$. These two files use the same numbered event labels as the full-family certificate in Appendix S2.27.6.

The program additionally checks Berge girth five from the incidence graph, computes every meet and join in the 76-event order, verifies the orthomodular law, and verifies the projector sum in Eq. (30) exactly in $\mathbb{Q}(\zeta)$. The nonexistence of an injective real representation is proved analytically in Sec. V A.

S3.4. Four-dimensional $H_{\mathrm{YO}}^{(4)}$ and H_{65}: completion and verification

The coordinates in Table S17 can be reproduced by the following finite selection rule. Normalize an integer vector by dividing by the greatest common divisor of its entries and choosing the sign so that its first nonzero entry is positive. Form the set of normalized nonzero vectors obtained from $\{-3,-2,\dots,3\}^4$, ordered by squared norm, then maximum absolute entry, then lexicographically. Process bridge edges in the order $e_{13},\dots,e_{24},e_1,\dots,e_{12}$. For each pair, choose η_A to be the first vector in this ordered pool orthogonal to its endpoints, and η_B the first such vector also orthogonal to η_A.

For all $(a,b) \in \{-7,-6,\dots,7\}^2 \setminus \{(0,0)\}$, form the pair in Eq. (S30) and normalize both vectors. Discard pairs that repeat an already placed ray or have an orthogonality to any already placed vertex other than the two endpoints. Among the remaining ordered pairs choose the lexicographically smallest score

$$\left(\max_{j,k} |(w_{m,j})_k|,\ \|w_{m,1}\|^2 + \|w_{m,2}\|^2,\ w_{m,1}, w_{m,2} \right). \tag{S43}$$

In Eq. (S43), the order on the last two entries is the usual lexicographic order on their integer coordinates. This rule yields exactly the 48 completion vectors in Table S17. It is a recipe for this specific example; the general existence assertion is established independently by Lemma 21.

The companion verifier parses the integrated coordinate and state tables, reproduces this completion rule and all 393,216 state extensions, and checks the maximal cliques, fractional-coloring certificate, atom-order failure, forced hub-completion zero, and warm-up data. It is included with the complete state file and invoked by the joint entry point in Sec. S3.7.

S3.5. Exact Cabello completion and forced-zero certificates

For each of the nine uncovered pairs in Eq. (S32), rational row reduction verifies that its two endpoint vectors span no third core ray. If two endpoint planes were orthogonal, their four endpoints would form another full basis. Exact four-clique enumeration gives only the original nine bases, all of whose pairs are already covered. This excludes such an orthogonal-plane pair. The integer vectors in Table S28 directly verify the resulting faithful completion: 36 distinct projective rays, eighteen full bases, and 108 orthogonal pairs, with every pair covered by exactly one listed basis. The nine-context graph and the full eighteen-ray orthogonality graph both fail four-colorability and admit five-colorings, as checked by exact finite search in the companion certificate. Lemma 21(2) transfers the chromatic value to the completion.

For the vertex-deleted core with c removed, rational row reduction finds exactly the span failures

$$v_{16} \in \mathrm{span}(v_1, v_2), \qquad v_{17} \in \mathrm{span}(v_7, v_8).$$

There is no orthogonal-plane-pair obstruction. The finite-avoidance choice in Lemma 21 can therefore avoid all extra incidences except those forced by these two spans; the explicit 45-ray inventory has exactly the four pairs printed in Sec. S1.17.2. The two fresh vectors completing $\{1,2\}$ must both be orthogonal to 16, and those completing $\{7,8\}$ must both be orthogonal to 17.

The twelve rows of Table S84 give a short complete proof of the resulting obstruction. Whenever 16 is selected, neither 1 nor 2 is selected. The context completing $\{1,2\}$ then needs one fresh value equal to one, while geometric exclusion with 16 requires both to be zero. Hence every geometric extension has $s(16)=0$. The same argument with $\{7,8\}$ gives $s(17)=0$. These implications use the displayed core states in addition to the span incidences; the span incidences alone would not imply them. The retained basis $\{10,11,16,17\}$ gives $s(10)+s(11)=1$. If $s(9)=1$, the other two hub bases give $s(10)=s(11)=0$, a contradiction. Thus 9, 16, and 17 have identical zero profiles.

Conversely, each of the eight indicated core rows avoids 16 and 17, so all its fresh-pair choices satisfy the four extra exclusions. Each misses eight bridges. This proves the exact count $8\,2^8=2{,}048$, as well as failure of geometric unitality and separation. The two excluded core rays remain impossible for every fresh-pair completion of these fixed planes; additional accidental orthogonalities cannot restore them. Table S89 supplies the individual span checks for all eighteen deletions. Their full core tables, exact integer completions, and extension counts are included in `data/cabello18-review-checks.json`.

To audit the ancestry result, enumerate every four-clique after adjoining either three-ray choice to all 65 original rays. In both cases there are exactly 35 bases, 195 geometric pairs, and 192 context-covered pairs; the maximum basis intersection is two. The first union contains the displayed eighteen-ray parity proof. The second union contains the twenty-ray proof with multiplicities $4,4,2,\ldots,2$ and eleven contexts. Each of its twenty ray deletions has a compatible assignment by Table S90. All coordinates, full basis lists, and the preserved H_{65} restriction are checked independently of the unrestricted ancestry existence proposition.

S3.6. Transverse extension and criticality certificates

For $d=3$, number the two copies of an n-vertex input as i and $n+i$, and the fresh transverse vertex as $2n+i$. The transverse contexts are $\{i,n+i,2n+i\}$ and the horizontal contexts are the two numbered copies of the original list. This explicitly specifies all 45 vertices and 33 contexts for $T(H_9)$, and all 75 vertices and 57 contexts for $T(H_{\mathrm{YO}}^{+})$.

For each deleted original context e, a strong three-coloring of $H\setminus\{e\}$ having two equal colors and a singleton color on e gives a state after deletion of one horizontal copy of e. Use the repeated-color indicator on the deleted copy and the singleton-color indicator on the intact copy. Both have exactly one active vertex in each of their required contexts and their supports are disjoint; the remaining transverse values are forced. For deletion in the other copy, interchange the two indicators. Table S92 supplies an explicit coloring for every required context of both inputs. The color strings use original vertex order, and their labels 0,1,2 may be permuted.

TABLE S92: Explicit deletion colorings with pattern (1,1,2) and the number of states after deleting the corresponding first-copy horizontal context.

Input	Deleted original context	Color string	Extension states
H_9	$\{1,2,3\}$	212010201201122	20
H_9	$\{1,11,12\}$	201020120121221	20
H_9	$\{2,8,15\}$	021020120121220	12
H_9	$\{3,4,5\}$	012120102012212	20
H_9	$\{4,10,13\}$	012102120121000	12
H_9	$\{5,6,7\}$	012010120121220	20
H_9	$\{6,12,14\}$	012102102012202	12
H_9	$\{7,8,9\}$	012010200121222	20
H_9	$\{9,10,11\}$	012010201121222	20
H_{YO}^{+}	$\{1,2,3\}$	200021210112020102011222 1	16
H_{YO}^{+}	$\{1,10,19\}$	210021201212020001011202 1	8
H_{YO}^{+}	$\{1,13,20\}$	120210201012120011200220 1	8
H_{YO}^{+}	$\{2,5,8\}$	021021012120201122211001 2	24
H_{YO}^{+}	$\{3,11,14\}$	012021102120200122210001 1	8
H_{YO}^{+}	$\{3,12,25\}$	021210102101222020210101 0	8
H_{YO}^{+}	$\{4,5,6\}$	021200210102121110220221 0	16
H_{YO}^{+}	$\{4,10,18\}$	021210201202121010120201 0	8

Input	Deleted original context	Color string	Extension states
H^+_{YO}	$\{4, 11, 24\}$	02120101212020112021100 02	8
H^+_{YO}	$\{6, 12, 17\}$	02121010210022211021010 12	8
H^+_{YO}	$\{6, 13, 22\}$	02101210212020011221000 12	8
H^+_{YO}	$\{7, 8, 9\}$	02101200212010111222001 2	16
H^+_{YO}	$\{7, 11, 15\}$	02120101210022112021100 12	8
H^+_{YO}	$\{7, 13, 21\}$	02101210212010011222000 12	8
H^+_{YO}	$\{9, 10, 23\}$	02121020110212101022020 10	8
H^+_{YO}	$\{9, 12, 16\}$	02121010210222201021010 10	8

Deleting one transverse context T_v instead permits an ordered pair of original state supports S_1, S_2 exactly when $S_1 \cap S_2 \subseteq \{v\}$. In these non-three-colorable inputs disjoint pairs do not exist, so the intersection must equal $\{v\}$; the isolated fresh vertex has two choices. For H_9 this gives four states for each $v = 1, \ldots, 12$ and twelve states for each $v = 13, 14, 15$. Together with Table S92, this proves criticality of all 33 contexts. For completed Yu–Oh it gives zero states at $v = 1, \ldots, 13$ and eight states at $v = 14, \ldots, 25$. Thus horizontal criticality does not imply criticality of this full transverse construction. The corresponding G_{32} extension has 45 vertices and 35 contexts and is stateless. Its distinct six state supports have pairwise intersection size one; for H_9 the distinct eight supports have intersection size one in 21 pairs and size three in seven pairs.

S3.7. Data and verification

The companion electronic package contains coordinate and context inventories, complete state and coloring enumerations, verification programs, exact parent inventories, and finite certificates. Vertex labels and state orderings agree with the tables in this supplement. The verification programs check the common Yu–Oh numbering, the published 49-ray parent map, the parent vectors and contexts, the full H_{65} state inventory, and cross-reference targets. The finite lower-bound certificate for the eighteen-state Hesse minimum is the exhaustive 540,672-candidate argument of Sec. S3.3.

Run the companion verification entry point from the project directory:

```
python verification/verify-integrated.py
```

The exact checks use rational or $\mathbb{Q}(\zeta)$ arithmetic, with $\zeta^2 + \zeta + 1 = 0$. Reports and hashes accompany the source.

The finite checks certify the instances and certificates they explicitly test. The analytic realization and existence results also rely on their proofs in the text.

S3.8. Historical Kochen–Specker gadget certificates (1965 Fig. 1 and 1967 Γ_1)

S3.8.1. The 1965 triangles and the completed historical TITS

Identifying the repeated occurrences of vertices 1 and 2 in Kochen–Specker's 1965 Fig. 1 gives

$$\{134, 325, 2\,9\,11, 11\,1\,10, 479, 5\,8\,10, 678\}. \tag{S44}$$

These are also the triples used in the displayed function definitions on p. 182 of Ref. [9]. Their four state supports are

$$\{1, 5, 6, 9\}, \quad \{2, 4, 6, 10\}, \quad \{3, 6, 9, 10\}, \quad \{4, 5, 6, 11\}.$$

The TIFS between 1 and 2 is nonvacuous. Directly, if both were true, contexts 134, 325, 2 9 11, and 11 1 10 would make 3,4,5,9,10,11 false; contexts 479 and 5 8 10 would then make both 7 and 8 true, contradicting 678.

Every state has 6 true and 7,8 false, so no three-coloring can exist: the three color-class states of any coloring would need to be disjoint. The graph is abstract rather than an injective ray example. In a purported injective realization in $\mathbb{C}^3$, vertices 1 and 9 are both orthogonal to 4 and 11. Distinct rays 4 and 11 span a plane, whose orthogonal complement is one-dimensional; thus rays 1 and 9 coincide. This proves the obstruction even over the complex field. No claim that the 1965 authors presented this graph as a Hilbert-space realization is intended.

For the 1967 Γ_1[10, pp. 68,70], the correspondence between its original vertices and $I_{16,9}$ is

Γ_1	a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9
$I_{16,9}$	10	9	1	7	3	8	2	11	4	24

The already complete triangles are $\{a_1, a_3, a_5\}$, $\{a_2, a_4, a_6\}$, and $\{a_7, a_8, a_9\}$. The six remaining edges

$$a_0a_1,\ a_0a_2,\ a_0a_8,\ a_3a_7,\ a_4a_7,\ a_5a_6$$

are completed by fresh vertices $23, 19, 18, 15, 14, 5$, respectively. The resulting nine contexts are exactly Eq. (S33). This also identifies its directed TITS $a_0 \to a_9$ with $10 \to 24$.

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